A Turkish Introduction to Minimalist Syntax

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Contents:

CHAPTER 1: What does grammar do?

This book is intended both as a second introduction to syntactic theory and as a resource for self-study. It aims to provide (a) an introduction to the basics of syntactic analysis using Turkish data and (b) an exploration of various properties of Turkish grammar with the tools of syntactic analysis. We will see how linguistic patterns in Turkish motivate several aspects of current syntactic theories of Minimalist variety and how such theories help us discover deep connections between different corners of Turkish grammar.

One might, of course, be under the impression that a surface description of Turkish grammar captures all that there is to know about this language: Determine the category of each linguistic expression (that is, determine whether it is a noun, or a verb, or a postposition etc.), explain how it is used inside phrases and sentences and be done with syntactic analysis! Luckily, this is far from the truth. As we shall see in this book, questions invited by the syntactic theory help us unearth many intricacies of Turkish grammar that would otherwise go unnoticed.

1.1.The purpose of grammar

When given a sequence of words, a speaker of Turkish typically has clear intuitions as to whether this sequence belongs to their language. Consider the list of expressions in (1), which are obtained by ordering the Turkish words *çalıştım* 'I worked'*, boyunca* 'throughout' and *gece* '(the) night'.

(1) a. boyunca çalıştım gece.
b. boyunca gece çalıştım.
c. çalıştım boyunca gece
d. çalıştım gece boyunca
e. gece çalıştım boyunca
f. gece boyunca çalıştım

When presented the list in (1), a speaker of Turkish tells us that while (d) and (f) are fine sentences that they might use at some point in their lives, the expressions in (a), (b), (c) and (e) are nothing but word salad. There is clearly something off about them. We put the asterisk symbol ('*') in front of any expression that the speaker we are working with (typically, our very selves!) reports as being 'bad' or 'unacceptable'.

(2) a. *boyunca çalıştım gece.
b. *boyunca gece çalıştım.
c. *çalıştım boyunca gece
d. çalıştım gece boyunca
e. *gece çalıştım boyunca
f. gece boyunca çalıştım

What is the nature of the difference between those expressions that are acceptable and those that are not? The central idea of syntactic analysis is that this is a distinction in grammaticality. Those expressions that are allowed by Turkish grammar are experienced as being 'acceptable' and those that are not explainable by the rules of grammar are judged as 'unacceptable'. Now it is important to make note of the fact that the idea that there is a one-to-one relation between grammaticality and acceptability is a simplifying assumption. There may be many factors other than grammar that enter into whether an expression is judged as acceptable or unacceptable. For instance, it is quite possible that the expression in (3) will be considered unacceptable by some Turkish speakers. (We first give a word-to-word translation of Turkish expressions, which is called *glossing*. For now, the glosses we provide do not contain information about the pieces that make up the words, but this will change in the later chapters. We also give the English translation of the expression under discussion inside single quotation marks.)

(3) tek eleman-ı tek eleman-ı **boş küme** olan küme olan küme
only member-its only member-its empty set being set being set
'the set whose only member is the set whose only member is the empty set'

When we are given a pen and a paper and shown the steps in which the expression in (3) can be constructed, we come to realize that it is a Turkish phrase like any other with a determinate meaning.

(4)

STEP-BY-STEP DERIVATION	MEANING
boş küme 'empty set'	{}
tek elemanı **boş küme** olan küme 'the set whose only member is the empty set'	{{}}
tek elemanı **tek elemanı boş küme olan küme** olan küme **'**the set whose only member is the set whose only member is the empty set'	{{{}}}

There is something about constructions that are embedded inside similar constructions that makes them difficult for us to process and interpret in our minds. Hence, it is quite likely that the initial reaction of Turkish speakers to (3) is indicative of this difficulty associated with processing rather than the grammaticality status of (3). There is a distinction to be made between the **competence** of a language user, i.e. the grammar they have internalized as part of their linguistic system, and **performance limitations** arising from systems outside of language such as memory and attention, which may make aspects of this competence unusable in real life. Such complicating factors are to be ignored in this book.

To summarize, one of the central tasks of syntactic analysis is to provide us with explicit grammar rules that differentiate Turkish expressions from non-Turkish ones. In fact, we can think of Turkish grammar as a machine that produces the answer "Yes!" when it is given a Turkish expression as input and "No!" when what it is given as input does not belong to Turkish. (The exclamation mark after "Yes" and "No" should indicate that this is an excitable machine.)

Considering now the set of expressions in (2), it is not difficult to find the grammar rule that distinguishes acceptable strings from unacceptable ones. For the examples in (2), it can be said that an expression is grammatical if the string *gece* 'night' is immediately followed the string *boyunca* 'throughout'. So simple, right? This simplistic world is not the world we are living in, however. It is not always clear what makes a sentence acceptable or unacceptable. Consider the sentences in (5).

(5) a. Meryem Rus edebiyatı ile.ilgili birçok kitab-ı okudu.
Meryem Russian literature about many book-ACC read
'Meryem has read many books on Russian literature.'
b. Meryem Rus edebiyatı ile.ilgili çoğu kitab-ı okudu.
Meryem Russian literature about most book-ACC read
'Meryem has read most books on Russian literature.'

These sentences differ only in the quantifier they contain: *birçok* 'many' vs *çoğu* 'most'. These quantifiers are similar in meaning in that they do not tell us anything about exactly how many books Meryem has read. As we shall see now, this similarity in meaning is not reflected in their syntactic behavior.

One property of Turkish we will discuss further in this book is that it is possible to drop the case marker on the object and still get an acceptable sentence. When we delete the accusative case marker (ACC) on the object in (5a), we find that the resultant sentence (6a) remains acceptable. This is not true for (5b), however. (The intended meaning of an expression is the meaning that it would be expected to have were it acceptable. 'Intended' is sometimes shortened to 'Int.')

(6)	a. Meryem Rus	edebiyatı ile.ilgili	birçok kitap	okudu.
	Meryem Russian	literature about	many book	read
	'Meryem has read many books on Russian literature.'			
	b. *Meryem Rus	edebiyatı ile.ilgili	çoğu kitap	okudu.
	Meryem Russian	literature about	most book	read
	Intended: 'Meryem has read most books on Russian literature.'			

The distinction between these two types of quantifiers has ramifications throughout grammar. Consider (7a) and (7b):

(7)	a. **Birçok** öğrenci	toplantı-ya	katıldı
	many student	meeting-to	attended
	'Many students	attended the meeting.'	
	b. **Çoğu** öğrenci	toplantı-ya	katıldı
	most student	meeting-to	attended
	'Most students attended the meeting'		

Now suppose that we wish to say that (7a) and (7b) are what we heard to be true, but we are not ready to assert them as true. To express this thought, we would need to embed these sentences inside the Turkish equivalent of the expression *I heard that*.. Doing so leads to various morphological changes on the subject containing the quantifier as well as on the predicate, which we are to study in Chapter 7. For now, let us just note that the quantified subjects inside the embedded sentences are marked with a suffix called the Genitive case marker (GEN).

(8)	a. Ben birçok öğrenci-**nin**	toplantı-ya	katıldığını	duydum
	I many student-**GEN**	meeting-to	attended	heard
	'I heard that many students attended the meeting.'			
	b. Ben çoğu öğrenci-**nin**	toplantı-ya	katıldığını	duydum
	I most student-**GEN**	meeting-to	attended	heard
	'I heard that most students attended the meeting.'			

When we drop the genitive marker on the embedded subjects, both sentences become completely unacceptable.

(9)	a. *Ben birçok öğrenci	toplantı-ya	katıldığını	duydum
	I many student	meeting-to	attended	heard
	Int. 'I heard that many students attended the meeting.'			
	b. *Ben çoğu öğrenci	toplantı-ya	katıldığını	duydum
	I most student	meeting-to	attended	heard
	Int. 'I heard that most students attended the meeting.'			

Here is the twist, though. There is a surprising asymmetry between the quantificational subject containing *birçok* 'many' and the one containing *çoğu* 'most' as far as their behavior in the embedded sentences is concerned. If we place the embedded subject in (9a) in the immediately preverbal position, the sentence is suddenly acceptable (10a). No such rescue strategy is available for the embedded subject in (9b), as can be seen in (10b).

(10) a. Ben toplantı-ya birçok öğrenci katıldığını duydum
I meeting-to many student attended heard
'I heard that many students attended the meeting.'
b. *Ben toplantı-ya çoğu öğrenci katıldığını duydum
I meeting-to most student attended heard
Int. 'I heard that most students attended the meeting.'

What is going on? From (5) to (10), we have seen a complex array of acceptability reports associated with sentences containing *birçok* 'many' and *çoğu* 'most' in the subject position. In this book, we are to show that all these contrasts we find in the acceptability judgments above boil down a single formal difference between these two quantifiers in Turkish, the consequences of which propagate throughout grammar. The conviction that underlying complexity of languages is a set of general, simple principles defining possible linguistic expressions is just an aspect of good science. In linguistics, this conviction has been given a special name, i.e. the Minimalist program. This book is minimalist not only in the types of tools that it relies on to make sense of linguistic patterns in Turkish but also in its emphasis that we should not be easily overwhelmed by the complexity of surface appearances.

There could actually be an easy way to solve the acceptability problem for Turkish, the problem that our grammar (our machine!) is expected to address. Let us imagine that there is a finite number of acceptable expressions in Turkish. We could then let our machine store all these expressions in its database. When given an input, the machine searches this database for a match, and if it finds one, it prints "Yes!". It prints "No!" if it fails to find the match.

The problem with this approach is that Turkish expressions are not finite in number, i.e. there is no natural number n such that Turkish has at most n expressions. To get a sense of this claim, consider the sequence of Turkish expressions in (11) :

(11)

PHRASES	MEANING
baba	father
büyük baba	grandfather
büyük büyük baba	great-grandfather
büyük büyük büyük baba	great-great-grandfather
…	

The three dots at the end indicate that we could go on like this *ad infinitum*. The general pattern associated with (11) is shown in (12), where the superscript *n* on the word *büyük* 'big' indicates that we are to repeat this word n times (e.g. $(\textit{büyük})^3$ = *büyük büyük büyük*). ℕ is the set of natural numbers (and, like all kind people, we take it that Zero is a natural number).

(12) For any $n \in \mathbb{N}$
$(\textit{büyük})^n$ *baba* is a Turkish expression.

What this means is that we can find a distinct Turkish expression for each natural number n, i.e. *(büyük)*^{n} *baba*. (For n = 0, we would have *baba*.) That is, there are (at least) as many Turkish expressions as there are natural numbers. Now, we know that there are infinitely many natural numbers (a type of infinity that mathematicians call 'countable infinity'). From this we conclude that there are (at least) infinitely many Turkish expressions. In fact, we do not even need phrases (as in (11)) to show that Turkish has infinitely many expressions; we can exhibit the infinity phenomenon with just words. When the suffix *-lık*, which means something like *container*, is added to the noun *kitap* 'book' in Turkish, we get a new noun, i.e. *kitaplık*, which means something like a container for books, a book-shelf. We can add this suffix again to the newly formed noun to form *kitaplıklık*, which means a container for containers for books, i.e. a container for book-shelves. This, we can do again and again and again.

(13)

WORDS	MEANING
kitap	book
kitap-lık	book-container
kitap-lık-lık	book-container-container
kitap-lık-lık-lık	book-container-container-container
…	

What do Turkish speakers know when they know that each of these infinitely many expressions belongs to Turkish? When we give them a word from the list, is it that they scan their memory for a match and report the result? This does not sound plausible since it would implicate that Turkish speakers store an infinite number of expressions inside their brains. It is more plausible that the storage capacity of our brain should be finite, and something finite cannot store an infinite list.

Let us take a closer look at this last claim with a somewhat contrived example illustrating its content. The average volume of the human brain is around 1300 cm^3. For simplicity, let us pretend that the human brain is a cube of 11 cm edge, where 11 cm × 11 cm × 11 cm = 1331 cm^3. Such a brain would consist of 1331 cubes of volume 1 cm^3. Now if each unit of memory were stored inside one of these 1 cm^3 cubes, then a human would be able to store 1331 units of information. Of course, it presumably takes much less than 1 cm^3 of the brain to store a piece of information. Indeed, let us choose the smallest unit *u* of length such that storing a memory takes up a volume of 1 u^3. (We shall ignore the issue of whether a memory is unitary or distributed across brain regions.) The only condition we impose on *u* is that there is a number such that multiplying this unit with that number gives us a length that is bigger than 1 cm. In other words, there is a $n \in \mathbb{N}$ such that $n \times u > 1 \times cm$. Since, by assumption, each unit of memory takes up a volume of 1 u^3, the brain can store as many units of memory as the result of 1331 cm^3 divided by 1 u^3. We know that there is a natural number n such that 1 × cm is less than n × u. What this means is that the volume of our brain is less than $1331 \times (n \times u)^3$, which is $1331 \times n^3 \times u^3$. Therefore, our brain can store less than $1331 \times n^3$ units of information. Since n is a natural number, $1331 \times n^3$ is also a natural number. That is, the storage capacity of our brain is bound by some natural number, i.e. it is finite.

Are memories really physical objects? Are they located in physical space? We do not really know. Psychologists talk of the memory engram, which is "a unit of cognitive information imprinted in a physical substance, theorized to be the means by which memories are stored", the all-knowing Wikipedia tells us. The word *substance* in this description suggests that perhaps memories do occupy physical space after all, in which case our attempt at articulating why a finite brain cannot store an infinite list might not be off the point. Regardless, it is intuitively perfectly clear that we cannot store every single Turkish expression in our brain.

Our brain is finite. Our language is infinite. What now? This is where the notion of grammar comes to the rescue. Grammar provides a finite characterization of an infinite language, thus solving the infinity problem. How does it do that? Grammar contains a set of rules which can be applied to their output indefinitely. A grammar with such rules is called a **recursive** grammar. For instance, we can express the infinite list in (11) just by two rules:

(14) Rule 1. *baba* is a Turkish expression.
Rule 2. If *X* is a Turkish expression, then *büyük* + *X* is also a Turkish expression.

Here the notation *büyük* + *X* means that the string *büyük* 'big' immediately precedes whatever string replaces the variable *X*. Observe that Rule 2 is a recursive rule. When it is given an input, which is a Turkish expression, it generates an output by adding the sting *büyük* in front of it, which is also a Turkish expression. We can re-apply Rule 2 to this newly formed output. Our claim is that Rule 1 and Rule 2 are sufficient to describe all the strings in the list in (11). Consider for instance the process of deciding whether *büyük büyük büyük baba* is a Turkish expression. This process would involve the following steps:

CLAIM: *büyük büyük büyük baba* is Turkish.	
PROOF: If *büyük büyük baba* is Turkish then *büyük büyük büyük baba* is Turkish	(Rule 2)
If *büyük baba* is Turkish then *büyük büyük baba* is Turkish.	(Rule 2)
If *baba* is Turkish then *büyük baba* is Turkish	(Rule 2)
baba is Turkish.	(Rule 1)
Hence, *büyük büyük büyük baba* is Turkish.	(By inference)

The upshot of this discussion is that language is infinite because grammar is a recursive system. The infinity of languages is also called the property of **productivity**. Natural languages are productive. Turkish is, too.

The example above involving an infinity of fathers may seem suspiciously simple. Can we really provide a recursive set of rules for other infinite lists inside Turkish? Consider the list concerning the empty sets we have seen in (3) and (4) before.

(15)

PHRASES	MEANING
boş küme	{}
tek elemanı boş küme olan küme	{{}}
tek elemanı tek elemanı boş küme olan küme olan küme	{{{}}}
tek elemanı tek elemanı tek elemanı boş küme olan küme olan küme olan küme	{{{{}}}}
…	

The pattern in this list of expressions can be expressed as in (16):

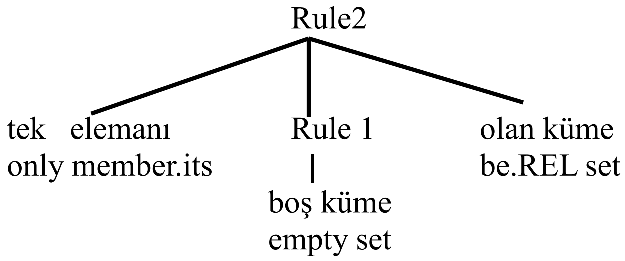
(16) For any $n \in \mathbb{N}$
(tek elemanı)^n *boş küme (olan küme)*^{n} is a Turkish expression.

This pattern is somewhat different from what we have seen up to now. There is some kind of dependency between the string *tek elemanı*, which precedes *boş küme*, and the string *olan küme*, which follows it. There must be an equal number of repetitions of each of them, whatever this number may be. An expression that does not follow this pattern is unacceptable. The rules in (17) can characterize the list in (16).

(17) Rule 1. *boş küme* is a Turkish expression.
Rule 2. If *X* is a Turkish expression,
then *tek elemanı* + *X* + *olan küme* is also a Turkish expression.

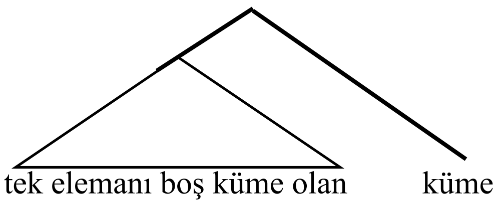
Suppose that we are asked to show that *tek elemanı boş küme olan küme* ‘the set whose only member is the empty set’ is a Turkish expression. By Rule 1, we are given that *boş küme* ‘empty set’ is Turkish. We can derive *tek elemanı boş küme olan küme* by applying Rule 2 to it. The tree representation of this process can be shown as in (18).

(18)



There is, however, something deeply wrong with this tree. We have a noun *küme* ‘set’ which is modified by the relative clause *tek alemanı boş küme olan* ‘whose only member is the empty set’. Like an adjective modifying a noun, this relative clause is likely to be a unit of sorts, merged with the noun that it modifies. Ignoring for now the structure inside the relative clause, the tree of this expression should look as in (19).

(19)



(17) may be a grammar distinguishing acceptable strings from non-acceptable ones. That does not mean that we have found the right grammar to do that. There is more to syntactic analysis than generating the strings. In other words, finding the recursive grammar that gets the strings right is not enough given that we need to get the underlying structure right, too. This is no trivial task. This is what this book intends to achieve as far as Turkish grammar is concerned.

1.2.Form and Meaning

We now have some understanding of what a theory of grammar in general, and a theory of Turkish grammar in particular, must be able to accomplish: (a) distinguish acceptable expressions from non-acceptable ones and (b) provide a finite understanding of the infinity of language. Thus far, we have only talked about the form of Turkish expressions. However, a theory of grammar also has a set of meaning-related tasks. For starters, there are cases in which a single form is paired with more than one meaning, a phenomenon known as **ambiguity**. The Turkish expression in (20) has two distinguishable meanings given below as its translations.

(20) kırmızıkalem kutusu
red pencil box
‘a box for red pencils’
‘a red box for pencils’

How is it that this form is paired with these two meanings? Intuitively, it seems that we should be able to associate two distinct structures with this nominal phrase. Perhaps, in one case the combination of *kırmızı* 'red' and *kalem* 'pencil' behaves as a unit of sorts so that we get the 'a box for red pencils' reading. In the second case, the adjective *kırmızı* 'red' modifies not the noun *kalem* 'pencil' but the phrase *kalem kutusu* 'pencil box', which means that we are talking about a red pencil-box. The type of ambiguity that comes out of different associative possibilities among words is called **structural ambiguity** as opposed to lexical ambiguity, which is about words having more than one meaning (e.g. *yüz* in Turkish means both 'face' and 'a hundred'). We will see how structural ambiguity can be expressed formally in Chapter 3.

To be clear, it is not the task of syntactic theory to explain how the form in (20) ends up having the meanings that it does. This is a task for another field of linguistics called **semantics**, which is the study of meaning of natural language expressions. Semantics is about how basic forms are mapped to basic meanings and complex forms to complex meanings. What we, as syntacticians, do is give our semanticist colleagues all the structures associated with an expression, which can be used to explain the ambiguity of this expression. Getting meanings out of strings is a long journey that requires a lot of collaboration between syntacticians and semanticists.

Not only do we need to explain why some strings have more than one meaning but we must also be able to understand why sometimes similar constructions do not give rise to ambiguity. Consider (21), which is obtained by replacing *kırmızı* 'red' in (20) with *her* 'every'.

(21) her kalem kutusu
every pencil box
'every pencil box'
cannot mean 'a box for every pencil'

There is no problem with putting *her* 'every' and *kalem* 'pencil' together; *her kalem* 'every pencil' is a perfectly fine Turkish phrase. Somehow when we try to merge this complex with the noun *kutusu* 'box', we seem to have a problem. What is this problem? This will be discussed in Chapter 3.

We have seen that there is a linguistic context such that *kırmızı* 'red' gives rise to ambiguity, but *her* 'every' does not. We will show that the asymmetry between these two expressions is related to their syntactic distribution in other contexts. Consider their behavior in the object position.

(22) a. Ali kırmızı gömleğ-**i** giydi
Ali red shirt-**ACC** wore
'Ali wore the red shirt'
b. Ali her gömleğ-**i** giydi
Ali every shirt-**ACC** wore
'Ali wore every shirt.'

When we drop the accusative case marker on the object in (22a) and (22b), the first sentence remains acceptable but the second sentence sounds quite bad.

(23) a. Ali kırmızı gömlek giydi

Ali	kırmızı	gömlek	giydi
Ali	red	shirt	wore

'Ali wore a red shirt'

b. *Ali her gömlek giydi

*Ali	her	gömlek	giydi
Ali	every	shirt	wore

Int. 'Ali wore every shirt.'

As far as the quantifier *her* 'every' is concerned, the obligatoriness of the accusative case marker in the object position seems to be related to the lack of ambiguity in nominal phrases. What is the exact nature of this relationship? Explaining how different corners of Turkish grammar are related to each other is something that we can expect a theory of Turkish grammar to accomplish. Indeed, we will get some understanding of how these disparate-looking phenomena are related to each other in Chapter 3 and Chapter 6.

Turkish is typically described as a **free word order** language in which a verb and its arguments, for instance, can occupy various positions inside the sentence. The sentence in (24), which contains the verb *ver-* 'give' and the nominal expressions *Zeynep*, *Ali'ye* and *kitabı* 'the book', can be ordered in 4 × 3 × 2 × 1 = 24 ways, all of which are acceptable. The preservation of acceptability under word permutations is sometimes called the scrambling property; Turkish is said to be a **scrambling** language.

(24)

Zeynep	Ali'ye	kitabı	verdi
Zeynep	Ali.to	book.ACC	gave

'Zeynep gave the book to Ali.'

Zeynep Ali'ye kitabı verdi.
Zeynep Ali'ye verdi kitabı.
Zeynep kitabı Ali'ye verdi.
Zeynep kitabı verdi Ali'ye.
Zeynep verdi Ali'ye kitabı.
Zeynep verdi kitabı Ali'ye.

Ali'ye Zeynep kitabı verdi.
Ali'ye Zeynep verdi kitabı.
Ali'ye kitabı Zeynep verdi.
Ali'ye kitabı verdi Zeynep.
Ali'ye verdi Zeynep kitabı.
Ali'ye verdi kitabı Zeynep.

Kitabı Zeynep Ali'ye verdi.
Kitabı Zeynep verdi Ali'ye.
Kitabı Ali'ye Zeynep verdi.
Kitabı Ali'ye verdi Zeynep.
Kitabı verdi Zeynep Ali'ye.
Kitabı verdi Ali'ye Zeynep.

Verdi Zeynep Ali'ye kitabı.
Verdi Zeynep kitabı Ali'ye.
Verdi Ali'ye Zeynep kitabı.
Verdi Ali'ye kitabı Zeynep.
Verdi kitabı Zeynep Ali'ye.
Verdi kitabı Ali'ye Zeynep.

A remarkable fact about all these sentences is that they pretty much mean the same thing. The order of arguments signals something about their information structural properties: arguments following the verb are typically backgrounded or presupposed, arguments immediately preceding the verb are highlighted or focused and arguments in the sentence-initial position tend to be interpreted as topic, i.e. as what the sentence is about. It remains true, however, that the meaning of the sentences is constant across permutations. In each case, there is an event of giving which took place before the speech time. The agent of this event is Zeynep, the theme in this event is the book under discussion and the recipient is Ali. It is tempting to conclude that word order permutations in Turkish have no effect on meaning. This is not true, however.

(25) a. Arı Ali'yi soktu
bee Ali.ACC sting.PST
'The bee stung Ali.'
b. Ali'yi arı soktu
Ali.ACC bee sting.PST
'Ali got stung by a bee.'

While (25a) is about what a specific bee did, (25b) is more about what Ali had to go through, a bee-stinging event. This difference in meaning can be made explicit by adding a phrase like *hem de iki defa* 'in fact twice' to the end of each sentence. In (26a), we get the meaning that it is the same bee that stung Ali. In (26b), on the other hand, it is quite likely that Ali got stung by different bees.

(26) a. Arı Ali'yi soktu – hem.de iki defa.
bee Ali.ACC sting.PST – in.fact two time
'The bee stung Ali – in fact, twice' (the same bee)
b. Ali'yi arı soktu – hem.de iki defa.
Ali.ACC bee stung – in.fact two time
'Ali got stung by a bee – in fact, twice' (possibly different bees)

We have a case in which a difference in word order between the arguments ends up having consequences for the meaning of the sentence. It is, then, our syntactic obligation to associate (25a) and (25b) with structures that can be used to express these distinct meanings. Regrettably, this book will have nothing to say about the nature of word order permutations we have seen in (24) that have no semantic consequences.

To conclude, in addition to explaining acceptability reports about Turkish and its infinity, syntactic theory is also responsible for giving us the ingredients to represent form-meaning relations such as ambiguity and word order effects. This is precisely what we will do in this book.

As you can see, we have a lot of work to do. It is time to dig in.[1]

References and Selected Readings:
In syntactic research, the word *minimalism* is used in two distinct senses: (1) as a methodological heuristic guiding research on language and (b) as a substantive claim about how human grammar is embedded in cognition (see Martin & Uriagereka, 2000 on different senses of minimalism). In the methodological sense, minimalism is the claim that the surface complexity of linguistic structures can be reduced to the interactions of a small number of simple rules. At times, sustaining theoretical simplicity/generality/symmetry comes at the cost of postulating linguistic objects that leave no visible track; that is, we may at times be required to postulate silent linguistic elements to achieve explanatory depth. (The discussion of the silent determiner in Turkish in Section 3 is a good

[1] A word must be said about writing conventions concerning Turkish examples. Turkish is a language that exhibits the assimilatory phonological process of vowel harmony, which means that a vowel agrees in frontness with the vowel preceding it (front vowels: *i, e, ü, ö*; back vowels: *ı, a, u, o*). Moreover, a high vowel is round (*u*, *ü*) if the preceding vowel is round (*u*, *ü*, *o*, *ö*). We express a high vowel with the capital letter "I", meaning that this vowel agrees with the preceding vowel in frontness and in roundedness (i.e. it has one of the following forms: *ı, i, u, ü*). A non-high vowel is expressed with the capital letter "A", meaning that it agrees with the preceding vowel only in frontness (i.e. it is externalized either as *a* or as *e*). When we write a consonant in parentheses, this suggests that it can be contextually deleted. For instance, the compound marker is written as –"*(s)I(n)*", which indicates that the consonants *s* and *n* can go missing in the appropriate context. Finally, the consonants that undergo phonological changes are also written in capital letters. In this context, it is important to remember that Turkish has a rule of final devoicing, due to which *b*, *d*, *c*, *ğ* turns into *p*, *t*, *ç*, *k* respectively in the final position of a phonological domain. The issues around Turkish morphophonological is obviously much more complex than what this footnote suggests. The reader interested in Turkish morphology and phonology may consult Erguvanlı-Taylan (2015).

example of this issue). In this textbook, we use minimalism in this *methodological* sense. In its substantive sense, minimalist research intends to show that the apparent complexity of linguistic phenomena is explained by the fact that language, simple though it is, must “talk” to the rest of the cognition, more specifically to the systems of thought and externalization. In other words, the visible complexity of language is a consequence of the requirements that systems external to language impose on language. Crucially, this complexity is not about the linguistic system as such (Chomsky, 1995).

The two textbooks that guided this work in terms of content are Adger (2003) and Sportiche et al. (2013). The inspiration to analyze constituency in Turkish without dwelling too much on what is called *constituency tests* came from the lecture notes of Pesetsky (2012) for the *Introduction to Linguistics* class taught at MIT. There are many other important syntax textbooks that the reader may consult (Haegeman, 1994; Bresnan, 2000; Sag et al. 2003; Carnie, 2012; Radford, 2012; for an overview of different approaches to syntax, see Müller 2023).

CHAPTER 2: In the beginning was the Morpheme

There is no limit as to how complex a linguistic expression can get. For any sentence you are given, you can produce a longer one by adding, for instance, *She said that …* in front of it. That is, linguistic expressions can be quite complex, and complex objects may be difficult to study. The good news about language is that its complexity can be broken down into basic pieces and their combinatorial possibilities. This section is about the pieces that make up the words in Turkish and the rules governing how these pieces are put together. The area of linguistics that focuses on the structure of words is called **morphology**. This section is about the morphology of Turkish.

2.1. Pieces of words

Consider the word *akıllı* 'wise' in Turkish. There can be different ways of determining its pieces. If we think of this word as a phonological entity and look for the syllables in it, we get the following three pieces: *a-kıl-lı*. We can also analyze this word into the units that have distinguishable meanings: *akıl-lı*, where *akıl* means something like 'reason' and *-lı* has a meaning akin to *possessing*. More generally, given a sentence, we can partition it into pieces that have distinguishable meanings. (1) exemplifies a sentence broken into its basic meaningful pieces.

(1)	Akıl-lı	düşman	akıl-sız	dost-tan	iyi-dir
	reason-with	enemy	reason-without	ally-from	good-FACT

'A wise enemy is better than a foolish ally.'

The smallest meaningful units in a language are called the **morphemes** of this language. There are two main types of morphemes: lexical morphemes and functional morphemes. **Lexical morphemes** denote things, properties, situations or events. A typical word in Turkish consists of a lexical morpheme to which a number of affixes have been added. Every morpheme that is not lexical is a **functional morpheme**. An **affix** is a **functional bound** morpheme, i.e. a functional morpheme that must be attached to something. Affixes are further categorized into **prefixes** and **suffixes** depending on whether they precede or follow the expression they are combined with. In Turkish, affixes are typically suffixes. There are also **functional free (non-bound) morphemes** such as postpositions and complementizers. We will meet them in later chapters.

Lexical morphemes are divided into three main categories: verbs, nouns and adjectives. We use various tests to distinguish these lexical morphemes from each other. A lexical morpheme is in the verb category if the infinitive affix *-mAk* can be added to it. This suffix cannot be added to other lexical morphemes.

(2)	Verbs:	**git**-mek	**kal**-mak	**gör**-mek	**bil**-mek
		go-INF	stay-INF	see-INF	know-INF
		'to go'	'to stay'	'to see'	'to know'
	Others:	*araba-mak	*ruh-mak	*muz-mak	*vergi-mek
		car-INF	soul-INF	banana-INF	tax-INF
		*kırmızı-mak	*kare-mek	*yeni-mek	*güzel-mek
		red-INF	square-INF	new-INF	beautiful-INF

We can also distinguish lexical morphemes in the noun category from those in the adjective category. When two nouns are combined, one of them must be affixed with the compound marker (CM). This is how we tell nouns from other lexical categories. When an adjective is combined with a noun, the noun is not affixed with the compound marker. Verbs, on the other hand, cannot be combined with nouns at all.

(3)	Nouns:	**ders** kitab-ı course book-CM 'a course book'	*ders kitap course book	**araba** lastiğ-i car tire-CM 'a car tire'	*araba lastik car tire
	Adjectives:	**kırmızı** kitap red book 'a red book'	*kırmızı kitab-ı red book-CM	**büyük** lastik big tire 'a big tire'	*büyük lastiğ-i big tire
	Verbs:	*çalış kitab-ı study book-CM	*çalış kitap study book	*kay lastiğ-i slide tire-CM	*kay lastik slide tire

A **word** in Turkish contains a lexical morpheme to which a number of suffixes have been added. If this number is zero, then this is a word that contains no affix; it is a lexical morpheme that also happens to be a word. A word may also contain a set of affixes, which are, as we have noted, functional bound morphemes. Bound morphemes in Turkish can be of two kinds: **derivational suffixes** and **inflectional suffixes**. An inflectional suffix typically plays a role in expressing a relation between distinct words. In Turkish, the subject agreement suffix on a predicate expresses a relation between the subject and the predicate and the case suffix on an object expresses a relation between the object and the verb. Agreement markers and case markers are inflectional suffixes. We will study them in later chapters.

Given a word in Turkish, you can strip away all the suffixes on it one by one to determine the **root** of the word. If there are no suffixes in a word, then the root of the word is the word itself. The **stem** of a word is that part of the word to which an affix has been added. Given a word that consists of a lexical morpheme (*lex*) to which two suffixes (*suf*) have been added, i.e. $lex\text{-}suf_1\text{-}suf_2$, we say that *lex* is the root of the word and *lex* and $lex\text{-}suf_1$ are the stems in the word. We use linguistic terms such *root* or *stem* for purposes of **description only**. They do not play any explanatory role in our analyses of Turkish expressions.

This chapter is all about lexical morphemes and the derivational suffixes attached to them. We will focus on the rules by which derivational suffixes are organized in a word.

2.2. Trees for words

Consider the word *anlayışsızlık* 'inconsiderateness'. This word consists of a verbal lexical morpheme *anla* 'understand/consider' and three derivational morphemes: *-(y)Iş*, *-sIz*, and *-lIK*. These derivational suffixes must come in a specific order. Every other order leads to unacceptability.

(4) a. anla-yış-sız-lık
b. *anla-yış-lık-sız
c. *anla-sız-ış-lık
d. *anla-sız-lık-ış
e. *anla-lık-ış-sız
f. *anla-lık-sız-ış

We need to understand why it is that of all these orders, only one is acceptable. Explaining acceptability through grammaticality is one of the major tasks of our theory after all. We first observe that *anla* is a verbal morpheme and *-yış* is a derivational suffix (DS) that attaches to verbs and only verbs.

(5) a. anla-yış b. *araba-yış c. *yeni-yiş
understand-DS car-DS new-DS
'(an) understanding'

We say that the derivational suffix *–(y)Iş* has **selects for** a verb. What is the category of the expression obtained after combining *anla* and *yış*, i.e. *anlayış*? Observe that the suffix *-sIz* attaches to nouns and nouns only.

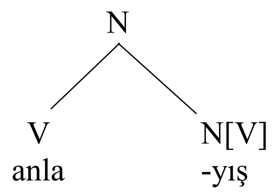
(6) a. *anla-sız b. kitap-sız c. *yeni-siz
understand-DS book-DS new-DS
'bookless'

Given that the derivational suffix *-sIz* can be attached to *anlayış* 'understanding', we conclude that the stem *anlayış* is of the noun category. For objects in the noun category, we sometimes use the term **nominal element**, or **N-element.**

(7) a. anla-yış b. anla-yış-sız
understand-DS understand-DS-DS
'understanding' 'one who lacks understanding'/'inconsiderate'

To summarize, *-(y)Iş* is a type of derivational suffix that attaches to verbs and the result of this attachment is a nominal element. We will talk about the attachment properties of suffixes in terms of **their selectional properties**. We say that the derivational suffix *–(y)Iş* selects for a verb and it is itself in the noun category. In our case, the derivation suffix *–(y)Iş* is **merged** with the verb *anla-* 'understand' to generate the expression *anlayış* 'understanding', which is a noun. The operation that is responsible for the merger of linguistic elements, i.e. the Merge operation, will be discussed soon.

We express the categorical and selectional properties of a suffix by first indicating its category and then listing the category that it must be merged with inside square brackets, e.g. *N[V]*. Upon the combination of the lexical verb *anla* 'understand' and the derivational suffix *–(y)Iş*, the nominal category of the suffix ends up determining the category of the whole expression. We can represent these observations in the format of a tree.

(8)


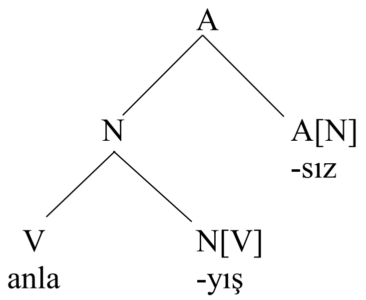
The upside-down tree we have in (8) consists of **nodes** that are **labelled** with various categories. The lines connecting these nodes are called **branches** of the tree. Using an analogy from family relations, we say that the node labeled with N is the **mother node** of the nodes below it. Similarly, *V* and *N[V]* are **daughters of** *N*, or, equivalently, **daughter nodes** of the node *N*. Moreover, we say that the nodes *V* and *N[V]* are **sister nodes**.

We are not done yet. We have already seen that the suffix *-sIz* must be combined with a nominal stem (i.e. a stem in the noun category). What is the categorical result of this combination? There is evidence that a word that ends in *–sız* is in the adjective category. For instance, when we combine *anlayışsız* 'inconsiderate' with the lexical noun *adam* 'man', we find that the combination is acceptable (hence, *anlayışsız* is not in the verbal category) and that the noun that *anlayışsız* is combined with is not marked with the compound marker, which suggests that this word is in the adjective category and not in the noun category.

(9) a. anla-yış-sız adam
understand-DS-DS man
'(an) inconsiderate man'
b. *anla-yış-sız adam-ı
understand-DS-DS man-CM

This means that the category of *–sIz* can be represented as *A[N]* and the tree representing the stem *anlayışsız* 'inconsiderate' should be as in (10).

(10)

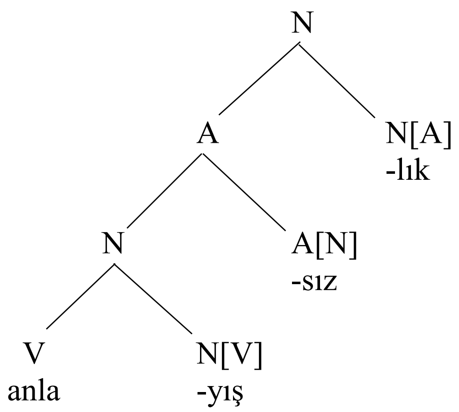


The derivational suffix *–lIK* combines with an adjective and generates a noun after its merger with the adjective it selects for. The evidence for the nominal status of a word that ends with *–lIK* comes from the fact that when combined with another noun, the compound marker is obligatory on the second noun. Moreover, the acceptability of (11a) provides evidence that *anlayışsızlık* 'inconsiderateness' is not in the verb category.

(11) a. anla-yış-sız-lık problem-i
understand-DS-DS-DS problem-CM
'the problem of inconsiderateness'
b. *anla-yış-sız-lık problem
understand-DS-DS-DS problem

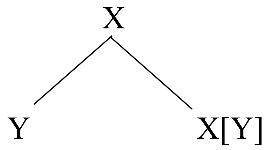
Here is the complete tree for the word *anlayışsızlık* 'inconsiderateness'.

(12)



One thing that we learn from this discussion is that combinatorial properties of suffixes are satisfied under sisterhood relation. The general schema for this relation can be shown as in (13). In syntactic research, the operation responsible for combining morphemes or words is called the Merge operation. Although there will be some variations on the schema in (13), we will see that the Merge operation will remain the main rule of structure building throughout this book.

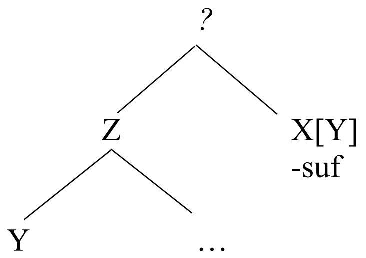
(13) The Merge Operation



We say that *X[Y]* takes *Y* as an argument; in other words, *Y* is an argument of *X[Y]*. We also say that *X* **is** the category of the merger of *Y* and *X[Y]* as it provides the label of the tree containing *Y* and *X[Y]* as sister nodes. Finally, as far as the Merge operation is concerned, the order between sister nodes does not really matter. The tree representing this operation is an unordered tree, which means that the order between sister nodes is of no consequence. This is unlike the trees we draw to represent Turkish expressions, where the order between sister nodes represents the linear ordering of expressions below each node.

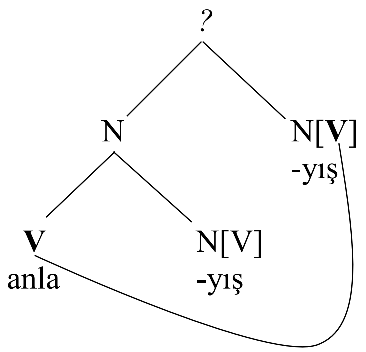
Determining the label of a mother node depends on daughter nodes and daughter nodes only. That is, deciding the label of a mother node is **a completely local decision** that is calculated on the basis of the labels of the sister nodes right below the mother node. To see what this **locality requirement** entails, imagine a tree in which instead of Y above we have a Z distinct from Y and has Y as a daughter node.

(14)



Is it possible to satisfy the selectional properties of the suffix *–suf* in the tree in (14)? There is reason to believe that it is not. In other words, there is reason to believe that (14) is not a possible configuration in which the selectional properties of the suffix *–suf* can be satisfied. Here is our argument. Consider a situation, given in (15), where we try to combine the stem *anlayış* 'understanding' with the suffix *–(y)Iş*. We draw a line indicating that the suffix *–(y)Iş* finds the node that it selects for not in its sister node but inside its sister node, instantiating the schema in (14).

(15)



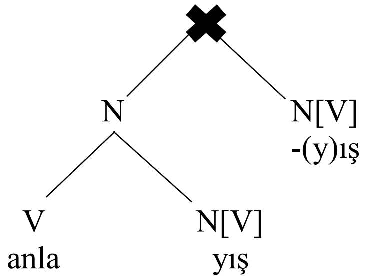
If this were a possible configuration, it would mean that we can repeat the suffix *–(y)Iş* on the lexical verb *anla-* as many times as we desire. Natural languages would be systems with **affix repetition property**, meaning that an affix that can be combined with a lexical morpheme can also be repeated on this lexical morpheme as many times as we desire. This is not at all what Turkish looks like.

16) a. anla
b. anla-yış
c. *anla-yış-ış
d. *anla-yış-ış-ış
e. *anla-yış-ış-ış-ış

Our observation concerning (16) can be summarized as in (17), where $\mathbb{N}$ / *{0}* expresses the set of natural numbers excluding Zero, i.e. the set of positive integers.

(17) For all n ∈ $\mathbb{N}$ / {0}
anla-yış-(ış)^n is not a Turkish word.

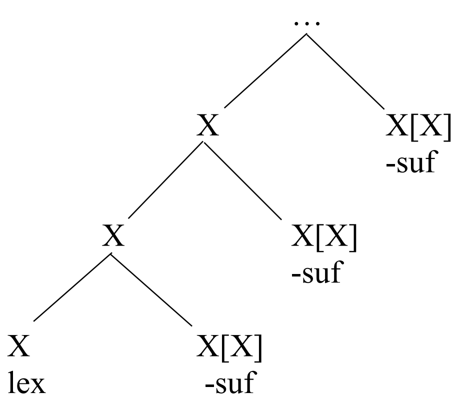
When it is not possible to draw a type of tree, we express this by putting a cross on the node where the Merge operation cannot be applied. It is not possible to draw the tree for **anlayışış* given that this expression cannot be made to conform to the schema in (13), which is a completely local schema.



2.3. Infinity one more time

It is possible to find a scenario in which affix repetition on a lexical morpheme happens to be possible. The schema in (13) relies on two variables, *X* and *Y*, that need not be distinct. When we let *Y* be identical to *X*, i.e. whenever *Y* = *X*, the selectional properties of the suffix with the category *X[X]* can be locally satisfied by its sister if the sister node is of the category *X*. We could, then, repeat this suffix as many times as we want and stay within Turkish. (18) is a schematic tree representing this possibility.

(18)



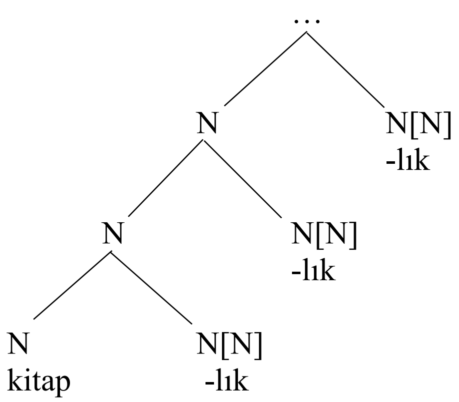
As a matter of fact, we have already seen an instantiation of this schema in Turkish morphology. Recall the whole issue of book-containers and the containers for book-containers etc. that we discussed in Chapter 1.

(19) WORDS MEANING

kitap	book
kitap-lık	book-container
kitap-lık-lık	book-container-container
kitap-lık-lık-lık	book-container-container-container
…	

The pattern above suggests that Turkish has a derivational suffix *–lIK*, which takes nouns as input gives nouns as output, i.e. *N[N]*. The tree below in (20) how such a suffix can be attached again and again to the newly formed nominal stems.

(20)



Observe that the output of the combination of *kitap* 'book' and the suffix *-lık*, i.e. *kitaplık* 'book-shelf', is of the right category to function as an argument to the same derivational suffix. In other words, when we have $X = Y = N$ in the schema (13), what we have is a recursive schema. This accounts for the fact that the infinity of expressions that satisfy the description in (21) are all Turkish words. Our grammar makes use only of finite rules to characterize an infinite set of Turkish expressions.

(21) For all $n \in \mathbb{N}$
kitap-(lık)^n is a Turkish word.

We have seen two distinct derivational suffixes that are of the form *–lIK*. In deriving the word *anlayışsızlık* 'inconsiderateness' from *anlayışsız* 'inconsiderate', the derivational suffix *–lIK* functions as a suffix from adjectives to nouns with the meaning 'property/status of…'. In deriving the word *kitaplık* 'book-shelf' from *kitap* 'book', the *–lIK* suffix takes a noun as argument and gives back another nominal element as its output. We take this to be a case of **homophony**, the association of distinct morphemes with the same phonological form. We distinguish between these two suffixes by adding index to them.

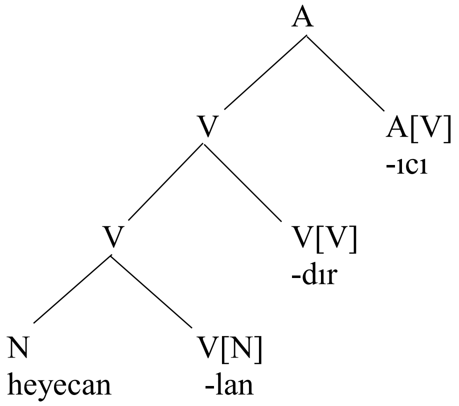
(22)

Suffix	Input	Output	Meaning
lIK_1	A	N	status/property of …
lIK_2	N	N	container of …

Homophony is frequently observed in the context of lexical morphemes. The noun *satır* in Turkish may both mean ‘a line (on a paper)’ and ‘a cleaver’. The table in (22) about the derivational *–lIK* suffixes implies that there is also homophony associated with functional morphemes. In fact, the semantic range of the suffix *–lIK* of the category *N[N]* may go beyond what we have shown in the table. When *–lIK* is added to the noun *ana* ‘mother’, we obtain a new noun *analık* meaning ‘foster mother’. The noun *aylık,* which is obtained by adding *–lIK* to the noun *ay* ‘month’, has the meaning of ‘(monthly) salary’. Finally, combining the noun *gece* ‘night’ with the suffix *–lIK* gives us the noun *gecelik* with the meaning ‘nightgown’. Unfortunately, this book is no position to offer insight on whether these interpretations associated with the derivational suffix *-lIK* of the category *N[N]* can be unified or whether they should be analyzed as further cases of homophony.

We have seen a derivational suffix that takes verbs as input in Turkish, i.e. *–(y)Iş*. There are also derivational suffixes that give verbs as outputs, i.e. verbalizing derivational suffixes. The word *heyecanlandırıcı* ‘exciting’ contains two such suffixes: *-lAn* from nouns to verbs and *–DIr* from verbs to verbs. The suffix *–lAn* attaches to the nominal root *heyecan* ‘excitement’ to generate the verbal stem *heyecanlan-* with the meaning *to get excited*. The suffix *–DIr*, from verbs to verbs, adds the meaning of causation, whereby the stem *heyecanlandır-* ends up having the meaning *to cause excitement*. The nominalizing suffix *–ICI* gives rise to the adjectival meaning *exciting* when combined with this verbal stem. The overall structure of *heyecanlandırıcı* ‘exciting’ can be expressed as in (23):

(23)



The analysis given above provides an account of why other combinations of these suffixes lead to unacceptability. Verifying this is one of the exercises that you will be asked to do at the end of this chapter.

(24)
a. heyecan-lan-dır-ıcı
b. *heyecan-lan-ıcı-dır
c. *heyecan-dır-lan-ıcı
d. *heyecan-dır-ıcı-lan
e. *heyecan-ıcı-lan-dır
f. *heyecan-ıcı-dır-lan

The table in (25) contains the derivational suffixes we have studied in this section. To this list, we have added the suffix *–lAş*, which turns adjectives into verbs: *beyaz* 'white', *beyazlaş* 'to turn white'. The table also contains the derivational morpheme *–ImsI*, a suffix from adjectives to adjectives: kırmızı 'red', kırmızı-msı 'reddish'. While this suffix is not frequently used in daily speech (and might not be part of grammar of some Turkish speakers), we have added it to the list for reasons of completeness.

(25)

Suffix	INPUT	OUTPUT
-Dir	V	V
-(y)Iş	V	N
-(y)IcI	V	A
-lAn	N	V
-lIK$_2$	N	N
-sIz	N	A
-lAş	A	V
-lIK$_1$	A	N
-ImsI	A	A

There is of course much more to be said about word formation in Turkish. The complexity of Turkish words is not the main focus of this book, however. What we have seen in this section should be enough to get us started for the syntactic analysis of complex expressions. In the next section, we will start putting words together.

Chapter 2 Exercises:

1. Explain why the expressions below are unacceptable. Try drawing trees for them and put a cross at the node where the Merge operation cannot be applied.

a. *heyecan-lan-ıcı-dır
b. *heyecan-dır-lan-ıcı
c. *heyecan-dır-ıcı-lan
d. *heyecan-ıcı-lan-dır
e. *heyecan-ıcı-dır-lan

2. Using the table in (25), draw trees for each word below (see (2) and (3) for the meaning of lexical morphemes).

a. heyecan-sız-lık
b. kırmızı-msı-laş-tır-ıcı
c. büyük-lük
d. çalış-tır-ıcı-sız
e. gid-iş-lik

3. Which of the following expressions is/are expected to be acceptable (see the table (22))? What should *kitap-lık-sız-lık* mean?

a. kitap-lık$_1$-sız-lık$_1$
b. kitap-lık$_1$-sız-lık$_2$
c. kitap-lık$_2$-sız-lık$_1$
d. kitap-lık$_2$-sız-lık$_2$

4. Suppose, for the sake of this exercise, that Turkish has a prefix *zis-* which takes an adjective as input and gives a noun as output. Which of the following expressions are expected to be acceptable? What would be the general rule about acceptability?

a. kitap
b. ziskitap
c. kitapsız
d. zisziskitap
e. ziskitapsız
f. kitapsızsız
g. ziszisziskitap
h. zisziskitapsız
i. ziskitapsızsız
j. kitapsızsızsız

5. Suppose, for the sake of this exercise, that Turkish has a prefix *kıl-* which takes a noun as input and gives a noun as output, just like lIK$_2$. How many distinct trees could be drawn for each of the expressions below?

a. kitap
b. kılkitaplık
c. kılkılkitaplık
d. kılkitaplıklık
e. kılkılkitaplıklık

CHAPTER 3: Putting words together

In the previous section, we focused on the combination of lexical morphemes with a set of derivational suffixes. We are now ready to go beyond the representation of words and take the next step in the analysis of Turkish. We will look at the syntactic representation of complex nominal phrases.

3.1. Nominal phrases in Turkish

Recall from the previous section that compounding in Turkish is a grammatical process that involves nouns and nouns only. A noun that is marked with the compound marker can only be merged with other nouns.

(1)	a. ders kitab-ı	b. *kırmızı kitab-ı	c. *çalış kitab-ı
	course book-CM	red book-CM	study book-CM
	'course book'		

This suggests that the compound marker in Turkish selects for two nouns, with which it is to be merged one by one. What is the category of the result of these operations? The examples in (2) show that a compound can be input to another layer of compounding. Since a noun with the compound marker can only be merged with another nominal element, we conclude that compounding ultimately results in a nominal expression.

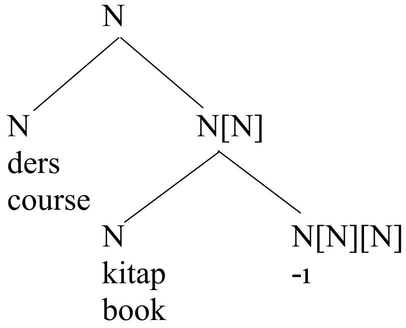
(2)	a. ders	kitab-ı			
	course	book-CM			
	'course book'				
	b. ders	kitab-ı	kapağ-ı		
	course	book-CM	cover-CM		
	'course book cover'				
	c. ders	kitab-ı	kapağ-ı	tasarım-ı	
	course	book-CM	cover-CM	design-CM	
	'course book cover design'				
	d. ders	kitab-ı	kapağ-ı	tasarım-ı	seçim-i
	course	book-CM	cover-CM	design-CM	choice-CM
	'the choice of course book cover design'				

The category of the compound marker in Turkish is given in (3). This is a kind of category that can be applied to its own output, another case of recursion. In this way, we can account for the potential infinity of the type of expressions in (2).

(3) The category of the compound marker is "N[N][N]"

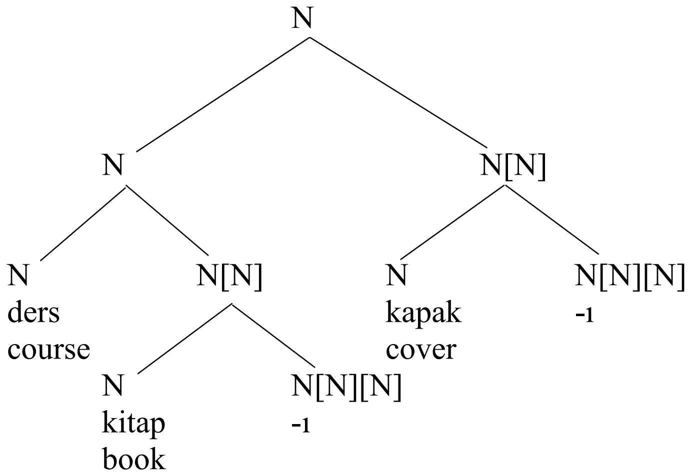
The syntactic representation of the nominal compound *ders kitabı* ‘course book’ is shown in (4)

(4)



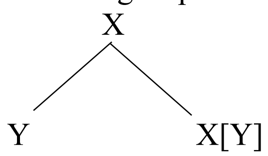
Given that the label of the tree (4) is in the category N, it may function as input to further compounding. The tree for *ders kitabı kapağı* ‘course book cover’ (2b), given in (5), exemplifies this possibility.

(5)



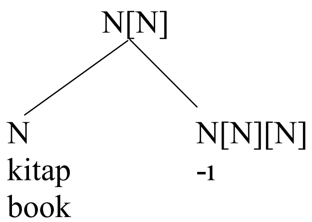
Looking at the tree above, one may get the impression that the merger of the compound marker, which is of the category *N[N][N]*, with a noun adds a layer of complexity that we have not seen before. It is important to note that, as far as the Merge operation is concerned, no such complexity has been introduced. This combination is nothing but a direct application of this operation.

(6) The Merge Operation



When we let *X = N[N]* and *Y = N*, we get the structure of the merger of a noun with a compound marker.

(7)



When two nouns are merged directly, the condition for the Merge operation is not met. A tree with two nouns as sisters cannot be labelled (an instance of ungrammaticality), the result of which is that such expressions are not acceptable to the speakers of Turkish.

(8) a.

N	N
ders	kitap
course	book

b.

N	N
kitap	kapak
book	cover

When an adjective is combined with a noun as in (9a) and (9c), however, it is the presence of the compound marker on the modified noun that leads to unacceptability.

(9)

a. **kırmızı**	kitap	b. *kırmızı	kitab-ı
red	book	red	book-CM
‘(a) red book’			
c. **büyük**	lastik	d. *büyük	lastiğ-i
big	tire	big	tire-CM
‘(a) big tire’			

This is one fact about adjective-noun combinations in Turkish that we need to account for, but it is not the only fact. When an adjective is adjacent to a nominal compound, the resulting expression is structurally ambiguous. For instance, an expression such as *kırmızı kalem kutusu* ‘red pencil box’ is typically understood to denote a type of pencil box which is red in color. It may also be used to talk about a box that is intended to contain red pencils for some reason. In (10), we provide a list of such structurally ambiguous expressions:

(10)

a. kırmızı	kalem	kutu-su
red	pencil	box-CM
‘a box for red pencils’		
‘a red box for pencils’		
b. büyük	araba	lastiğ-i
big	car	tire-CM
‘a tire for big cars’		
‘a big tire for cars (of any size)’		
c. siyah	telefon	kılıf-ı
black	phone	case-CM
‘a case for black phones’		
‘a black case for phones’		
d. renkli	oje	kataloğ-u
colored	nail.polish	catalogue-CM
‘a catalogue for colored nail polish’		
‘a colored catalogue for nail polish’		

We need to provide two distinct tree representations for each nominal expression in (10) to account for structural ambiguity. Moreover, as a methodological heuristic, we want to keep the central structure building operation Merge as simple as possible. We take it that it is a desirable property of any formal theory that it should rely on rules that are as simple and general as possible and reduce the visible complexity of natural phenomena to the repeated application of basic rules. This is one sense in which the analyses developed in this book can be said to be minimalist.

In proposing an analysis for adjectives in Turkish, we take inspiration from languages in which adjectives and nouns are combined with the help of a functional morpheme. In Persian, for instance, the combination of nouns with adjectives is always mediated with a morpheme called the **linker**. As can be seen in (11b), there are as many linkers in such constructions as there are adjectives.

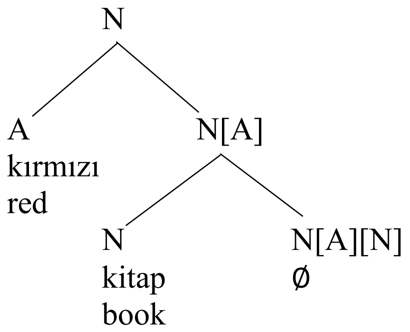
(11) a. mard-e pir
man-LINK old
'(an) old man'
b. gol-e sorx-e xošbu
flower-LINK red-LINK fragrant
'(a) fragrant red flower'

We suggest that in Turkish, too, there is a silent linker that is responsible for the merger of adjectives with nouns. This functional morpheme is firstly merged with the noun and then with the adjective and it produces a nominal category as the output. Independent evidence for the nominal status of adjective-noun combinations will be discussed shortly.

(12) The category of the linker morpheme is "N[A][N]"

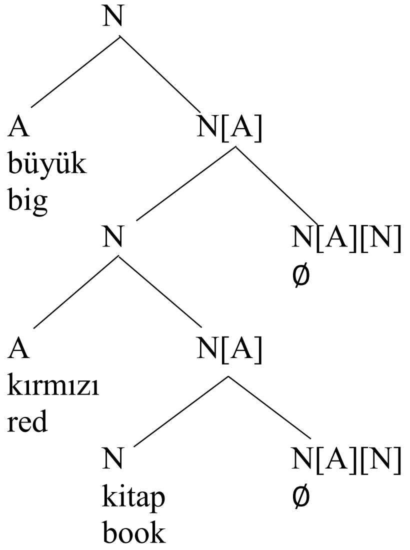
The syntactic representation of an adjective-noun combination in Turkish is shown in (13).

(13)



The construction in (13) can function as input to another linker morpheme. That is, a construction containing a linker can be embedded inside another similar construction.

(14)



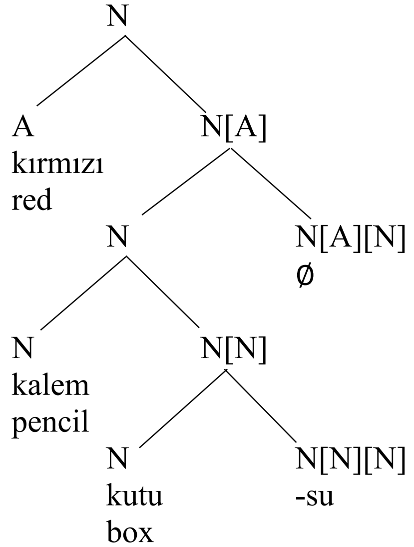
In this way, we can provide an explanation for the fact that there is in principle no limit as to how many adjectives can modify a nominal expression.

(15)

a.	masa					
	table					
b.	kırmızı	masa				
	red	table				
c.	yeni kırmızı		masa			
	new red		table			
d.	yuvarlak	yeni	kırmızı	masa		
	round	new	red	table		
e.	pahalı	yuvarlak	yeni	kırmızı	masa	
	expensive	round	new	red	table	
f.	güzel	pahalı	yuvarlak	yeni	kırmızı	masa
	beautiful	expensive	round	new	red	table

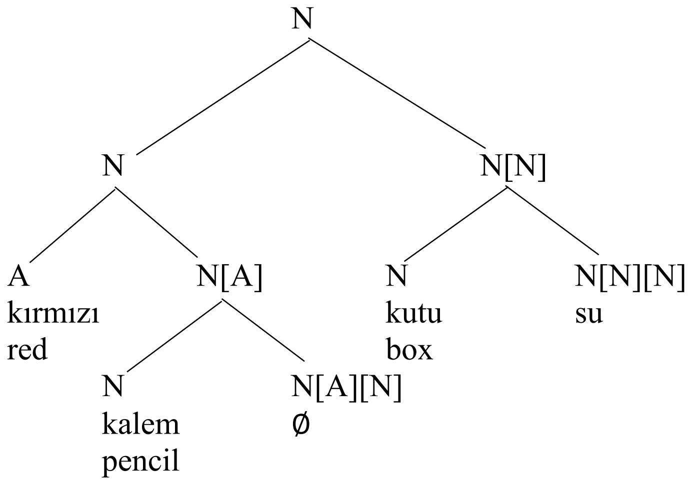
Going back to the issue of structural ambiguity, we see that the reading under which an adjective modifies the whole nominal compound can be expressed with a tree representation as in (16). In this tree, we first form the nominal compound and then add the adjective to this tree with the help of the linker. As a result, the color red is understood to be a property of the pencil box and not only of the box.

(16)



We could also merge the adjective *kırmızı* ‘red’ and noun *kalem* ‘pencil’ to first obtain *kırmızı kalem* ‘red pencil’. This expression can then undergo compounding with the noun *kutusu* ‘box-CM’. The fact that the adjective-noun combination in (13) can be input to compounding provides us with independent evidence that the merger of adjectives with noun should be in the noun category.

(17)



This tree should be interpretable as denoting boxes for red pencils. Showing the details of how this interpretation is to be derived out of the tree in (17) is a task that we must leave to semanticists. What we have shown is that the rule schema that our grammar makes use of provides a natural understanding of what such an ambiguity may boil down to. As promised, our grammar provides the ingredients for an account of structural ambiguity in Turkish.

3.2. The silent determiner in Turkish

In the previous section, we have shown how our grammar handles structural ambiguity. There are other cases in which we might expect to observe structural ambiguity, but we do not find any. Consider (18), which is obtained by replacing the adjective *kırmızı* 'red' in (10a) with the quantifier *her* 'every'.

(18) her kalem kutu-su
every pencil box-CM
'every pencil box'
cannot mean 'a box for every pencil'

This is not a peculiarity of the quantifier *her* 'every' in Turkish. We find a similar pattern with the quantifier *çoğu* 'most'.

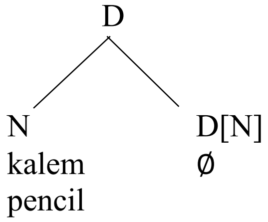
(19) çoğu kalem kutu-su
most pencil box-CM
'most pencil boxes'
cannot mean 'a box for most pencils'

Similarly, the demonstratives in Turkish, *bu* 'this', *şu* 'that' and *o(n)* 'yonder', show the same semantic behavior in this syntactic context.

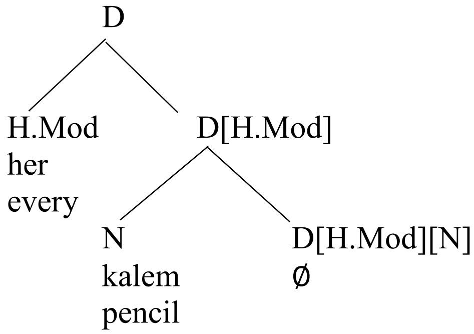
(20) a. bu kalem kutu-su
this pencil box-CM
'this pencil box'
cannot mean 'a box for this pencil'
b. şu kalem kutu-su
that pencil box-CM
'that pencil box'
cannot mean 'a box for that pencil'
c. o kalem kutu-su
yonder pencil box-CM
'that pencil box yonder'
cannot mean 'a box for that pencil yonder'

To explain the absence of the second reading in (18) to (20), we must make sure that phrases such as *her/çoğu/bu/şu/o kalem* 'every/most/this/that/yonder pencil(s)' cannot undergo compounding. Let us call these quantifiers and demonstratives **high modifiers**, implying, and it is true, that there is also such a thing as a low modifier. It is clear that high modifiers cannot be analyzed as adjectives. If they were adjectives, we would expect them to give rise to structural ambiguity preceding compounds, which they do not. We must somehow guarantee that phrases containing high modifiers are not labeled with *N*. How are we to do this?

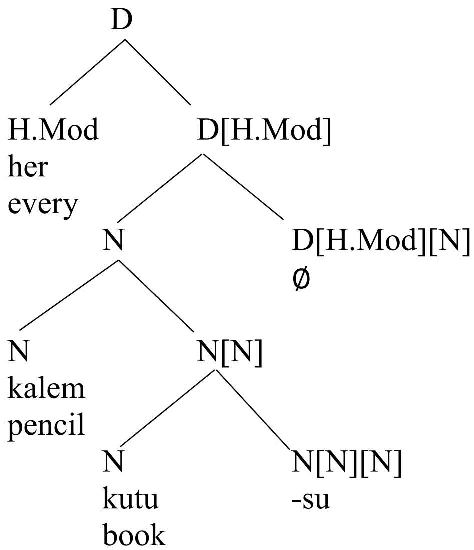
Let us explore the hypothesis that the combination of a high modifier with a noun is not labeled with *N* but with another category, the determiner category (shortened to *D*). In implementing this idea, we will assume that there is a silent determiner in Turkish, represented with *Ø,* that selects for nouns. We will provide independent evidence for the presence of a silent determiner in Turkish later in this section.

(21)


In those cases where there is a high modifier preceding a nominal element, the categorical status of the silent determiner is slightly adjusted so as to select for the high modifier, shortened to *H.Mod*, after it has selected for the noun, i.e. *D[H.Mod][N]*. The syntactic representation of *her kalem* 'every pencil' is given in (22).

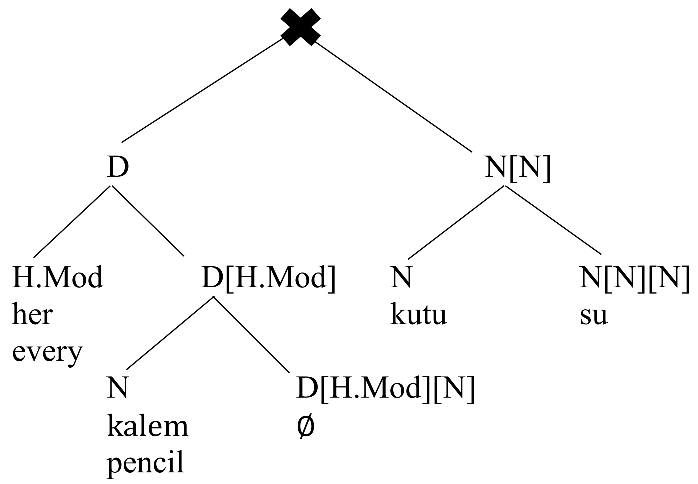
(22)


We will soon provide independent evidence for the determiner status of expressions containing high modifiers. Before we do that, however, let us see how exactly the tree in (22) accounts for the absence of ambiguity in (18). Given the categories we have, it is possible to draw the tree that is responsible for the reading in which we are talking about every one of the pencil boxes.

(23)


When we try to merge *her kalem* ‘every pencil’ with the noun *kutusu*, however, we find that the condition for the application of the Merge operation is not met.

(24)



The nominal element *kutusu* selects for another nominal element but what it finds instead is a node labeled with D, a D-element. The application of the Merge operation is thus rendered impossible, which leads to ungrammaticality. This ungrammaticality is reflected in the absence of the second reading.

We now see that there are two types of phrases in Turkish that involve nominal elements: those that are in the category N and those in the category D. Bare nouns, compounds and nouns modified with adjectives are in the noun category. The combination of nouns with high modifiers is represented with trees that are labeled with the category D. Are there any reasons, other than ambiguity, to believe that Turkish expressions must indeed be analyzed into these two main categories?

As a matter of fact, there is. The main evidence for this distinction comes from the syntactic behavior of distinct types of phrases in the object position. Consider the list of sentences in (25). In the first three sentences in this list, the objects, which are **bolded**, are members of the noun category (25a-c). The objects in the remaining sentences in this list (25d-g) are in the determiner category.

(25) a. Ali **şapka-yı** giydi.
Ali hat-ACC wear-PST
'Ali wore the hat.'

b. Ali **kırmızı** **şapka-yı** giy-di.
Ali red hat-ACC wear-PST
'Ali wore the red hat.'

c. Ali **kovboy** **şapka-sın-ı** giy-di.
Ali cowboy hat-CM-ACC wear-PAST
'Ali wore the cowboy hat.'

d. Ali **her** **şapka-yı** giy-di.
Ali every hat-ACC wear-PST
'Ali wore every hat.'

e. Ali **çoğu** **şapka-yı** giy-di.
Ali most hat-ACC wear-PST
'Ali wore most hats'

f. Ali **bu/şu** **şapka-yı** giy-di.
Ali this/that hat-ACC wear-PST
'Ali wore this/that hat.'

g. Ali **o** **şapka-yı** giy-di
Ali yonder hat-ACC wear-PST
'Ali wore that hat yonder.'

In Chapter 1, we have discussed the possibility of dropping the accusative case marker in Turkish. When we delete the accusative case markers in (25), the sentences with an object in the noun category remain acceptable (26a-c). The sentences that contain an object in the determiner category become completely unacceptable, however (26d-g).

(26) a. Ali **şapka** giydi.
b. Ali **kırmızı şapka** giydi.
c. Ali **kovboy şapka-sı** giydi.

d. *Ali **her şapka** giydi.
e. *Ali **çoğu şapka** giydi.
f. *Ali **bu/şu şapka** giydi.
g. *Ali **o şapka** giydi.

Here is our discovery: As far as high modifiers are concerned, the lack of ambiguity in the context of nominal compounds is related to the obligatoriness of case marking in the object position. These two apparently distinct phenomena are consequences of trees that involve strong modifiers being labeled with D. As we promised in the beginning of the book, we are starting to see how different corners of Turkish grammar are related to each other in ways that can be only unearthed by detailed syntactic analysis.

We have taken the obligatoriness of case marking in the object position to be indicative of the determiner status of a syntactic element. Let us make this assumption explicit.

(27) A syntactic element in the determiner category must have case.

How should we formalize this claim in our theory of grammar? Answering this question requires an understanding of how features such as number, person and case are related to the determiner category. This will be the topic of next chapter where we will make our assumptions about features and categories in Turkish more explicit.

With the diagnostic in (27) at our disposal, we can probe the category of names and pronouns.

(28) a. Ali **Mustafa-yı** gör-dü.
Ali Mustafa-ACC see-PST
'Ali saw Mustafa.'
b. Ali **ben-i** /**sen-i** /**on-u** gör-dü
Ali I-ACC /you-ACC /he-ACC see-PST
'Ali saw me/you/him.'
c. Ali **biz-i** /**siz-i** /**onlar-ı** gör-dü
Ali we-ACC you.PLU-ACC they-ACC see-PST
'Ali saw us/you/them.'

When we delete the accusative case markers in (28), the expressions we get in (29) are completely unacceptable.

(29) a. *Ali **Mustafa** gör-dü.
b. *Ali **ben** /**sen** /**o** gör-dü
c. *Ali **biz** /**siz** /**onlar** gör-dü

This suggests that names and pronouns are also in the determiner category. Unlike the silent determiner, however, they need not be merged with nouns.

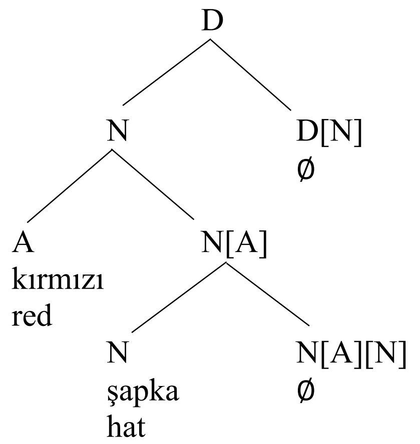
(30) The category of pronouns and names is "D".

There still is a problem, though. We know that bare nouns, compounds and nouns modified by adjectives need not be marked with the accusative case marker (31a-c). This is explained by the fact that they are not in the determiner category. Curiously, however, they *can* be case-marked (31d-f).

(31) a. Ali **şapka** giy-di.
Ali hat wear-PST
'Ali wore a hat.'
b. Ali **kırmızı** **şapka** giy-di.
Ali red hat wear-PST
'Ali wore a red hat.'
c. Ali **kovboy** **şapka-sı** giy-di.
Ali cowboy hat-CM wear-PAST
'Ali wore a cowboy hat.'
d. Ali **şapka-yı** giy-di.
Ali hat-ACC wear-PST
'Ali wore the hat.'
e. Ali **kırmızı** **şapka-yı** giy-di.
Ali red hat-ACC wear-PST
'Ali wore the red hat.'
f. Ali **kovboy** **şapka-sın-ı** giy-di.
Ali cowboy hat-CM-ACC wear-PAST
'Ali wore the cowboy hat.'

How should we make sense of the data in (31d-f)? A straightforward answer suggests itself once we remember that there is a silent determiner in Turkish which labels the tree with the category D. Since every syntactic element in this category must have case, the objects of in (31d-f) are marked with the accusative case. The syntactic representation of the object in (31e) is given in (32). (The exact mechanism by which verbs assign case to their objects will be discussed in Chapter 5.)

(32)



There are important semantic differences between case-marked and non-case-marked objects. Suppose that someone asks us what Ali did today and we give one of the answers in (33).

(33) a. Ali kitap oku-du
Ali book read-PST
'Ali read (a) book(s).'
b. Ali kitab-ı okudu
Ali book-ACC read-PST
'Ali read the book.'

While (33a) is a fine answer with the meaning that Ali did some book-reading today, (33b) would be strange way of responding to this question unless there is some specific book that Ali was expected to read. In other words, the presence of the accusative marker leads to an interpretation of specificity for the object, which is lacking in (33a). Given this, one may think that the silent determiner in Turkish is very much like the determiner *the* in English. Perhaps, it is a silent version of this definite determiner. While there are similarities between these two determiners in terms of their meaning, there are also some important grammatical differences. The definite determiner in English cannot be used together with the indefiniteness marker 'a(n)' (34c). The silent determiner in Turkish, on the other hand, is compatible with the indefiniteness marker *bi'* 'a' in Turkish (35c).

(34) a. a book
b. the book
c. *the a book

(35) a. bi' kitap
a book
b. kitab-ı
book-ACC
'the book'
c. bi' kitab-ı
a book-ACC
'a specific book'

The silent determiner in Turkish does not seem to encode definiteness. We will call it **a specificity marker**, with the understanding that specificity and indefiniteness are semantically compatible categories. Since our main concern here is the structural properties of Turkish, we will not focus on the semantic details of this claim in this book.

3.3. An anomaly about some low modifiers

The merger of the high modifiers such as *çoğu* 'most' and *her* 'every' with nominal elements results in a syntactic tree labeled with D. This, we have suggested, accounts for the obligatoriness of the case marking when arguments involving these modifiers are in the object position. Not all quantifiers share this property, however. Numerals such as *bir* 'one', *iki* 'two', vague quantifiers such as *birçok* 'many', *birkaç* 'several', *biraz* 'some' as well as the indefinite marker *bi'* 'a(n)', which can be thought of as some kind of existential quantifier, can be (36a-c) but need not be (37a-c) case-marked in the object position. We will call these quantifiers **low modifiers**.

(36) a. Meryem dün birçok kitab-**ı** oku-du
Meryem yesterday many book-**ACC** read-PST
'Yesterday, Meryem read many books.'
b. Meryem dün dört kitab-**ı** oku-du
Meryem yesday four book-**ACC** read-PST
'Yesterday, Meryem read (the) four books.'
c. Meryem dün bi' kitab-**ı** oku-du
Meryem yesterday a book-**ACC** read-PST
'Yesterday Meryem read a (specific) book.'
(37) a. Meryem dün birçok kitap oku-du
Meryem yesterday many book read-PST
'Yesterday, Meryem read many books.'
b. Meryem dün dört kitap oku-du
Meryem yesterday four book read-PST
'Yesterday, Meryem read four books.'
c. Meryem dün bi' kitap oku-du
Meryem yesterday a book read-PST
'Yesterday, Meryem read a book.'

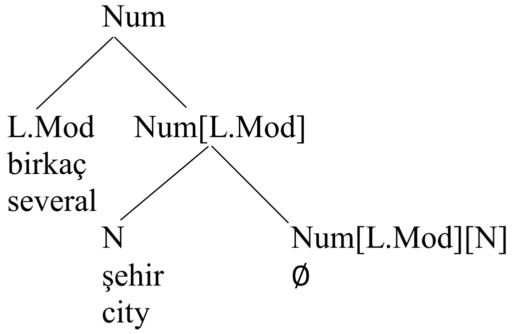
How should we analyze low modifiers? One might be tempted to think of them as adjectives, say as quantificational adjectives. If so, the absence of the accusative case marker on the objects in (37a-c) follows from the fact that the combination of these modifiers with nouns is labelled with N and not with D. There is reason to believe that such an analysis is not on the right track. Consider (38a-c).

(38) a. birkaç şehir haritası
several city map-CM
'several city maps'
cannot mean 'a map for several cities'
b. birçok çocuk oyuncağı
many kid toy-CM
'many toys for kids'
cannot mean 'a toy for many kids'
c. üç öğretmen raporu
three teacher reports
'three reports by teachers'
cannot mean 'a report by three teachers'

Recall that the combination of adjectives with compounds gives rise to structural ambiguity. If these modifiers are adjectives, then their combination with compounds should also lead to ambiguity. As can be seen in (38a-c), this is not what we find. That is, low modifiers behave neither as high modifiers (36a-c) nor as adjectives (38a-c). They are of a distinct category. What is it?

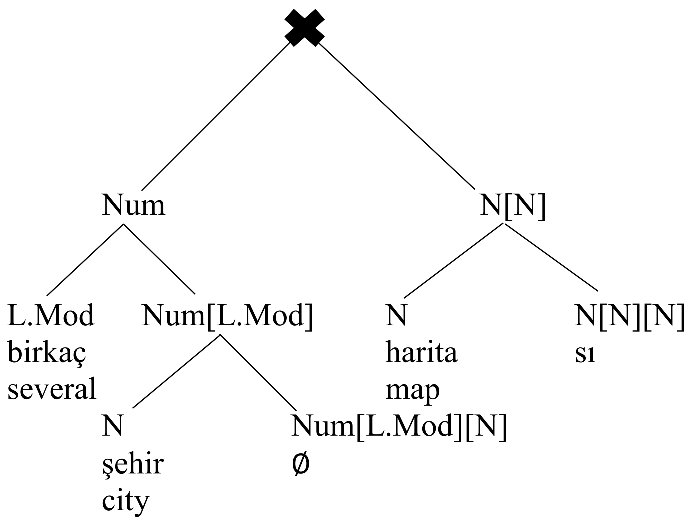
Low modifiers are contained in number-related constructions and such constructions should be labeled neither with N nor with D. We suggest that low modifiers, *L.Mod* for short, are selected by a number morpheme which labels the node containing a low modifier with the category *Num*. The syntactic representation of *birkaç şehir* 'several cities', compatible with this idea, in given in (39).

(39)



The phrase *birkaç şehir* 'several cities', which is of the Num category, cannot undergo compounding, which accounts for the lack of ambiguity in (38a). The same is true of all examples in (38).

(40)

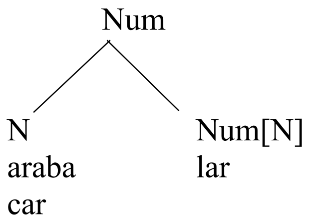


The category *Num* may also be made use of in representing morphological number marking in Turkish. The interpretation of a noun in Turkish depends on the presence or the absence of the plural marker *–lAr.*

(41) a. araba-Ø-da
car-SG-LOC
'in the car'
b. araba-lar-da
car-PL-LOC
'in the cars'

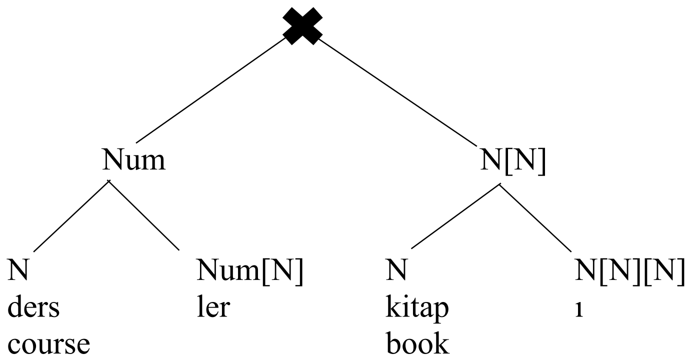
The contrast in (41) is a contrast in number properties of nouns, which, we suggest, is associated with the Num category. The syntactic representation of a noun marked with the plural marker *–lAr* can be represented as in (42).

(42)



This analysis of the plural marker in Turkish leads to the prediction that nouns marked with the plural marker should not undergo compounding.

(43)

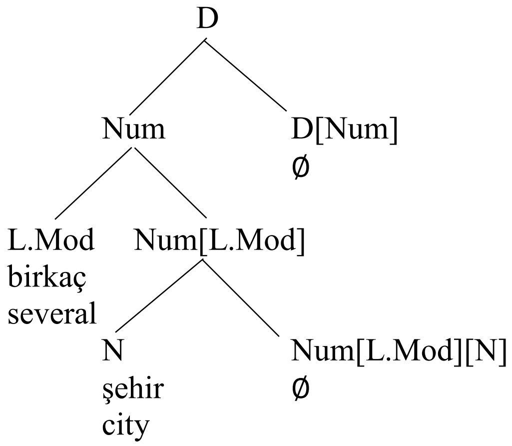


This is, indeed, a good prediction.

(44) a. *ders-ler kitab-ı
course-PL book-CM
Int. ‘a book for (various) courses’
b. *kalem-ler kutu-su
pencil-PL box-CM
Int. ‘a box for pencils’
c. *araba-lar lastiğ-i
car-PL tire-CM
Int. ‘a tire for cars’
d. *telefon-lar kılıf-ı
phone-PL case-CM
Int. ‘a case for phones’
r. *oje-ler kataloğ-u
nail.polish-PL catalogue-CM
Int. ‘a catalogue for (types of) nail polish’

There is one final issue we must address before we can bring this section to an end. Objects involving low modifiers need not be case-marked since they are not of category D. This suggests that in those cases where they are case-marked, such objects must be selected by the silent determiner in Turkish. Below, we draw the tree representation of a case-marked expression that contains a low modifier. (The mechanism behind case assignment will be introduced in Chapter 4 and Chapter 5).

(45)



We need to revise the entry for the silent determiner in Turkish so that it is merged not with a syntactic element labeled with N but with one labeled with Num. In fact, let us take this opportunity to summarize all the categories that we have discussed in this section.

(46) The category of the compound marker is “N[N][N]”
The category of the linker is “N[A][N]”
The categories of the number morphemes are
“Num[N]” and “Num[L.Mod][N]” (whichever works)
The category of low modifiers is “L.Mod”
The categories of the silent determiner are
“D[Num]” and “D[H.Mod][Num]” (whichever works)
The category of high modifiers is “H.Mod”
The category of pronouns and names is “D”

Given these categories, we can represent the complex expression *bu üç kırmızı kalem* ‘these three red pencils’ as in (47).

(47) D

H.Mod
bu
this

D[H.Mod]

Num

D[H.Mod][Num]
Ø

L.Mod
üç
three

Num[L.Mod]

N

Num[L.Mod][N]
Ø

A
kırmızı
red

N[A]

N
kalem
pencil

N[A][N]
Ø

The reader will recognize that every binary branching subtree in this tree is an instantiation of the schema for the Merge operation.

Let us finally remind ourselves of the types of morphemes that we have seen so far in the book.

Morphemes

Lexical Morphemes
Nouns
Verbs
Adjectives

Functional Morphemes
High Modifiers
- High Quantifiers
- Demonstratives

Low modifiers
- Numerals
- Vague quantifiers
- Indefiniteness marker

Specificity Marker
Pronouns
Case markers
Number markers
Compound marker
Linker
Derivational Suffixes

This is what we have learnt so far about morphemes in Turkish. In the next section, we will take a closer look at the internal makeup of morphemes. We will study features associated with categories of morphemes.

Exercises:

1. Draw syntactic trees for the following expressions. If an expression is structurally ambiguous, draw all of the possible trees.

a. birkaç	öğrenci	koltuğ-u			
several	student	chair-CM			
b. bu üç	kırmızı kalem	kutu-su			
this three	red pencil	box-CM			
c. çoğu	güzel	yeni	yuvarlak	masa	
most	beautiful	new	round	table	
d. her	telefon	kılıf-ı	reng-i	seçim-i	
every	phone	case-CM	color-CM	choice-CM	
e. her	kare	kırmızı	masa	örtü-sü	
every	square	red	table	cover-CM	

2. Below are some judgments associated with various Turkish expression. Explain whether our theory of grammar, the rules of which are given in (46), can explain these judgments or not. Assume that the rules in (46) all the rules of nominal syntax in Turkish.

a. *her	birçok	öğrenci
every	many	student
b. *çoğu	üç	öğrenci
most	three	student
c. *her	çoğu	öğrenci
every	most	student
d. uzun	her	öğrenci
tall	every	student
e. bu	üç	öğrenci
this	three	student
f. her	bi’	öğrenci
every	a	student
g. *çoğu	bi’	öğrenci
most	one	student
h. *telefon	siyah	kılıf-ı
telephone	black	case-CM

3. Below are some Turkish expressions. Explain whether they are expected to be acceptable or unacceptable given the theory of grammar we have discussed in this chapter. (Optional: You can consult a speaker of Turkish for their judgments concerning these expressions. Note that our theory of grammar need not be completely successful.)

a. öğrenci	birçok	koltuğ-u
student	many	chair-CM
b. kırmızı	her	araba
red	every	car
c. kalem	kırmızı	kutu-su
pencil	red	box-CM
d. ucuz	çoğu	araba
cheap	most	car
e. o	Ali	film-i
that	Ali	movie-CM

CHAPTER 4: Features on Categories and Agreement

In the last two chapters, we focused on morphemes without worrying about their possible internal makeup. It seems that there is more to be said about morphemes than just their category. We need to take a closer at the concept of **features**, which are the building blocks of morphemes.

4.1. Features inside categories

Consider the pronouns *ben* 'I', *biz* 'we', *sen* 'you' and *siz* 'you all'. As far as Chapter 3 is concerned, these are just four different morphemes with nothing in common other than being pronouns. Such a holistic approach misses some important generalizations, however. There is a property shared between *biz* 'we' and *siz* 'you all' to the exclusion of *ben* 'I' and *sen* 'you', i.e. they share the number feature. They denote a plurality of individuals containing the speaker and the addressee respectively. Similarly, there is a property shared between *ben* 'I' and *biz* 'we' to the exclusion of *sen* 'you' and *siz* 'you all', i.e. they share the person feature. Features such as number and person have their role to play in Turkish grammar. For instance, the agreement marker on the predicate co-varies with the number and person properties of the pronominal subject.

(1) a. **Ben** Ankara-ya git-ti-**m**
I Ankara-DAT go-PST-1SG
'I went to Ankara'
b. **Sen** Ankara-ya git-ti-**n**
you Ankara-DAT go-PST-2SG
c. **O** Ankara-ya git-ti-**Ø**
s|he/it Ankara-DAT go-PST-3SG
d. **Biz** Ankara-ya git-ti-**k**
we Ankara-DAT go-PST-1PL
e. **Siz** Ankara-ya git-ti-**niz**
you.all Ankara-DAT go-PST-2PL
f. **Onlar** Ankara-ya git-ti-**ler**
they Ankara-DAT go-PST-3PL

Now, one may wish to argue that the agreement suffix is indicative of the presence of the pronominal morphemes. That is, it is not the features associated with the morphemes that trigger agreement on the predicate but the morphemes themselves. There is, however, reason to believe that agreement involves more than reference to morphemes. In (2a) and (2b), we find first-person plural agreement on the predicate in the absence of the first-person plural pronoun. Agreement seems to be sensitive to the person feature on *ben* 'I' and the number feature of the combination of *ben* 'I' and *sen* 'you', which suggests that agreement is sensitive to featural makeup of morphemes and not to morphemes themselves.

(2) a. Ben ve sen dışarı çık-tı-**k**
I and you out leave-PST-**1PL**
'You and I went out.'
b. Ben ve o dışarı çık-tı-**k**
I and s|he out leave-PST-**1PL**
'S|he and I went out.'
c. Sen ve o dışarı çık-tı-**nız**
you and s|he out leave-PST-**2PL**
'You (singular) and s|he went out.'

The pronouns in Turkish express two types of information: number and person. This system can be represented with the table in (3).

(3) Pronouns in Turkish

	1st person	2nd person	3rd person
Singular (S)	ben	sen	o(n)
Plural (P)	biz	siz	onlar

We have already seen that pronouns are in the D category. We will take the number and person features associated with category *D* to be encoded on *D* as a list. For now, this list contains number {S, P} and person {1, 2, 3} features associated with each pronoun. The representation of pronouns can be shown as in (4):

(4)

D(S, 1) ben	D(S, 2) sen	D(S, 3) o(n)
D(P, 1) biz	D(P, 2) siz	D(P, 3) onlar

A second property of pronouns is that they must be case-marked. However, unlike number and person, case is not an inherent property of pronouns. Case is assigned to pronouns by other categories such as verbs and postpositions. The case marker on a pronoun changes as a function of the verb it is combined with.

(5)
a. Ali on-u gör-dü-Ø
Ali s|he-ACC see-PST-3SG
'Ali saw him/her/it.'
b. Ali on-a bak-tı-Ø
Ali s|he-DAT look-PST-3SG
'Ali looked at him/her/it.'
c. Ali ora-da yaşa-dı-Ø
Ali there-LOC live-PST-3SG
'Ali resided there.'
d. Ali on-dan kork-tu-Ø
Ali s|he/it-ABL fear-PST-3SG
'Ali feared him/her/it.'

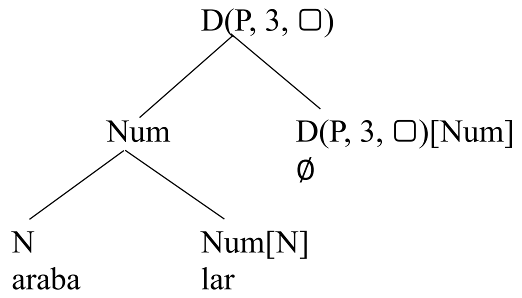
We will add the case feature as the third member of the list associated with determiners, but unlike number and person, there is no value assigned to it, a fact we represent using "▢". The value of the case feature on the pronoun will be assigned during the syntactic computation. Here is the revised representation of the pronouns in Turkish.

(6)

D(S, 1, ▢) ben	D(S, 2, ▢) sen	D(S, 3, ▢) o(n)
D(P, 1, ▢) biz	D(P, 2, ▢) siz	D(P, 3, ▢) onlar

The plural noun *arabalar* 'cars' with a silent determiner can be represented as in (7). The person feature of a silent determiner is always specified as being the third person. The plural number feature on the category *D(P, 3, ▢)* reflects the number feature under the category *Num[N]*, i.e. both are plural. In what follows, as a simplifying assumption, we take it that the number feature on the *D* category reflects the number feature associated with the Num category that it selects for. (In British English, one gets to say things like *The jury disagree on this issue*, where the subject is in the singular but the agreement marking on the verb is that of a plural subject. We will ignore such complications.)

(7)



While the representation of categories with feature lists on them is more complex than before, no complication has been introduced to our grammar as far as the Merge operation is concerned. This simple rule is still operative in determining which syntactic representations are legitimate. The labeling of mother nodes in (7) is nothing but the application of the Merge operation.

4.2. The Agreement operation

We are ready to provide the syntactic representation of the grammatical phenomenon known as **agreement**. This is a syntactic process that involves feature covariation between distinct elements in a tree, as illustrated by possessive constructions in Turkish. In (8), the genitive marked *Meryem* is said to be the possessor and *arabalar* 'cars' is the possessee (i.e. that which is possessed).

(8)	Meryem-in	araba-lar-ı
	Meryem-GEN	car-PL-3SG
	'Meryem's cars'	

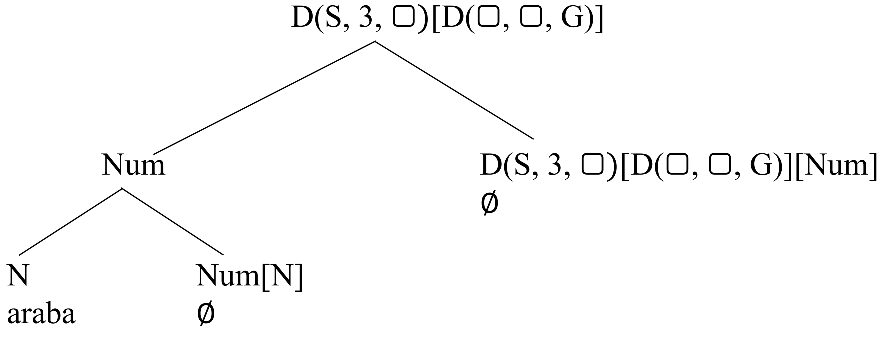
In such constructions, the morphological form of the agreement marker on the possessee, as well as the form of the genitive case marker, depends on the number and person features of the possessor.

(9) a. **ben-im** araba-**m**
I-GEN.1SG car-1SG
'my car'
b. **sen-in** araba-**n**
you-GEN.2SG car-2SG
'your car'
c. **on-un** araba-**sı**
s|he/it-GEN.3SG car-3SG
'his/her/its car'
d. **biz-im** araba-**mız**
we-GEN.1PL car-1PL
'our car'
e. **siz-in** araba-**nız**
you-GEN.2PL car-2PL
'you all's car' (a plurality of individuals involving the addressee)
f. **onlar-ın** araba-**ları**
they-GEN.3PL car-3PL
'their car'

There are (at least) two properties of such constructions that we need to explain. First, we need to understand how number and person features of the possessor control agreement that appears on the noun *araba* 'car'. Second, we need to understand how the possessor gets to be marked with the genitive case. This requires a revision of the kind of categories that can be associated with the silent determiners. We shall provide two distinct categories for the silent determiners in Turkish, one for the simple case where there is no possessor (as in (7)) and another for the silent possessive determiner (as in (8) and (9)). In this chapter, our focus is the syntactic representation of constructions with the silent possessive determiner.

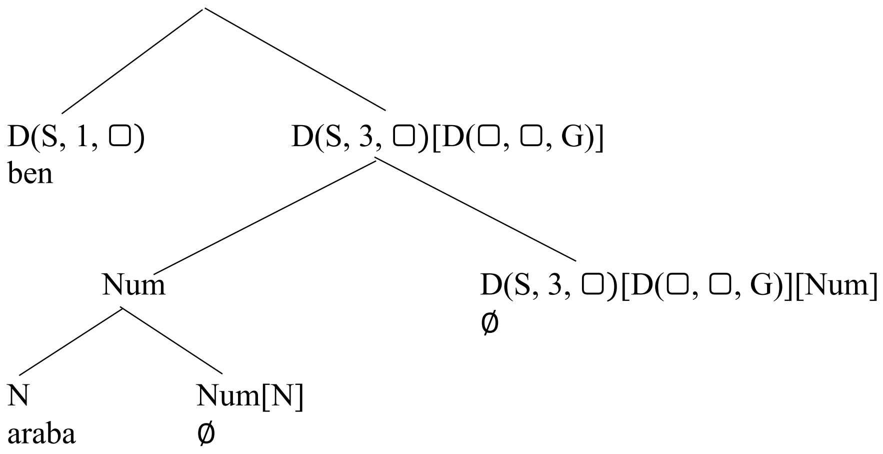
(10) The category of the silent determiner in Turkish is "D(NUM, 3, □)[Num]",
where *NUM* is one of *{S, P}* (reflecting the feature on the selected category *Num*).
The category of the silent possessive determiner in Turkish is
"D(NUM, 3, □)[D(□, □, G)][Num]"
where *NUM* is one of *{S, P}* (reflecting the feature on the selected category *Num*)

Let us see now how this categorical specification enables us to account for the set of expressions in (9), more specifically (9a). We start with the representation of the noun *araba* 'car' with the possessive determiner.

(11)


Now comes the tricky part. When we try to merge this tree with a pronoun, we find that the conditions for the application of the Merge operation are not met.

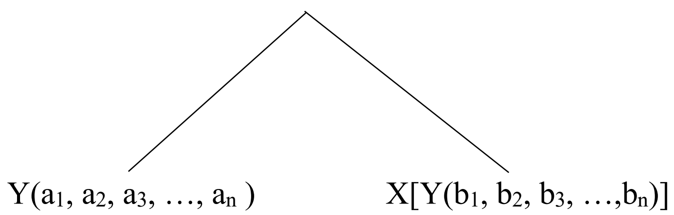
(12)



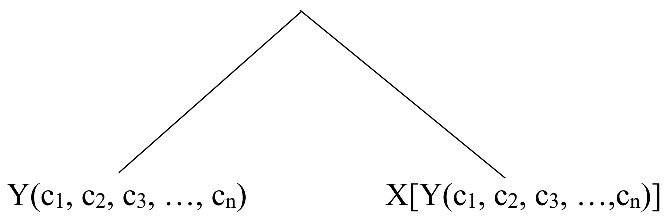
Since *D(S, 1, □)* and *D(□, □, G)* (inside *D(S, 3, □)[D(□, □, G)]*) are not identical categories, we cannot apply the Merge operation. Such discrepancies between sister nodes are precisely why there is a grammatical phenomenon called Agreement. This configuration will induce agreement between *D(S, 1,□)* and *D(□, □, G)* so that they share features and the conditions for the application of the Merge operation is met. This agreement is achieved by filling unvalued features on each list with the valued feature on the other list. This is what the Agreement operation, defined in (13), does.

(13) for any j such that $1 \leq j \leq n$
The Agreement Operation
is defined when at least one of a_j and b_j is "□"
when defined,

INPUT



OUTPUT

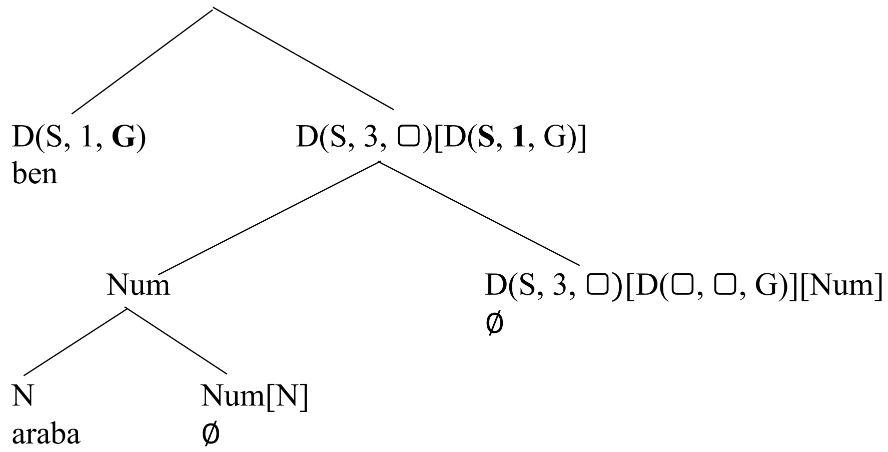


where $c_j \neq □$ and
$c_j = a_j$ or $c_j = b_j$

Intuitively, the Agreement operation is responsible for the swapping of features between lists on the sister nodes, the result of which is that the lists end up being identical to each other. The identity of the feature lists on the sister nodes is a precondition for the application of the Merge operation, which is the central operation for building structure.

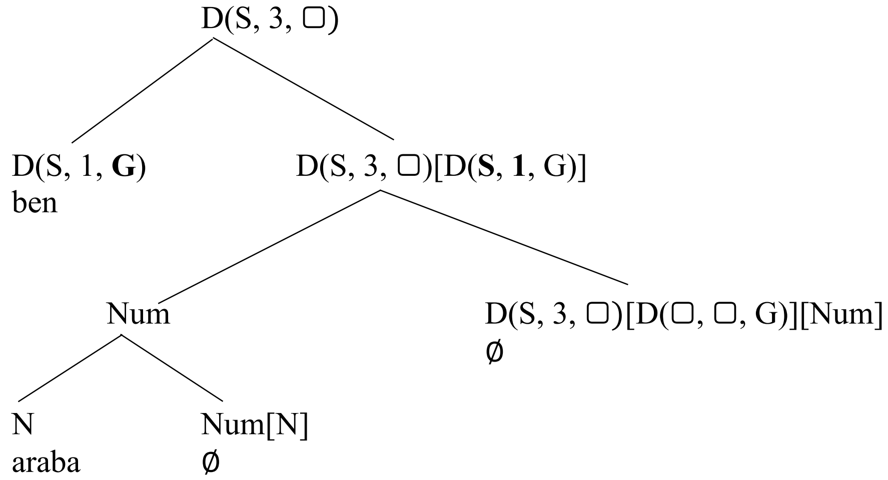
Applying the Agreement operation to the tree in (12), we obtain the tree in (14), where the features that are copied from the sister node are bolded.

(14)



This tree now meets the condition for the application of the Merge operation, thanks to which it now has the label *D(S, 3, □)*.

(15)



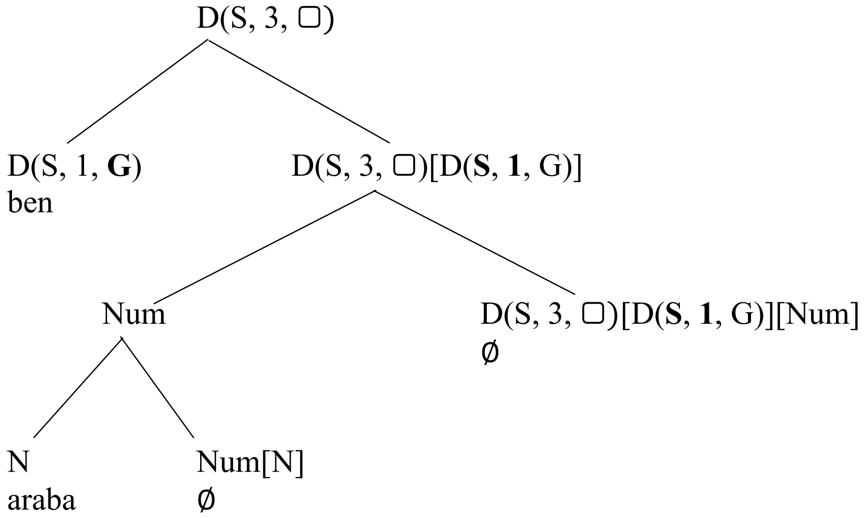
We assume that once the features on a category have been changed, the features on every category that was identical to it must also be changed. Let us call it the Copy Rule. We define it as follows:

(16) The Copy Rule

whenever X ≠ X’

In the tree in (15), letting X = *D(S, 3, ▢)[D(▢, ▢, G)]*, X' = *D(S, 3, ▢)[D(**S**, **1**, G)]* and Y = *Num* , we obtain X'[Y] = *D(S, 3, ▢)[D(**S**, **1**, G)][Num]* to replace X[Y] = *D(S, 3, ▢)[D(▢, ▢, G)][Num]* in the daughter node. The result of the application of the Copy Rule is shown in (17).

(17)



We must introduce a set of rules of realization for the genitive case marker on pronouns. The rules of realization for genitive case in (18) give us a way of expressing how it is that the form of the genitive suffix depends on the pronoun it appears on. The dots around the categories in (18) indicate that there may be other categories in the same terminal node. As far as pronouns are concerned, however, they are no such categories (i.e. "…" is empty).

(18) The rules of realization for the genitive case markers
… D(S, 1, **G**) … ⇔ -(I)m
… D(S, 2, **G**) … ⇔ -(I)n
… D(S, 3, **G**) … ⇔ -(n)In
… D(P, 1, **G**) … ⇔ -(I)m
… D(P, 2, **G**) … ⇔ -(I)n
… D(P, 3, **G**) … ⇔ -(n)In

What about the agreement suffix on the possessee? The list of rules in (19) expresses the realization of possessive agreement in Turkish. When a terminal node, i.e. a node with no branches, contains one of the categories of the form *D(NUM, 3, ▢)[D(**NUM**, **PERS**, G)],* where ***NUM*** is one of {S, P} and ***PERS*** is one of {1, 2, 3}, we apply the following rules for the morphological realization of the agreement suffix. The category *D(**NUM**, **PERS**, G*) must be an argument of *D(NUM, 3, ▢)*, where NUM is either S or P. The dots around *D(NUM, 3, ▢)[D(**NUM**, **PERS**, G)]* indicate that the relevant terminal node may contain other categories than *D(NUM, 3, ▢)[D(**NUM**, **PERS**, G)]*.

(19) The rules of realization for the possessive agreement suffixes
where NUM is one of {S, P},
… D(NUM, 3, ▢)[D(**S**, **1**, G)] … ⇔ -(I)m
… D(NUM, 3, ▢)[D(**S**, **2**, G)] … ⇔ -(I)n
… D(NUM, 3, ▢)[D(**S**, **3**, G)] … ⇔ -(s)I(n)
… D(NUM, 3, ▢)[D(**P**, **1**, G)] … ⇔ -(I)mIz
… D(NUM, 3, ▢)[D(**P**, **2**, G)] … ⇔ -(I)nIz
… D(NUM, 3, ▢)[D(**P**, **3**, G)] … ⇔ -lArI

The resultant tree after the insertion of the genitive case marker and the possessive agreement marker is given in (20).

(20)

D(S, 3, ▢)

- D(S, 1, **G**)
  ben-**im**
- D(S, 3, ▢)[D(**S**, **1**, G)]
  - Num
    - N
      araba
    - Num[N]
      ∅
  - D(S, 3, ▢)[D(**S**, **1**, G)][Num]
    ∅-m

4.3. Possessives inside possessives

The analysis of possessive constructions described in the previous section makes two important syntactic predications. The first one is that since possessive constructions are in the category D, it is required that they be case-marked. This is an accurate prediction.

(21) a. Ali araba-yı sat-tı-∅
Ali car-ACC sell-PST-3SG
'Ali sold the car.'
b. Ali araba sat-tı-∅
Ali car sell-PST-3SG
'Ali sold (a) car(s)'

(22) a. Ali Zeynep-in araba-sın-ı sat-tı-∅
Ali Zeynep-GEN car-POSS-ACC sell-PST-3SG
'Ali sold Zeynep's car.'
b. *Ali Zeynep-in araba-sı sat-tı-∅
Ali Zeynep-GEN car-POSS sell-PST-3SG
Int. 'Ali sold Zeynep's car.'

Secondly, the possessive construction in (20) can be input to other possessive construction, which means that we can embed possessive constructions inside possessive constructions (another case of recursion). This is also true.

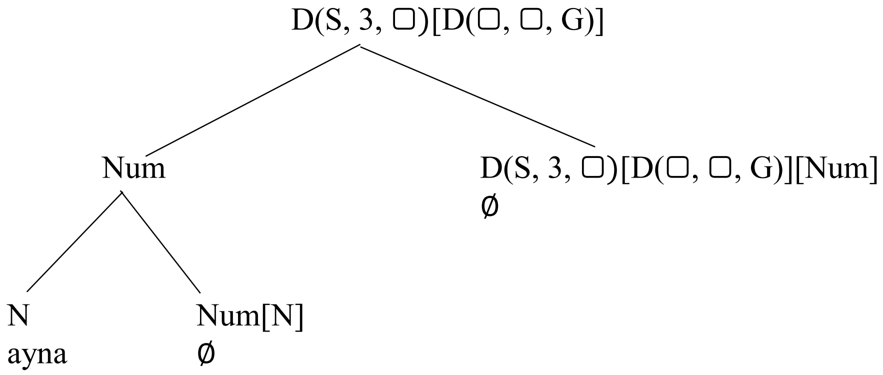
(23) a. Meryem
b. Meryem-in arkadaş-ı
Meryem-GEN friend-3SG
'Meryem's friend'
c. Meryem-in arkadaş-ın-ın arkadaş-ı
Meryem-GEN friend-3SG-GEN friend-3SG
'Meryem's friend's friend'
d. Meryem-in arkadaş-ın-ın arkadaş-ın-ın arkadaş-ı
Meryem-GEN friend-3SG-GEN friend-3SG-GEN friend-3SG
'Meryem's friend's friend's friend'

Considering the tree in (20), let us see what happens when it is embedded inside another possessive construction. Our aim is to draw a syntactic tree for the expression in (24).

(24)	ben-im	araba-m-ın	ayna-s-ı
	I-GEN.1SG	car-1SG-GEN	mirror-3SG
	‘the (rear-view) mirror of my car’		

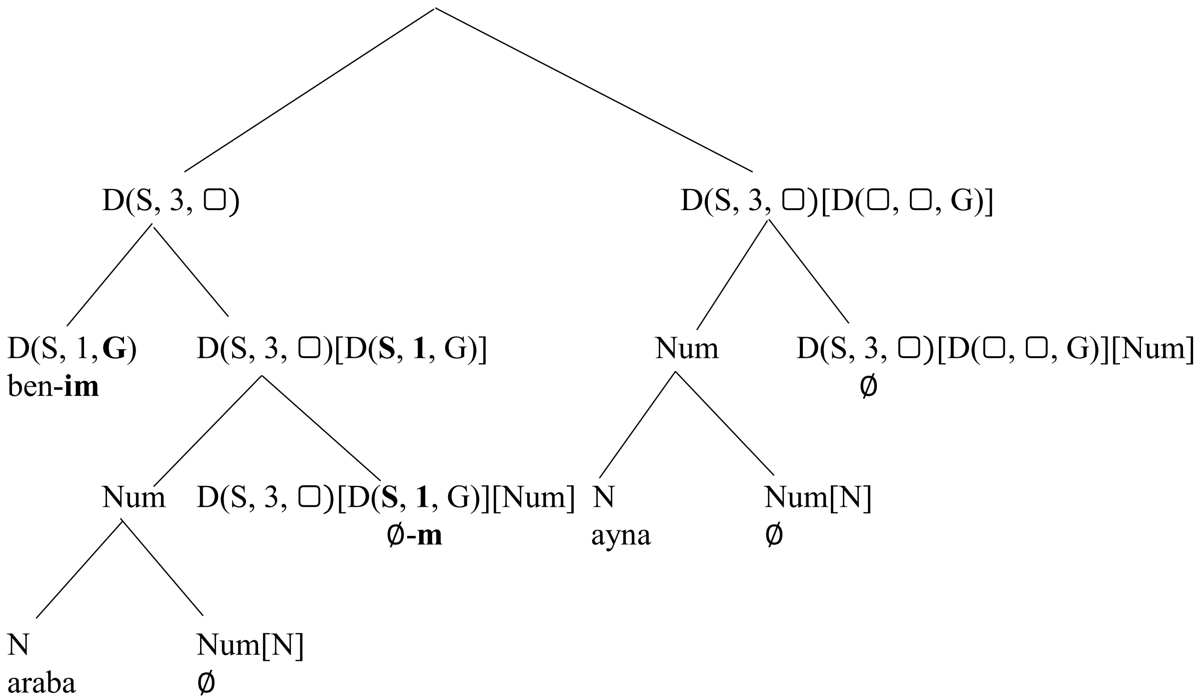
We start with the merger of *ayna* ‘mirror’ with the possessive determiner, which is given in (25):

(25)



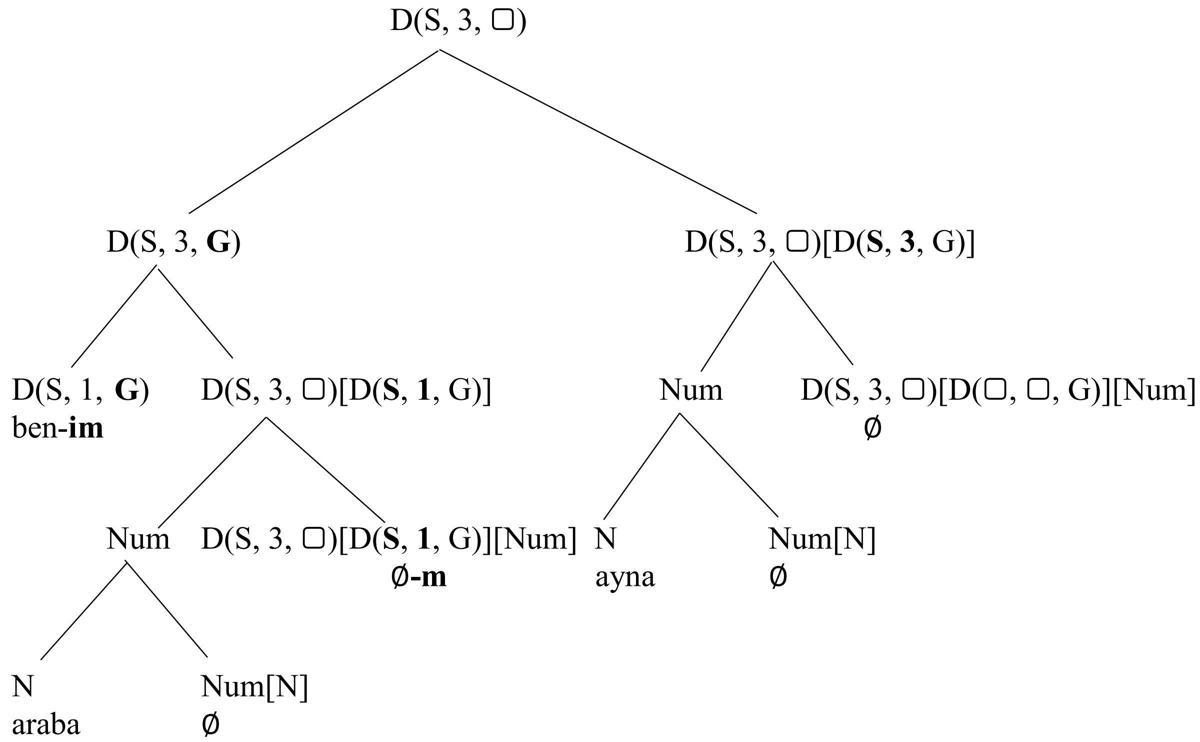
This tree is then to be merged with the possessive construction in (20). We obtain the syntactic representation in (26).

(26)



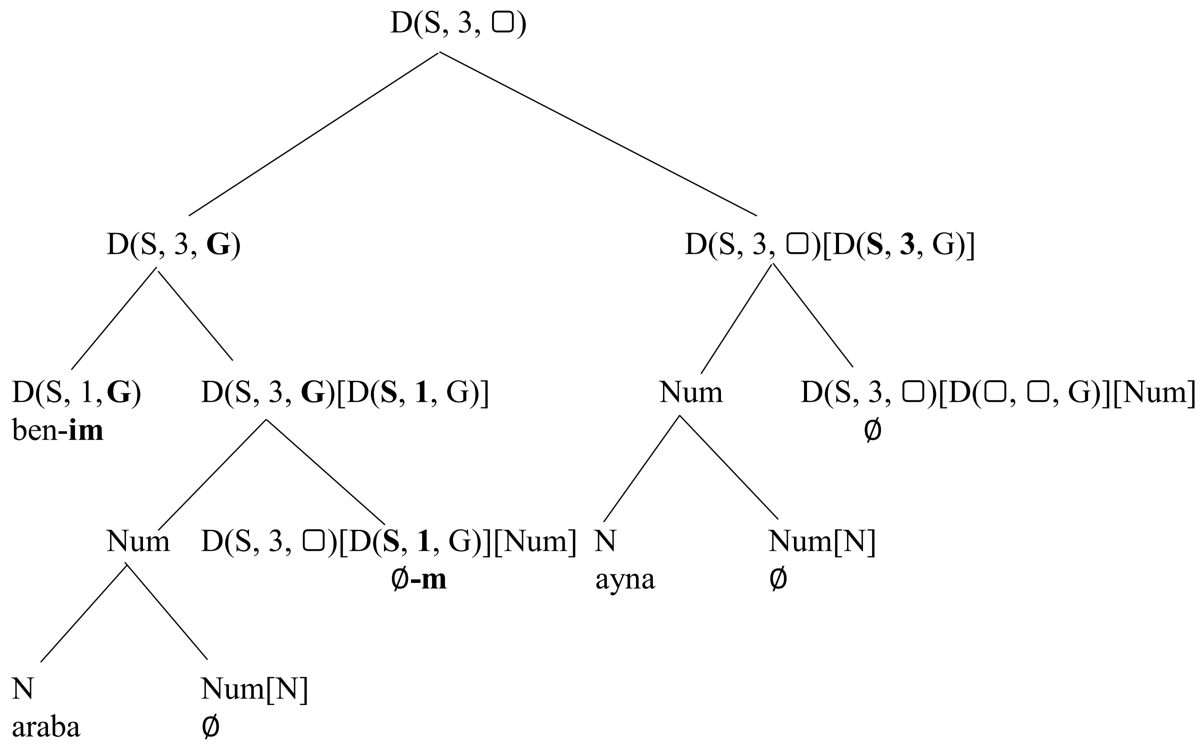
At this moment in the derivation, conditions for the application of the Agreement operation are met. We do “feature-swapping” between *D(S, 3, ▢)* and *D(▢, ▢, G)* to obtain *D(S, 3,* ***G****)* and *D(****S****,* ***3****, G*) respectively. This enables the application of the Merge rule so that the whole tree is now labeled as D(S, 3, ▢).

(27)



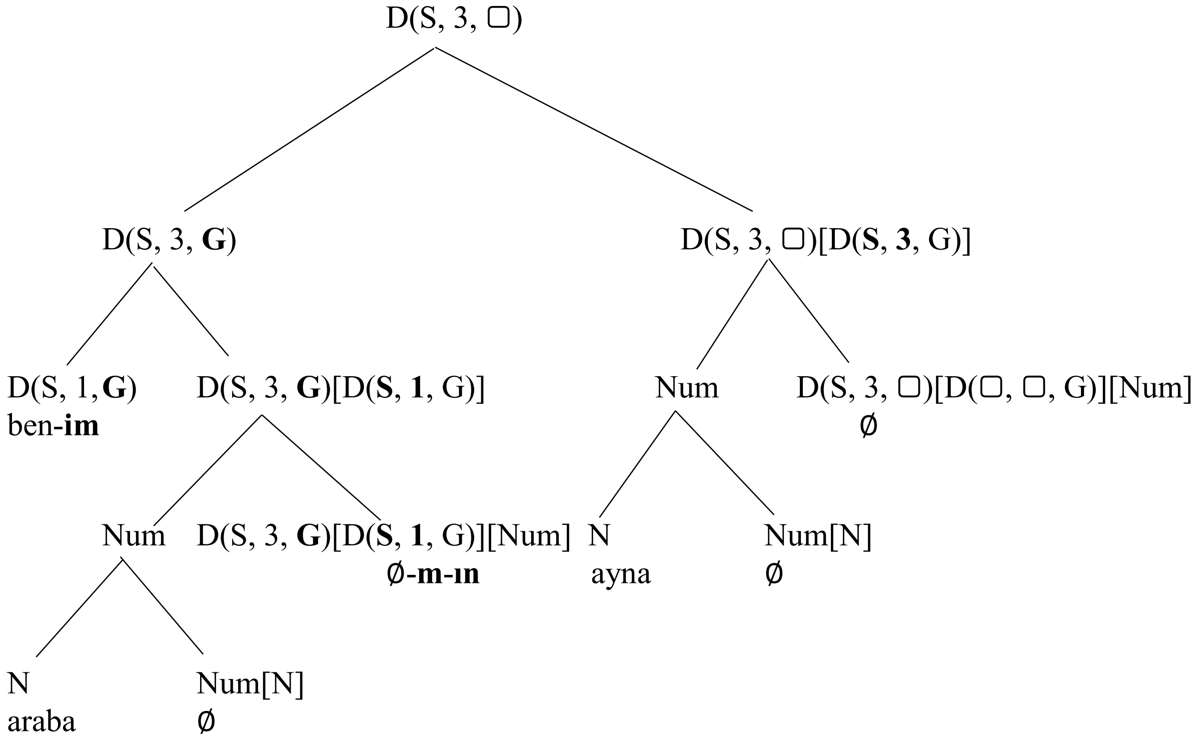
We apply the Copy rule on the left-hand side of the tree. We let X = *D(S, 3, ▢)*, X’ = *D(S, 3,* ***G)*** and Y = *D(****S****,* ***1****, G)* to obtain X’[Y] = *D(S, 3,* ***G****)[D(****S****,* ***1****, G)]* and we copy this category on the tree.

(28)



We should apply the Copy rule one more time. We let X' = *D(S, 3, **G**)[D(**S**, **1**, G)]* and Y = *Num* to obtain X'[Y] = *D(S, 3, **G**)[D(**S**, **1**, G)][Num]*, which is to replace X[Y] = *D(S, 3, ▢)[D(**S**, **1**, G)][Num]* on the tree in (29).

(29)



The insertion of the genitive suffix in (29) is governed by the rule in (30), which is repeated from (18). A more complete analysis of the realization of case markers will be discussed in Chapter 5.

(30) … D(NUM, 3, **G**) … ⇔ (n)In, where *NUM* is one of *{S, P}*

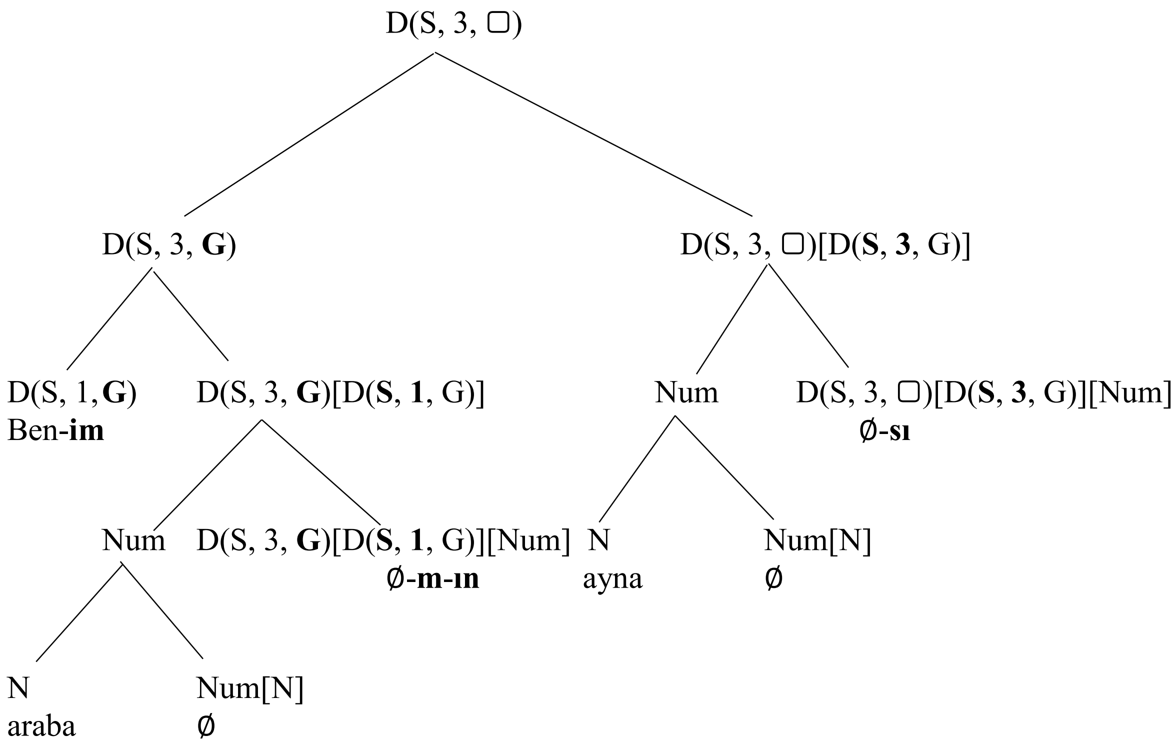
Observe that there are two overt morphemes under the category *D(S, 3, **G**)[D(**S**, **1**, G)][Num]*. The reason is that this category satisfies two distinct rules of realization. In addition to the rule of realization for the genitive suffix given in (30), this category also satisfies the rule of realization for the first-person singular possessive agreement given in (31), which is repeated from (19).

(31) … D(NUM, 3, ▢)[D(**S**, **1**, G)] … ⇔ -(I)m, , where *NUM* is one of *{S, P}*

The order of these two suffixes reflects the order in which valuations take place on the terminal category *D(S, 3, ▢)[D(▢, ▢, G)][Num]*, which is null. Each time this category is provided with new features, we insert overt morphemes on this node concerning the newly added features. In other words, *each step of feature valuation on a terminal node is followed by the application of a rule of realization for morphemes expressing the newly added features*. At the first step of feature valuation, we have *D(S, 3, ▢)[D(**S**, **1**, G)][Num]*, which means that we can, at this point, insert the first-person singular possessive suffix, the rule for which is given in (31). At the second round of valuation, the unvalued feature on *D(S, 3, ▢)* is valued as the genitive case, i.e. *D(S, 3, **G**)[D(**S**, **1**, G)][Num]*, which is when we can insert the genitive case morpheme, the rule for which is given in (30), as a result of which the genitive suffix follows the first-person singular possessive agreement suffix. As far as inflectional morphemes are concerned, morpheme insertion can only take place after the Copy rule has finished its job on the terminal nodes. In other words, we must wait for syntax to provide us with values of features before we can insert agreement or case morphemes. This is known as the **realizational** view of inflectional morphology, which is the kind of morphology that involves categories such as agreement and case marking. In this view, categories are basic units of syntactic computation and what morphemes do is associate categories with phonological elements, including the empty string.

We are not done with the tree in (29) yet. We need to apply the Copy rule on the right-hand side of the tree, too. Letting X' = *D(S, 3, ▢)[D(**S**, **3**, G)]* and Y = *Num*, we have X'[Y] = *D(S, 3, ▢)[D(**S**, **3**, G)][Num]* to replace X[Y] = *D(S, 3, ▢)[D(▢, ▢, G)][Num]*.

(32)



We can insert the third-person singular possessive agreement suffix under this category since the condition for the realization of this suffix is now met. (Note that (33) is repeated from (19))

(33) … D(NUM, 3, ▢)[D(**S**, **3**, G)] … ⇔ -(s)I(n), where *NUM* is one of *{S, P}*

This is our story for possessive constructions in Turkish and the role agreement plays in it. We have some understanding of why the Agreement operation is crucial for syntactic representations. In some contexts, it is precisely the availability of agreement that enables the Merge operation to go through. For instance, we cannot derive trees for possessive constructions with the Merge operation only. We would be stuck at the point where we try to merge the possessor with the possessee. In fact, as we will see in the next section, we cannot even form sentences with the Merge operation only: Agreement is a must. The same cannot be said for the Copy rule, however. The Copy rule has nothing to do with the Merge operation. In fact, it is fair to ask why the Copy rule should exist at all and whether we could find a way to map our trees into a sequence of morphemes without relying on this rule. This is a minimalist exercise that we do not have anything to contribute to.

Exercises:

1. Draw syntactic trees for the following expressions. If an expression is structurally ambiguous, draw all of the possible trees.

a. öğrenci-nin	kitap-lar-ı		
student-GEN	book-PL-3SG		
b. sıkıcı	kitap-ın	uzun	başlığ-ı
boring	book-GEN	long	title-3SG
c. büyük	araba	lastiğ-in-in	reng-i
big	car	tire-CM-GEN	color-3SG
d. Meryem-in	arkadaş-ın-ın	arkadaş-ın-ın	arkadaş-ı
Meryem-GEN	friend-3SG-GEN	friend-3SG-GEN	friend-3SG
e. telefon	kılıf-ı-nın	boy-u	
phone	case-CM-GEN	size-3SG	

2. Given (10), it is not possible to draw syntactic trees for the following expressions, which are all acceptable. Propose additional categories for the silent determiners (i.e. add new categories to (10)) so that it is possible to draw trees for the expressions below and then draw the trees.

a. ben-im	her	kitab-ım		
I-GEN.1SG	every	book-1SG		
'every book of mine'				
b. sınıf-ın	çoğu	uzun	öğrenci-si	
class-GEN	most	tall	student-3SG	
'most tall students in the class'				
c. bu sıkıcı ders-in	şu	öğrenci-si		
this boring class-GEN	that	student-3SG		
'that student of this boring class'				
d. bu ders kitab-ı-nın	her	bölüm-ün-ün	her	başlığ-ı
this course book-CM-GEN	every	section-3SG-GEN	every	title-SG
'every title of every section of this course book'				

CHAPTER 5: A first look at sentences

We now know how to combine syntactic categories with each other (the Merge operation) and how to express covariation of features amongst categories (the Agreement operation). We have all the tools needed to represent the structure of sentences in Turkish. This is what we are to do in this section.

5.1. Sentences as tensed expressions

There are several properties of sentences that we should be able to represent in our trees. One is that the person and number features of the subject are reflected on the tense marker. That is, there is agreement in person and number features, sometimes called phi-features or φ-features, between the subject and the predicate.

(1)	a. **Ben**	Ankara'ya	git-ti-**m**
	I	Ankara-DAT	go-PST-1SG
	'I went to Ankara'		
	b. **Sen**	Ankara-ya	git-ti-**n**
	you	Ankara-DAT	go-PST-2SG
	c. **O**	Ankara-ya	git-ti-**Ø**
	s|he/it	Ankara-DAT	go-PST-3SG
	d. **Biz**	Ankara-ya	git-ti-**k**
	we	Ankara-DAT	go-PST-1PL
	e. **Siz**	Ankara-ya	git-ti-**niz**
	you.all	Ankara-DAT	go-PST-2PL
	f. **Onlar**	Ankara-ya	git-ti-**ler**
	they	Ankara-DAT	go-PST-3PL

A second property of sentences we need to understand is that typically the case marker on the non-subject argument in the sentence depends on the verb. In other words, as far as case assignment is concerned, there is a close relation between the verb and the object. Below we provide examples in which verbs assign accusative (A), dative (D), locative (L) and ablative (B) case to the pronouns they are merged with.

(2)	a. Ali	on-**u**	**gör**-dü-Ø
	Ali	s|he/it-**ACC**	**see**-PST-3SG
	'Ali saw him/her'		
	b. *Ali	on-a	gör-dü-Ø
	Ali	s|he/it-DAT	see-PST-3SG
	Int. 'Ali saw him/her.'		
	c. Ali	on-**a**	**bak**-tı-Ø
	Ali	s|he/it-**DAT**	**look**-PST-3SG
	'Ali looked at him/her.'		
	d. *Ali	on-u	bak-tı-Ø
	Ali	s|he/it-ACC	book-PST-3SG
	Int. 'Ali looked at him/her.'		

(3)	a. Ali	ora-**da**	**yaşa**-dı-Ø
	Ali	there-**LOC**	**live**-PST-3SG
	'Ali resided there'		
	b. *Ali	ora-yı	yaşa-dı-Ø
	Ali	there-ACC	live-PST-3SG
	c. *Ali	ora-ya	yaşa-dı-Ø
	Ali	there-DAT	live-PST-3SG
	d. *Ali	ora-dan	yaşa-dı-Ø
	Ali	there-ABL	live-PST-3SG
(4)	a. Ali	on-**dan**	**kork**-tu-Ø
	Ali	s|he/it-**ABL**	**fear**-PST-3SG
	'Ali feared him/her.'		
	b. *Ali	on-u	kork-tu-Ø
	Ali	s|he/it-ACC	fear-PST-3SG
	c. *Ali	on-a	kork-tu-Ø
	Ali	s|he/it-DAT	fear-PST-3SG
	d. *Ali	on-da	kork-tu-Ø
	Ali	s|he/it-LOC	fear-PST-3SG

In the previous section, we saw that the genitive case is assigned to the possessor under sisterhood relation. The assignment of case features by verbs is calculated under the same configuration. In (2) to (4), for instance, the pronoun that is merged with some verb is assigned the case feature that is specified for this verb. The derivation starts when we put the object and the verb together.

(5)

D(S, 3, □)	V[D(□, □, A)]
o(n)	gör
s|he/it	see

In this merger, the condition for the application of the Merge operation is not yet met. We must apply the Agreement operation first to get a label for the whole tree as *V*.

(6)

V

D(S, 3, **A**)	V[D(**S**, **3**, A)]
on-**u**	gör
s|he/it-ACC	see

The assignment of other case features to the pronouns in other cases can be analyzed in the same manner.

(7) a. V

D(S, 3, **D**) V[D(**S**, **3**, D)]
on-**a** bak

b. V

D(S, 3, **L**) V[D(**S**, **3**, L)]
ora-**da** yaşa

c. V

D(S, 3, **B**) V[D(**S**, **3**, B)]
on-**dan** kork

The rules of realization relevant for the marking of case on pronouns can be given as follows: (Reminder: **N** = Nominative, **A** = Accusative, **G** = Genitive, **D** = Dative **L** = Locative and **B** = Ablative). The dots around categories indicate the possibility of other categories in the same terminal node. As far as the pronouns are concerned, there are no such categories (i.e. "…" is empty).

(8) The rules of realization for case markers
(for genitive case, see (18) in Chapter 4)
where NUM is one of {S, P} and PERS is one of {1, 2, 3}
… D(NUM, PERS, **N**) … ⇔ Ø
… D(NUM, PERS, **A**) … ⇔ -(y)I
… D(NUM, PERS, **D**) … ⇔ -(y)A
… D(NUM, PERS, **L**) … ⇔ -DA
… D(NUM, PERS, **B**) … ⇔ -DAn

Given the tree representations in (6) and (7), we could expect verbs in Turkish to show agreement with the object. While Turkish has no marking of object agreement on the verb, there are languages that do. For instance, in Cuzco Quechua, a member of the Quechuan language family spoken in Peru, Bolivia and Argentina, there are both object and subject agreement on the predicate. We take this to mean that it is a morphological property of Turkish that there are no morphemes expressing the presence of object agreement on the verb. (In other words, there are no rules of realization for object agreement in Turkish). It is to be noted that this is not a fundamental fact about human grammar.

(9) a. maylla-**wu**-rqa-n
wash-1SG-PAST-3SG
'S/he washed me.'
b. maylla-**wu**-rqa-nki
wash-1SG-PAST-2SG
'You washed me.'

We are now ready to discuss the merger of the subject into the tree. There are two puzzles about subjects in Turkish that we need to understand: (a) How is it that the person and number features of the subject are realized on the tense marker? (b) How is it that the subject in Turkish is always in the nominative (N) case? To address these questions, we need to determine the syntactic category of the tense markers in Turkish.

The examples we have seen thus far may have given the impression that the tense morpheme is always a suffix on the lexical verb, but that is not true. The tense morpheme and the lexical verb are morphologically separable categories. For instance, when the imperfective aspectual marker *-Iyor* appears on the verb, the tense morpheme is no longer part of the word that contains the verb. (We will discuss the syntactic representation of Aspect in the next section.)

(10) a. Ali kitab-ı oku-yor idi
Ali book-ACC read-IMPF PST
'Ali was reading the book.'
b. Ali kitab-ı oku-du
Ali book-ACC read-PST
'Ali read the book.'

In the presence of the imperfective aspect marker, we see that the tense morpheme and the lexical verb are morphologically separable, which suggests that the category tense should be represented independently of the lexical verb to which it may be attached.

We should also note that the past tense morpheme *-DI* in Turkish merges with verbs and verbs only. In other words, the tense category selects for a verbal category.

(11) a. Ben ağla-dı-m
I cry-PST-1SG
'I cried'
b. *Ben hemşire-di-m
I nurse-PST-1SG
Int. 'I was a nurse.'
c. *Ben hasta-dı-m
I sick-PST-1SG
Int. 'I was sick.'

Tense first merges with a verbal element and then with an element in the determiner category, which is the subject, as a result of which it assigns the nominative case to the subject. The merger of tense with the subject involves the application of the Agreement operation, which accounts for the agreement marker on the tense morpheme.

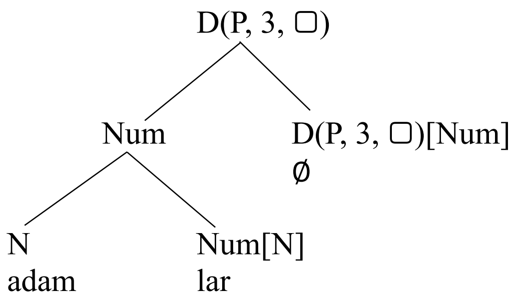
(12) (To be revised in (27))
The category of the past tense marker in Turkish is "TPAST[D(□, □, N)][V]".

Let us now provide a step-by-step derivation of (13) in Turkish.

(13)	Bi’	kadın	adam-lar-a	bak-tı-∅
	a	woman	man-PL-DAT	look-PST-3SG
	‘A woman looked at the men’			

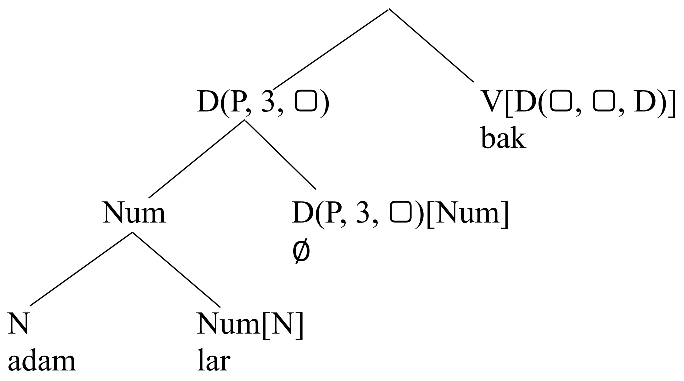
We start by constructing the object of the sentence.

(14)



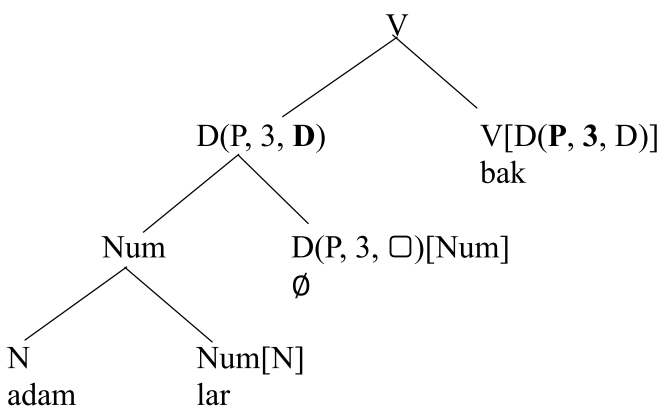
At this point in the derivation, we do not know which case is to be assigned to this object. That depends on the verb that it is to be merged with. In the sentence we are interested in, the object in (14) is merged with the lexical verb *bak* ‘look’ which assigns the dative case.

(15)

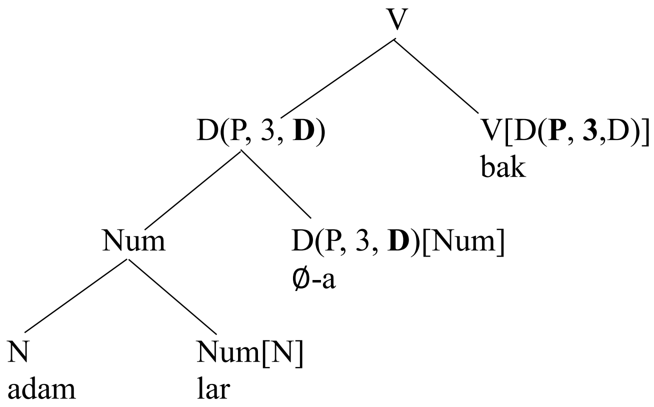


We must apply the Agreement operation between the object and the verb so that the Merge operation can determine the category of the tree.

(16)

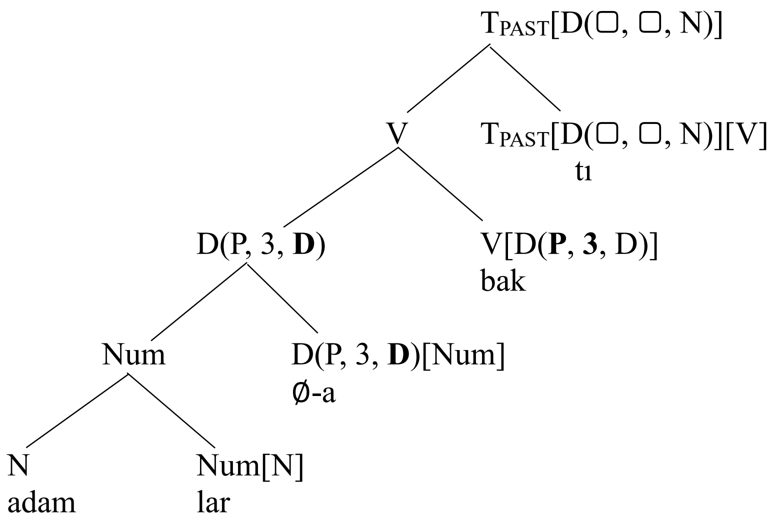


We apply the Copy Rule so that the dative case feature can appear on a terminal node, a node with no branches, and we then insert the dative marker. The rules of realization for case markers were discussed in (8).



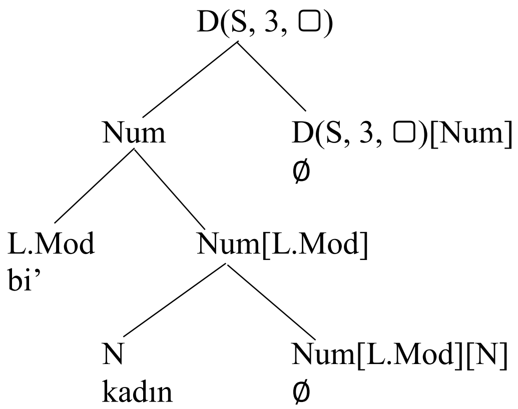
So far so good! We then add the tense category to the tree, which, we have seen, selects for a verb.

(17)



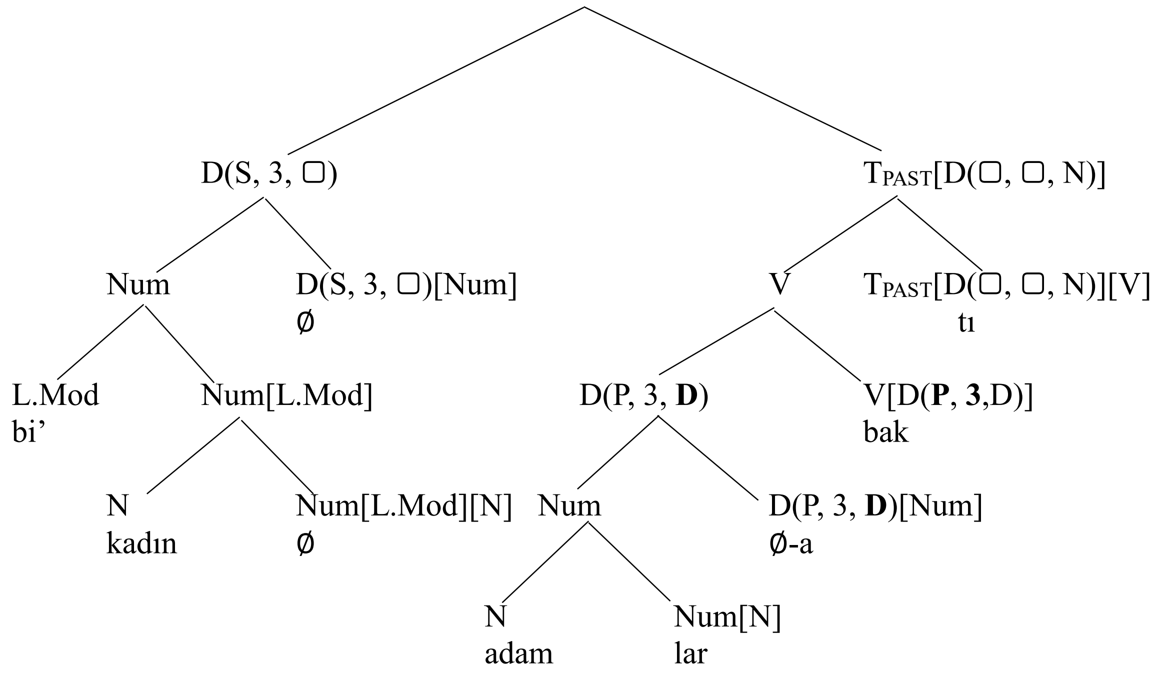
We now construct the subject that is to be merged with the tree in (17)

(18)



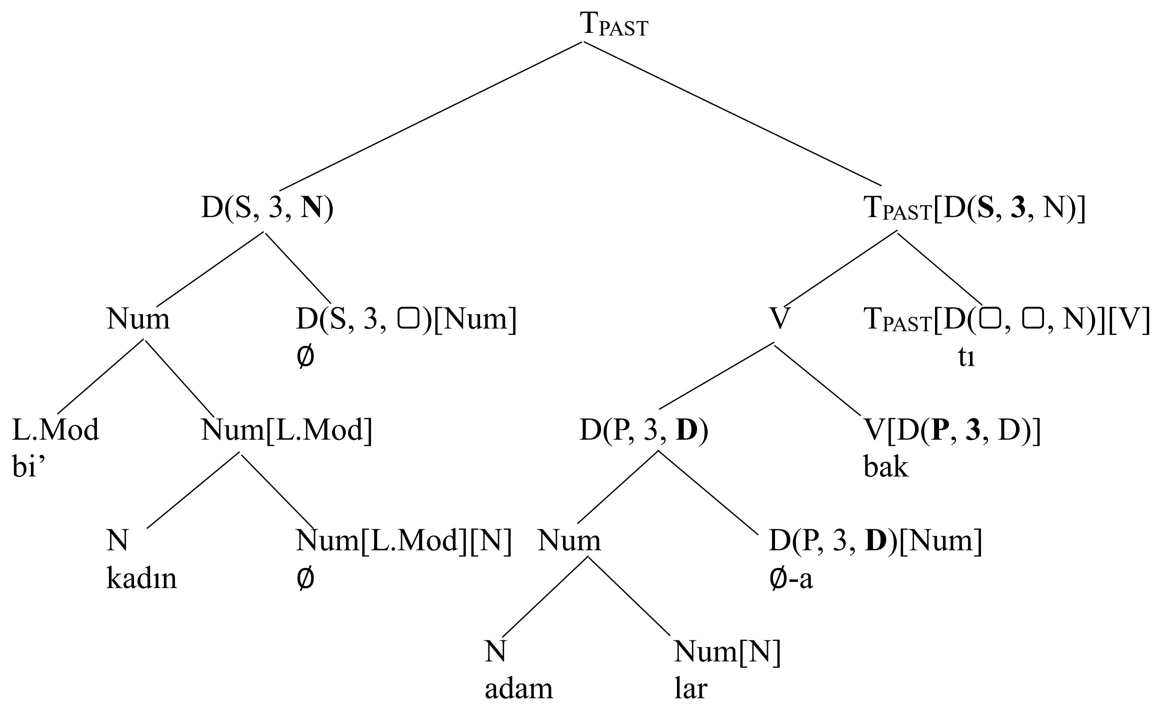
We merge the subject with the rest of the tree only to find out that there is another Agreement operation that we must apply before we can label the whole tree.

(19)



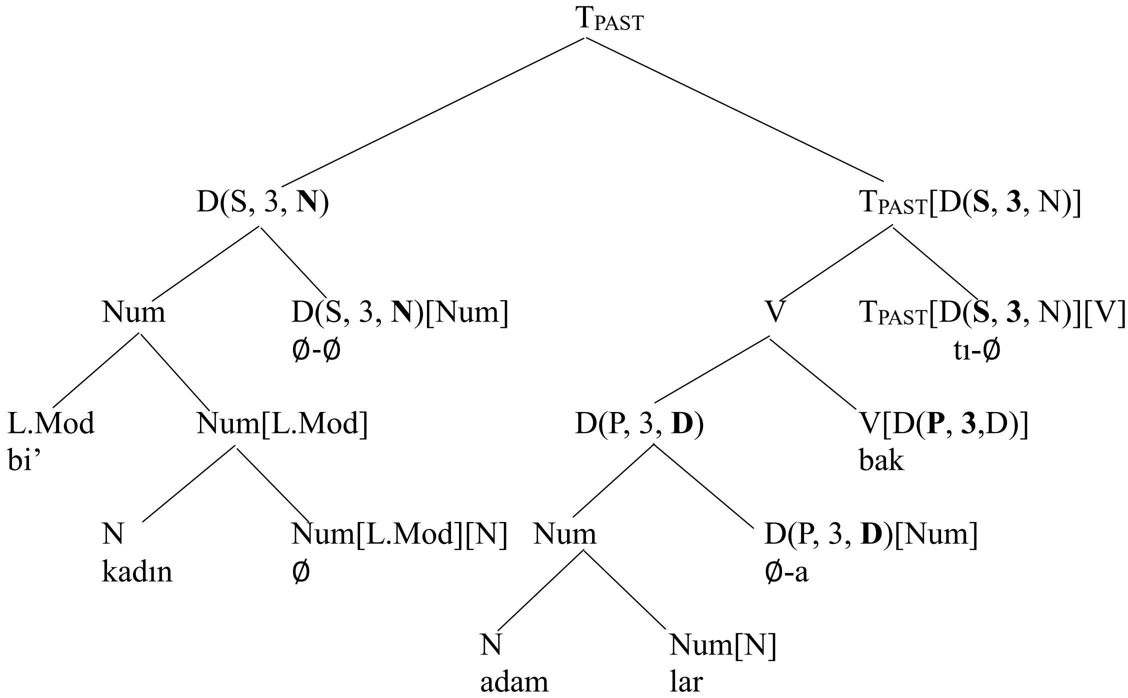
We apply the Agreement operation between sister nodes at the top of the tree, as a result of which we can label the tree as the category TPAST.

(20)



We are *almost* done. The careful reader will recognize that we need to apply the Copy rule once on the left-hand side and once on the right-hand side of the tree.

(21)



We supplement the tree in (21) with the realization rules for the agreement morphemes given in (22). (The rules of realization for case markers were discussed in (8))

(22) The rules of realization for subject agreement

… $\mathrm{T_{PAST}}$[D(**S**, **1**, N)] … ⇔ -m
… $\mathrm{T_{PAST}}$[D(**S**, **2**, N)] … ⇔ -n
… $\mathrm{T_{PAST}}$[D(**S**, **3**, N)] … ⇔ ∅
… $\mathrm{T_{PAST}}$[D(**P**, **1**, N)] … ⇔ -k
… $\mathrm{T_{PAST}}$[D(**P**, **2**, N)] … ⇔ -niz
… $\mathrm{T_{PAST}}$[D(**P**, **3**, N)] … ⇔ -ler

5.2. The syntax of Aspect in Turkish

Tense is typically understood to specify a temporal interval with respect to the speech time, which is *now*. The past tense specifies a temporal interval before the speech time, the present tense at the speech time and the future tense, if there is such a category (see below), after the speech time. Tense is not the only category concerning temporal properties of an event. There is also Aspect. In the typical case, Aspect informs us about whether the event in question is a completed event (perfective) or an on-going one (imperfective). In Turkish, whether an event is competed or not is morphologically encoded as a contrast concerning the presence or the absence of the imperfective aspect marker. When the aspectual marker *–(I)yor* is marked on the verb, the event denoted by the verb is understood to be an ongoing event. The absence of this morpheme is understood to convey that the event is a completed one.

(23) a. Zeynep kitab-ı taşı-yor idi-Ø
Zeynep book-ACC carry-IMPV PST-3SG
'Zeynep was carrying the book.'
b. Zeynep kitab-ı taşı-Ø-dı-Ø
Zeynep book-ACC carry-PRV-PST-3SG
'Zeynep carried the book.'

When the imperfective marker appears on the verb, the tense marker has the special form *idi* rather than the tense suffix *–DI*, as we see in (24a). In fact, in the context of the perfective aspect marker the tense marker cannot have this form.

(24) *Zeynep kitab-ı taşı-Ø idi-Ø
Zeynep book-ACC carry-PRV PST-3SG

To explain this asymmetry, we suggest that the perfective aspect in Turkish projects a verbal aspectual category, which we denote as Asp_V.

(25) The category of the perfective morpheme in Turkish is "Asp_V[V]"

The imperfective aspectual marker *-Iyor* differs from the perfective marker in that it lacks this verbal property. It takes a verb as input and gives an aspectual category as output. The term **participle** is sometimes used to express non-verbal aspectual categories.

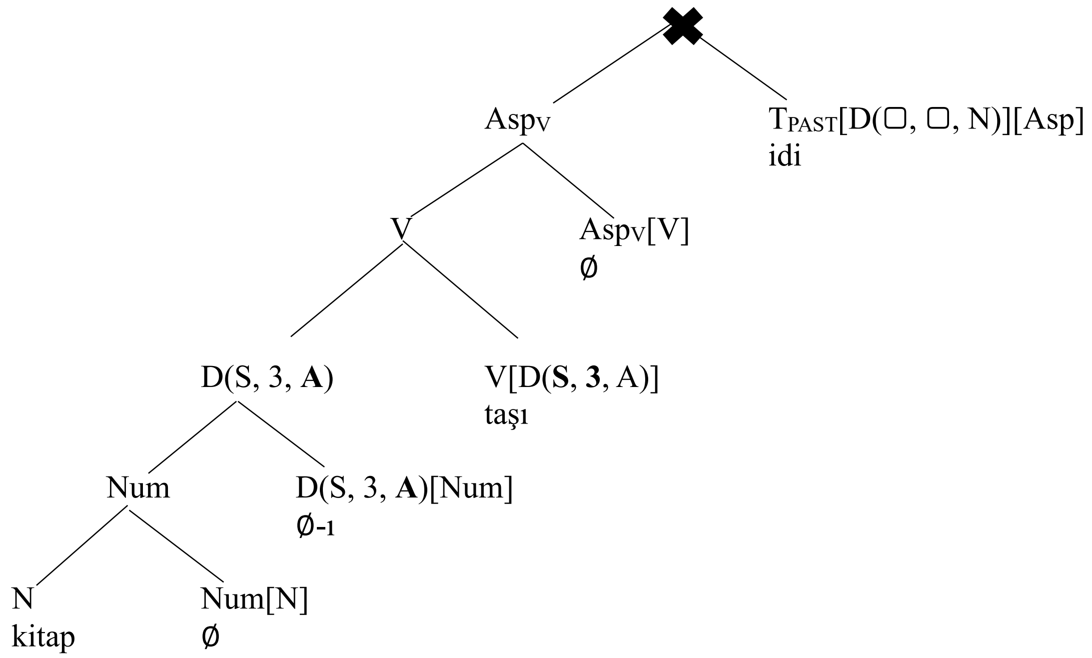
(26) The category of participles in Turkish is "Asp[V]"

We also need to revise the category of past tense morphemes *–DI* and *idi* to guarantee that they take Aspectual categories as their argument rather than lexical verbs. The past tense marker *–DI* is merged with a verbal aspectual category (as in (23b)), which we have shortened to Asp_V. The past tense morpheme *idi*, on the other hand, takes Aspect as input (as in (23a)) and gives us a category of tense, i.e. $T_{PAST}[D(\square, \square, N)]$.

(27) The category of the past tense marker *-DI* is "$T_{PAST}[D(\square, \square, N)][Asp_V]$"
The category of the past tense marker *idi* is "$T_{PAST}[D(\square, \square, N)][Asp]$"

Under these assumptions, the unacceptability of (24) is explained by the fact that the past tense marker *idi* cannot be merged with a verbal aspectual marker.

(28)



In the exercises at the end of this section, you will be asked to draw syntactic trees for (23a) and (23b). Let us, for now, focus on a variant of (23a) in which the past tense marker is missing.

(29) Zeynep kitab-ı taşı-yor
Zeynep book-ACC carry-IMPV
'Zeynep is carrying the book.'

We suggest that in such constructions the absence of the past tense morpheme implies the presence of the present tense morpheme, which is null. Independent evidence for the existence of such a category comes from the fact that the kind of agreement markers we find in the context of present tense differs from the kind of agreement markers we find on the past tense marker.

(30)

k-paradigm of agreement			**z-paradigm of agreement**		
Ben	Ankara-ya	git-∅-ti-**m**	**Ben**	Ankara-ya	gid-iyor-∅-**um**
I	Ankara-DAT	go-PRV-PST-1SG	I	Ankara-DAT	go-IMPV-PRES-1SG
'I went to Ankara.'			'I am going to Ankara.'		
Sen	Ankara-ya	git-∅-ti-**n**	**Sen**	Ankara-ya	gid-iyor-∅-**sun**
you	Ankara-DAT	go-PRV-PST-2SG	you	Ankara-DAT	go-IMPV-PRES-2SG
O	Ankara-ya	git-∅-ti-**∅**	**O**	Ankara-ya	gid-iyor-∅-**∅**
s|he	Ankara-DAT	go-PRV-PST-3SG	s|he	Ankara-DAT	go-IMPV-PRES-3SG
Biz	Ankara-ya	git-∅-ti-**k**	**Biz**	Ankara-ya	gid-iyor-∅-**uz**
we	Ankara-DAT	go-PRV-PST-1PL	we	Ankara-DAT	go-IMPV-PRES-1PL
Siz	Ankara-ya	git-∅-ti-**niz**	**Siz**	Ankara-ya	gid-iyor-∅-**sunuz**
you.all	Ankara-DAT	go-PRV-PST-2PL	you.all	Ankara-DAT	go-IMPV-PRES-2PL
Onlar	Ankara-ya	git-∅-ti-**ler**	**Onlar**	Ankara-ya	gid-iyor-∅-**lar**
they	Ankara-DAT	go-PRV-PST-3PL	they	Ankara-DAT	go-IMPV-PRES-3PL

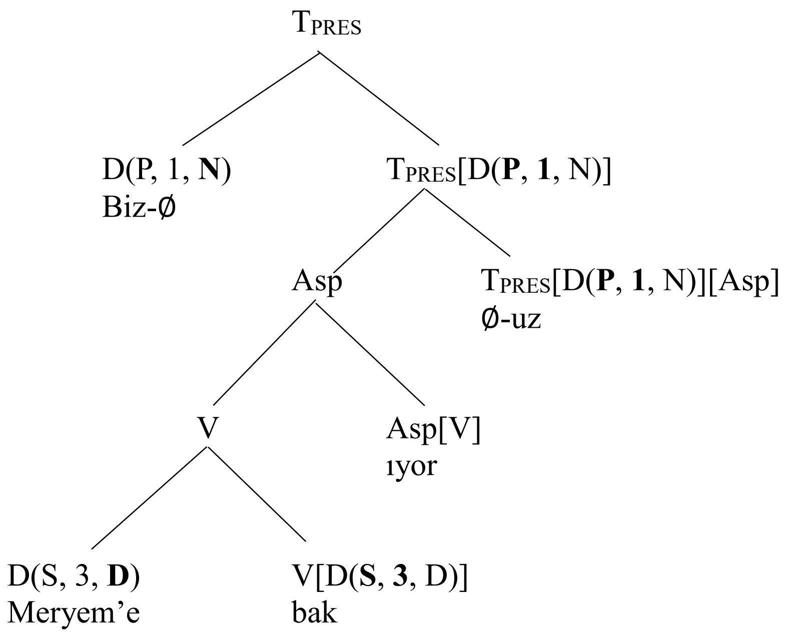
The agreement markers associated with the present tense are said to form the **z-paradigm of agreement** and the ones associated with the past tense are called the **k-paradigm of agreement**. These descriptive terms reflect the fact that the most salient difference between these two types of agreement markers is in the realization of the first-personal plural agreement (circled in (30)). The categorical specification of the present tense morpheme can be expressed as in (31).

(31) where ASP is one of {Asp, Asp_V}
The category of the null present tense morpheme is "TPRES[D(▢, ▢, N)][ASP]"

The syntactic representation of the sentence in (32) can be given as in (33).

(32) Biz Meryem'e bak-ıyor-Ø-uz
we Meryem-DAT look-IMPV-PRES-1PL
'We are looking at Meryem'

(33)



The rules of realization for the z-paradigm of agreement are given in (34). For the k-paradigm of agreement, we make use of the rules we have introduced in (22).

(34) The rules of realization for the z-paradigm of agreement
… TPRES[D(**S**, **1**, N)] … ⇔ -(y)Im
… TPRES[D(**S**, **2**, N)] … ⇔ -sIn
… TPRES[D(**S**, **3**, N)] … ⇔ Ø
… TPRES[D(**P**, **1**, N)] … ⇔ -(y)Iz
… TPRES[D(**P**, **2**, N)] … ⇔ -sInIz
… TPRES[D(**P**, **2**, N)] … ⇔ -lAr

Our analysis of tense in Turkish has consequences for the representation of future-oriented temporal relations. When we look at the agreement marker in such construction, we find that it is the agreement marker that appears on the present tense marker, i.e. the z-paradigm of agreement. This suggests that *–(y)Acak* in Turkish should analyzed as a prospective aspect marker (PROS) rather than a tense marker.

(35)	a. Ben	Meryem-i	ara-yacağ-∅-**ım**
	I	Meryem-ACC	call-PROS-PRES-1SG
	'I will call Meryem.'		
	b. Sen	Meryem-i	ara-yacak-∅-**sın**
	you	Meryem-ACC	call-PROS-PRES-2SG
	c. O	Meryem-i	ara-yacak-∅-**∅**
	s|he	Meryem-ACC	call-PROS-PRES-3SG
	d. Biz	Meryem-i	ara-yacağ-∅-**ız**
	we	Meryem-ACC	call-PROS-PRES-1PL
	e. Siz	Meryem-i	ara-yacak-∅-**sınız**
	you.all	Meryem-ACC	call-PROS-PRES-2PL
	f. Onlar	Meryem-i	ara-yacak-∅-**lar**
	they	Meryem-ACC	call-PROS-PRES-3PL

We can make sense of this approach when we consider the division of labor between Tense and Aspect. It is sometimes assumed that Tense directly anchors an event or a state with respect to the speech time. There is, however, reason to believe that the relation between the speech time and the event time is less direct than that. Instead, Tense may be taken to introduce a new temporal dimension known as the reference time. The present tense, for instance, introduces a time of reference, which is anchored in the present and it is Aspect, not Tense, that relates this reference time to the event time. In other words, the function of the prospective aspect is to relate the event time (the event of us calling Meryem) to this reference time (the present) introduced by Tense. The presence of the prospective aspect implies that the event time follows the reference time, which entails that the event of us calling Meryem takes place later than the reference time determined by Tense, which is the speech time due to the present tense. To sum up,

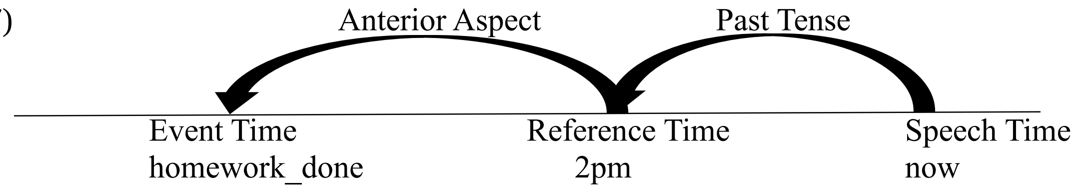
> Tense introduces the reference time, which is either before the speech time (the past tense) or at the speech time (the present tense)
> Aspect relates this reference time to the event time, which is either before the reference time (the anterior aspect, as we shall see below) or after the reference time (the prospective aspect). Aspect also indicates whether an event in a completed (the perfective aspect) or an on-going one (the imperfective aspect).

Given that there exists a prospective aspect marker, which indicates the event time follows the reference time introduced by Tense, we may also expect to find an anterior aspect marker, which requires that the event time precede the reference time. Indeed, the aspectual marker *–mIş* in Turkish does seem to have this interpretation.

(36)	Öğlen 2-de	Zeynep ödev-i	bitir-miş	idi-∅
	noon 2-LOC	Zeynep homework-ACC	complete-ANT	PST-3SG
	'Zeynep had (already) already finished her homework by 2 pm.'			

The past tense marker indicates that the reference time is in the past, which is specified as being 2 pm by the adverbial *öğlen 2'de* 'at 2 pm'. The presence of the anterior aspect marker *-mIş* entails that Zeynep had finished the homework before this reference time, i.e. the event time precedes the reference time. Here is a schematic representation of the relation between Tense and Aspect in this sentence.

(37)



Something interesting happens when the anterior aspect is used in the present tense. In such constructions, the reference time is the speech time (due to the present tense) and the event time precedes the reference time, which means that the event time precedes the speech time. As expected, such constructions give rise to an interpretation of an event taking place in the past. This is not the only interpretation we get, however. We are also informed that the speaker has a kind of access to the event that is not direct. This indirect access may be via hearsay as in (38) or it may be via an inference given the current state of affairs as in (39) . The nature of this additional meaning associated with the anterior aspect is a topic that is currently under exploration.

(38) Context: Zeynep's mother informs you that Zeynep completed her homework. You say:

Zeynep	ödev-i	bitir-miş-Ø-Ø
Zeynep	homework-ACC	complete-ANT-PRES-3SG

'Reportedly, Zeynep finished homework.'

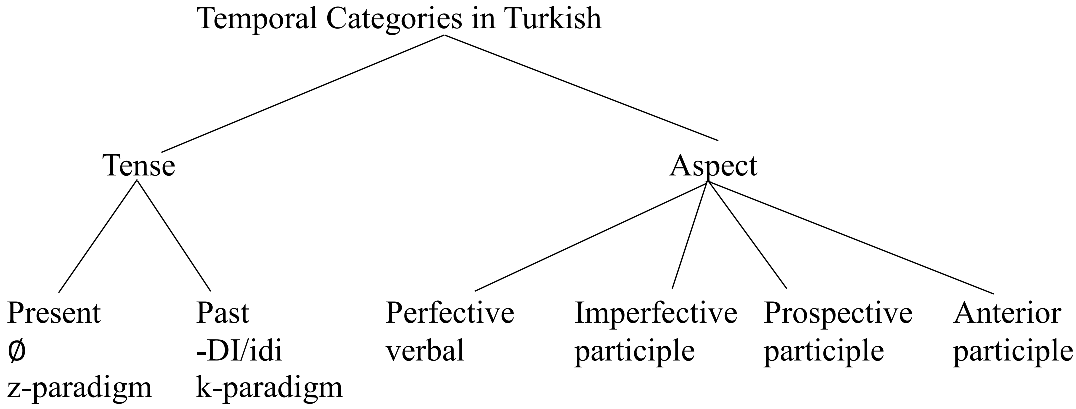
(39) Context: You wake up in the morning and see that the streets are wet now.

Dün	gece	yağmur	yağ-mış-Ø-Ø
Yesterday	night	rain	rain-ANT-PRES-3SG

'(It seems that) it rained yesterday night.'

Here is our summary of Turkish temporal system.

(40)



A word must be said about the morphological realization of the past tense marker *idi*. At times, this morpheme behaves as a bound morpheme that can be attached to participles and other categories. The bound-like behavior of an otherwise free morpheme is sometimes called **cliticization.**

(41)	UNDERLYING FORM			CLITICIZED SURFACE FORM
	a. Ali gel-iyor	idi-Ø	⇒	Ali gel-iyor-du-Ø
	Ali come-IMPV	PST-3SG		Ali come-IMPV-PST-3SG
	'Ali was coming.'			
	b. Ali gel-miş	idi-Ø	⇒	Ali gel-miş-ti-Ø
	Ali come-ANT	PST-3SG		Ali come-ANT-PST-3SG
	'Ali had already come.'			
	c. Ali gel-ecek	idi-Ø	⇒	Ali gelecek-ti-Ø
	Ali come-PROS	PST-3SG		Ali come-PROS-PST-3SG
	'Ali would come.'			

In the alternative surface form, the past tense marker *idi* starts to look a lot like the past tense marker *-DI*, but it is important to notice that they are distinct categories and to distinguish between them when needed. To do so, we shall introduce a third category in Turkish, which is **modality**.

Modality is about possibility and necessity and various specifications of these two categories. The inflectional suffix *–(y)AbIl* is a verbal modal category that can be used to express ability, permission and possibility.

(42) a. Ali yürü-yebil-Ø-di-Ø
Ali walk-ABIL-PRV-PST-3SG
'Ali managed to walk.'
b. Ali yürü-yebil-iyor idi-Ø
Ali walk-ABIL-IMPF PST-3SG
'Ali was able to walk.'

Another modality marker in Turkish is the suffix *-mAlI*, which is about obligations, rules and other necessities. Unlike the possibility suffix *–(y)Abil*, the necessity marker can only be used with the imperfective aspect, which, we shall assume, is null in this context. As a result, it can only be used with the past tense marker *idi*.

(43) a. *Ali yürü-meli-di-Ø
Ali walk-NEC.IMPF-PST-3SG
Int. 'Ali had to to walk.'
b. Ali yürü-meli idi-Ø
Ali walk-NEC.IMPF PST-3SG
'Ali was required to walk.'

The cliticization of the past tense marker *idi* in (43b) is given in (44).

(44) Ali yürü-meli-ydi-Ø
Ali walk-NEC.IMPF-PST-3SG
'Ali was required to walk.'

The form of the past tense marker *idi* in (44) involves the consonant "y", which is possibly the consonantal version of the first vowel "i" in *idi*. This means that the clitic form of *idi* is actually *(y)DI*, where the first consonant is pronounced only when the verbal complex to which it attaches ends in a vowel. This is unlike the verbal past tense marker *-DI*, which has the same form irrespective of whether the verbal complex ends in a vowel or consonant.

(45) a. Ali gül-Ø-dü-Ø
Ali smile-PRV-PST-3SG
'Ali smiled.'
b. Ali büyü-Ø-dü-Ø
Ali grow-PRV-PST-3SG
'Ali grew up.'
c. *Ali büyü-ydü

To sum up, there are cases where the cliticized version of the participle past tense marker *idi* looks like the verbal past tense marker *-DI*, but there is reason to believe they are still underlyingly different. It is important not to be taken by surface appearances.

An interesting issue at this point is how to integrate the category of modality into our grammar. This is not a task we will undertake in this book, however.

Exercises:

1. Draw syntactic trees for (24a) and (24b) as well as for the following expressions. If an expression is structurally ambiguous, draw all of the possible trees.

a. Biz-Ø bi' öğretmen-Ø-i gör-ecek idi-k
we-NOM a teacher-SG-ACC see-PROS PST-1PL
b. İyi üniversite öğrenci-si-Ø-Ø ders kitab-ı kapağ-ın-Ø-ı
good university student-CM-SG-NOM course book-CM cover-CM-SG-ACC
seç-iyor-Ø-Ø
choose-IMPV-PRES-3SG

2. Postpositional phrases are of the category P and postpositions select for an element of the category D and provide it with the case value (similar to verbs). Provide a category for the postpositions *için* ‘for’, *doğru* ‘towards’, *kadar* ‘up to’, *beri* ‘since’, *göre* ‘according to’ and *ötürü* ‘because of’ and draw syntactic trees for the following expressions.

a. sen-in için
you-GEN for
‘for you’

b. biz-e doğru
we-DAT towards
‘towards us’

c. o okul-a kadar
that school-DAT up.to
‘up to that school’

d. geçen yaz-dan beri
last summer-ABL since
‘since last summer’

e. ben-im kırmızı kitab-ım-a göre
I-GEN.1SG red book-1SG-DAT according.to
‘according to my red book’

f. üniversite öğrenci-sin-in arkadaş-ın-dan ötürü
university student-CM-GEN friend-3SG-ABL because.of
‘because of the friend of the university student’

References and Selected Readings: See Demirok & Sağ (2023) on the formal semantics of aspect and tense in Turkish. The distinction between the event time and the reference time was formulated by Reichenbach (1947), see also Klein (1994). The examples of Quechua object agreement are from Myler (2017). See also Yavaş (1980), Kornfilt (1996), Kelepir (2001), Sezer (2001) and Enç (2004) on the complexities of Turkish verbal inflection.

CHAPTER 6: Internal Merge of Objects

We now have some understanding of how to represent sentences syntactically. It is time to look at the syntactic representation of various forms of arguments inside sentences. What is the difference between a case-marked object and a non-case-marked object? In trying to answer this question, we shall make use of an operation called Internal Merge, where we apply the Merge operation to the syntactic structure X and to a structure contained in X. This adds a layer of complexity to our analyses that we have not seen before.

6.1. The representation of objects

We have noted earlier that it is sometimes possible to drop the accusative case marker in Turkish. In (1b), the presence of the accusative case marker indicates that we are talking about a single, specific book. Its absence in (1a) implies that we are talking about an arbitrary book or books.

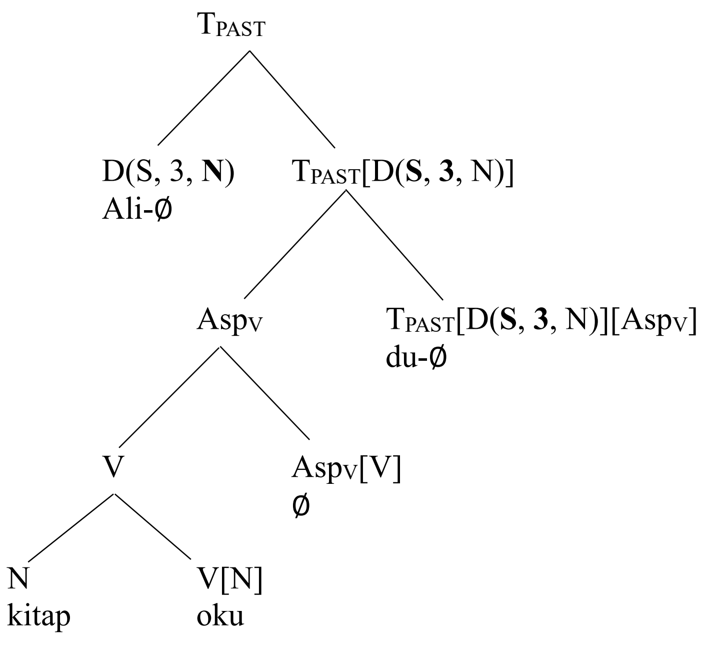
(1) a. Ali kitap oku-du-Ø
Ali book read-PST-3SG
'Ali read (a) book(s)'
b. Ali kitab-ı oku-du-Ø
Ali book-ACC read-PST-3SG
'Ali read the book.'

The lack of the case marker in (1a) suggests that the object is not in the category D, which is compatible with the object lacking specificity reading. There is no number specification for the noun *kitap* 'book', either. Ali may have gone through a single book or, in fact, many books. We take this to mean that there is no number projection either. In other words, the verb *oku* 'read' in (1a) merges with an element in the N category.

(2) The category of the verb *oku* 'read' is "V[N]" or "V[D(□, □, A)]" (whichever works)

The tree representation of (1a) is given in (3) :

(3)



Given the entry in (2) for verbs, we can also provide an explanation for the acceptability of (4a) and (4b) as well as the unacceptability of (4c).

(4) a. Ali ders kitab-ı oku-du-Ø
Ali course book-CM read-PST-3SG
'Ali read (a) course book(s)'
b. Ali bilimsel kitap oku-du-Ø
Ali scientific book read-PST-3SG
'Ali read (a) scientific book(s).'
c. *Ali bilimsel oku-du-Ø
Ali scientific read-PST-3SG

We cannot, however, provide an explanation for the fact that number-related expressions in the object position are also acceptable in the absence of any case marking.

(5) a. Ali üç kitap oku-du-Ø
Ali three book read-PST-3SG
'Ali read three books.'
b. Ali birçok kitap oku-du-Ø
Ali many book read-PST-3SG
'Ali read many books.'
c. Ali birkaç kitap oku-du-Ø
Ali several book read-PST-3SG
'Ali read several books.'

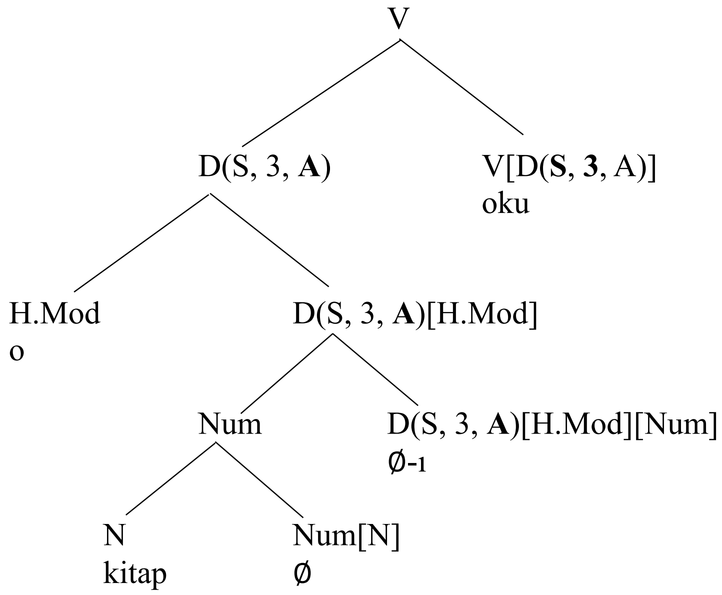
To do so, we need to let the verb *oku* 'read' select for an object in the Num category, too. That is, the lexical entry for *oku* 'read' should be re-formulated as in (6).

(6) The category of the verb *oku* 'read' is "V[N]", "V[Num]" or "V[D(□, □, A)]" (whichever works)

We now understand better why it is that, unlike low modifiers, the presence of a high modifier in the object position makes case marking obligatory. The tree that contains a high modifier is labeled with the category D, thanks to which it is input to the Agreement operation with the lexical verb and obtains its case feature after this operation. The bottom part of the tree representation for (7a) is shown in (8). A good exercise at this point would be to complete this tree. (In *Chapter 4* Exercise 3, you were asked to provide the rules of realization for case markers on the silent determiner that selects for a high modifier. The rule for the silent determiner in Turkish is either "D(NUM, 3, □)[Num]" or "D(NUM, 3, □)[H.Mod][Num]" (whichever works). See the answer key for Chapter 4)

(7) a. Ali o kitab-ı okuyor idi-Ø
Ali that book-ACC read-IMPV PST-3SG
'Ali was reading that book.'
b. *Ali o kitap oku-yor idi-Ø
Ali that book read-IMPV PST-3SG

(8)



We can also now integrate adverbs into our syntactic representations. There are some similarities between adjectives modifying nouns and adverbs modifying verbs. Just like we can add as many adjectives as we desire to a noun, we can also insert as many adverbs as we desire to a sentence.

(9) a. Ali kitap oku-yor idi-∅
Ali book read-IMPV PST-3SG
'Ali was reading books.'

b. Ali yavaş-ça kitap oku-yor idi-∅
Ali slow-ly book read-IMPV PST-3SG
'Ali was reading books slowly.'

c. Ali saat-ler-dir yavaş-ça kitap oku-yor idi-∅
Ali hour-PL-for slow-ly book read-IMPV PST-3SG
'Ali was reading books slowly for hours.'

d. Ali yine saat-ler-dir yavaş-ça kitap oku-yor idi-∅
Ali again hour-PL-for slow-ly book read-IMPV PST-3SG
'Ali was again reading books slowly for hours.'

e. Ali muhtemel-en yine saat-ler-dir yavaş-ça kitap oku-yor idi-∅
Ali probable-ly again hour-PL-for slow-ly book read-IMPV PST-3SG
'Probably Ali was again reading books slowly for hours.'

We have seen that nouns do not select for adjectives, which is why nouns can be used in the absence of any adjectives. Similarly, there is nothing in a verb that requires that it be used with an adverb, i.e. verbs do not select for adverbs. We suggest that, similar to adjectives, adverbs are introduced to the structure via linkers. Independent evidence for this claim comes from Mandarin Chinese, where we find the linker *DE* appearing between adverbs and verbs.

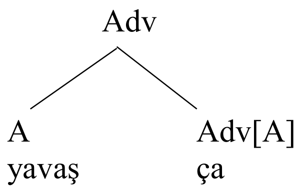
(10) Lisi gaoxing-de bangzhu le wo
Lisi happy-LINK help PERF 1SG
'Lisi helped me happily'

We propose the following category for the adverbial linker in Turkish.

(11) The category of the adverbial linker is "V[Adv][V]"

In Turkish adverbs are typically built out of adjectives with the suffix *-cA*.

(12)

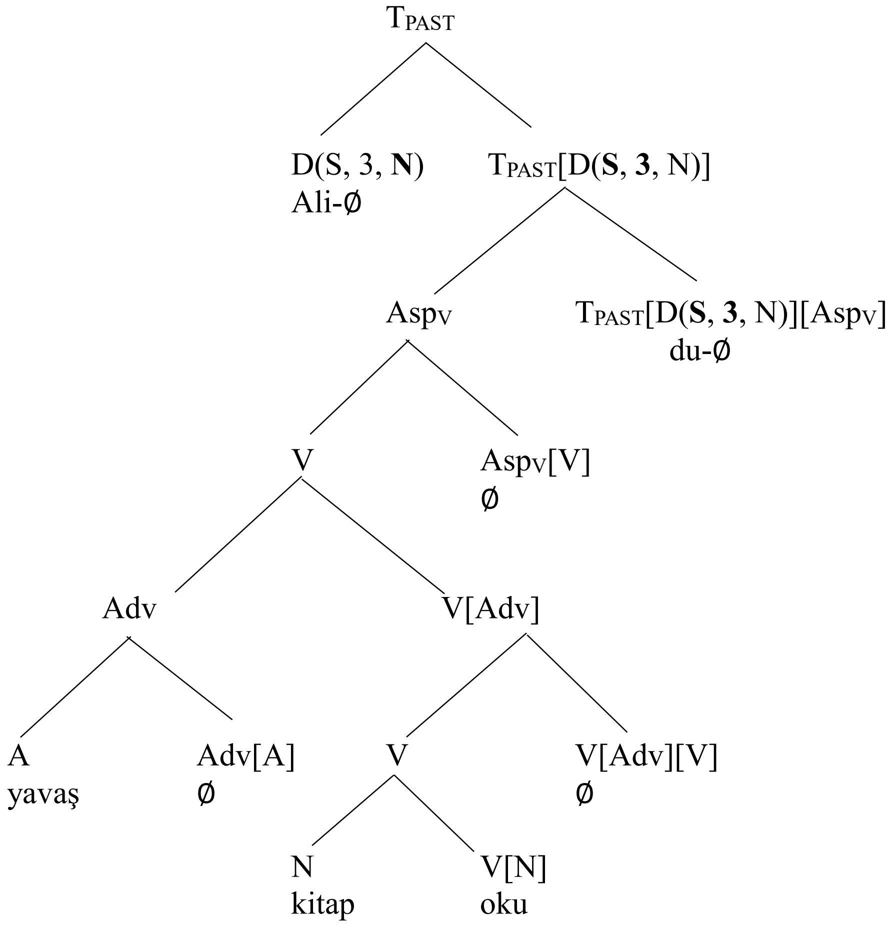


This suffix may, at times, go missing, as in (13). We will keep assuming that such expressions are adverbials. In other words, the expression yavaş ‘slow’ in (13) is to be analyzed as an adverb.

(13) Ali yavaş kitap oku-du-∅
Ali slow book read-PST-∅
‘Ali read (a) book(s) slowly.’

The syntactic representation of the sentence in (13) is given in (14)

(14)

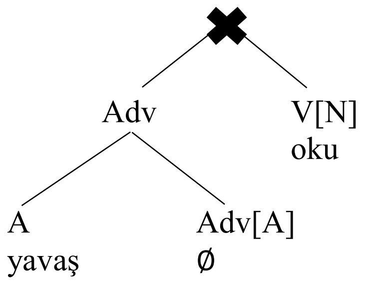


Changing the order of the object and the adverb in (13) leads to unacceptability.

(15) *Ali kitap yavaş oku-du-∅
Ali book slow read-PST-3SG

This is as it should be, given that merging a lexical verb with an adverb is syntactically ill-formed.

(16)

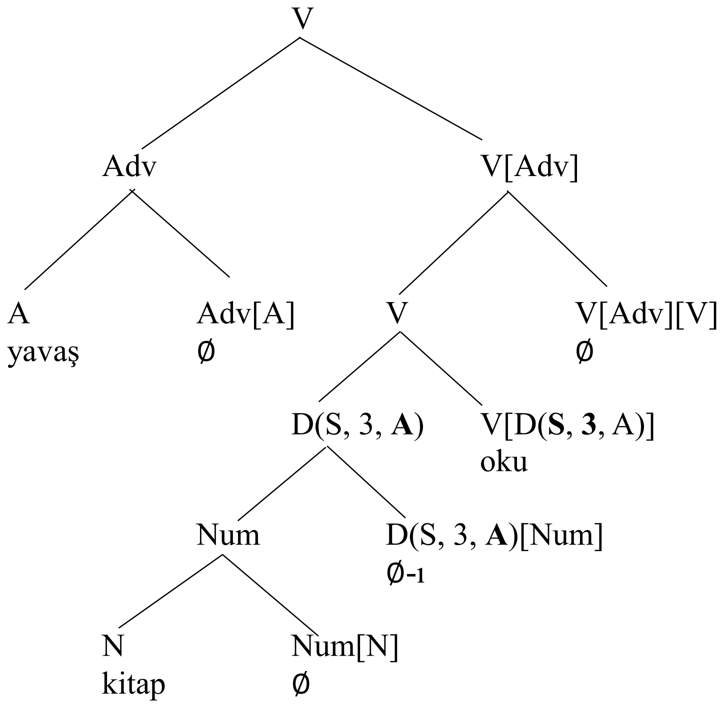


When we turn to case-marked objects, however, we seem to have a problem. This is the topic of the next section.

6.2. An anomaly about case-marked objects

Case-marked objects are introduced to the structure in the same position as non-case marked objects, which is where they enter into the Agreement relation with the verb and get their case feature. This should mean that adverbs must be merged with the V node that is obtained after the merger of the verb with the case-marked object.

(17)



We then expect adverbs to precede case-marked object. The problem is that they do not.

(18) *Ali yavaş kitab-ı oku-du-∅
Ali slow book-ACC read-PST-3SG
Int. 'Ali read the book slowly.'

What we find, instead, is that the case-marked object *kitab-ı* 'the book' must precede the adverb *yavaş* 'slowly'.

(19) Ali kitab-ı yavaş oku-du-∅
Ali book-ACC slow read-PST-3SG
'Ali read the book slowly.'

What are we to do now? The position of the object in (19) cannot be the position at which the object is introduced to the structure, which is the position at which the object is assigned case. Where in the tree is the object, then?

Let us also take note of the fact that case-marking interacts with Aspect. In the absence of the accusative case, (20a) is not understood to be a completed event but an ongoing one. The presence of the accusative case marker suggests that we are talking about a completed event.

(20) a. Ali kitap oku-du-∅
Ali book read-PST-3SG
'Ali read (a) book(s).'
b. Ali kitab-ı oku-du-∅
Ali book-ACC read-PST-3SG
'Ali read the book (to completion).'

There are two properties of case-marked objects in Turkish that we must be able to reconcile. As far as case is concerned, the object and the verb are in a close relationship. This must be reflected in our syntactic representation. However, we must also make sense of the fact that case-marked objects precede adverbs and interact with Aspect.

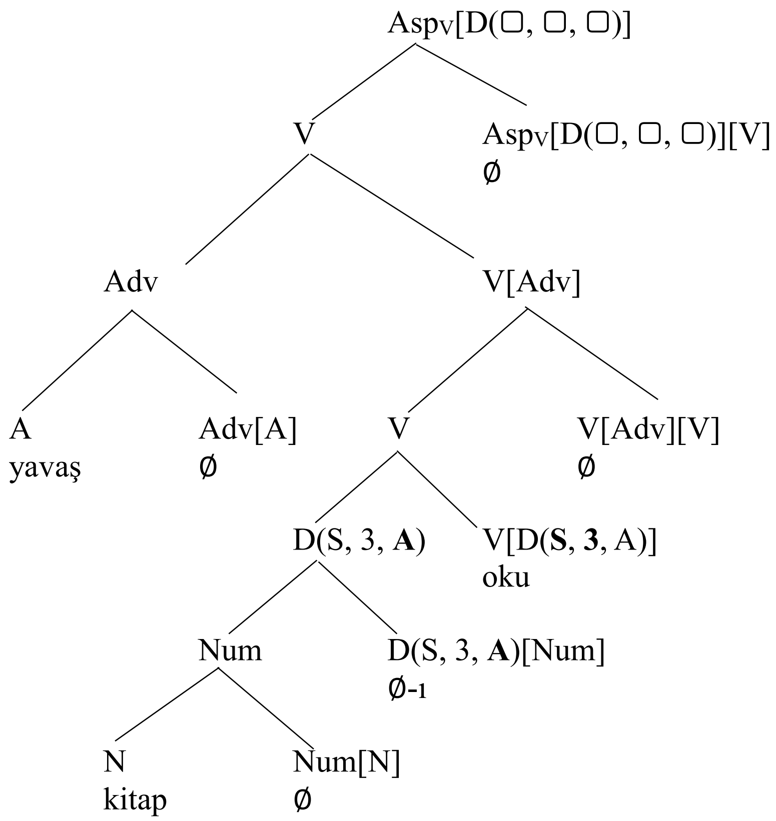
Here is our solution: The case-marked object in (20b) occupies two distinct positions at the same time. It is a node with two distinct mothers. It is the sister node of the category V, which is where it gets to be assigned its case. It is also sister node to the category Aspect, which is how it gets to interact with the aspectual properties of the sentence. This is also the position in which the object gets to be overtly expressed, which is why it precedes the adverb in (19).

Here is what such an analysis looks like. We first let the Perfective morpheme and the participles have the category given in (21).

(21) The category of the perfective morpheme in Turkish is
"Asp$_V$[D(□, □, □)][V]" if there is a case-valued object in the tree
"Asp$_V$[V]" otherwise
The category of a participle in Turkish is
"Asp[D(□, □, □)][V]" if there is a case-valued object in the tree
"Asp[V]" otherwise

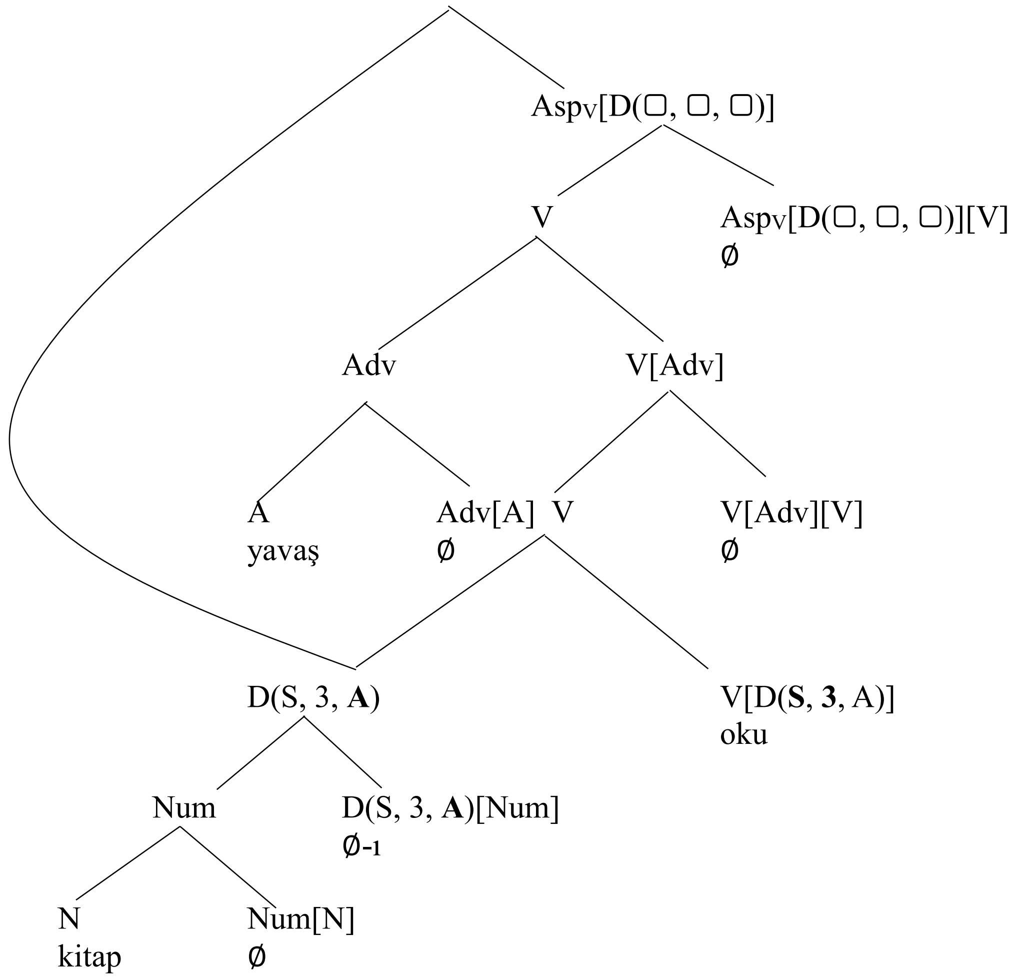
Adding the aspect marker onto the tree in (17), we obtain the following configuration:

(22)



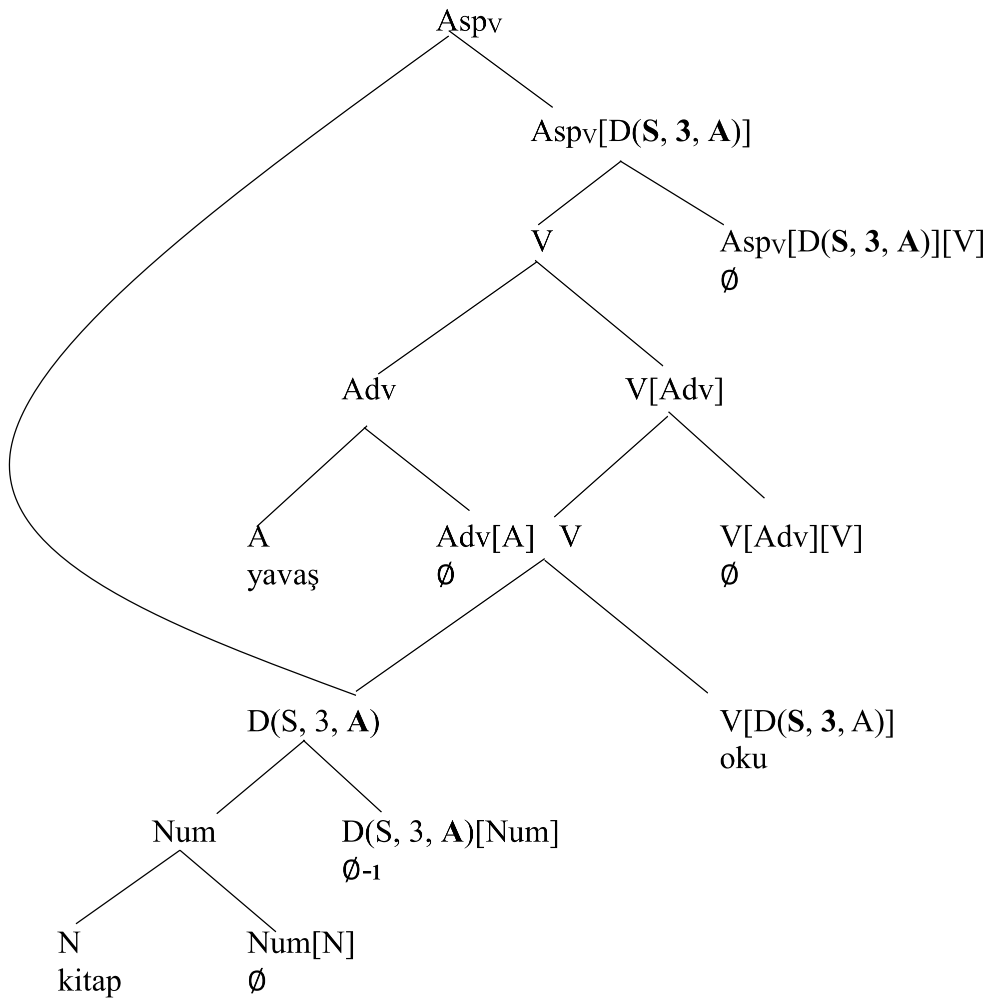
At this moment, we need an element of the determiner category that has a case feature. Such an element does exist inside the tree, i.e. *D(S, 3, A)*. So we will let the category $Asp_V[D(\square, \square, \square)]$ search into the tree and find this D-element with the accusative case feature. This is the first time in our syntactic rules where we seem to need access to more information than what is in the daughter nodes. This operation is called Internal Merge, where we merge a tree with a subtree inside it. Applying this operation to (22), we get:

(23)


Asp$_V$[D(▢, ▢, ▢)]
V
Asp$_V$[D(▢, ▢, ▢)][V]
∅
Adv
V[Adv]
A
yavaş
Adv[A]
∅
V
V[Adv][V]
∅
D(S, 3, **A**)
V[D(**S**, **3**, A)]
oku
Num
D(S, 3, **A**)[Num]
∅-$_1$
N
kitap
Num[N]
∅

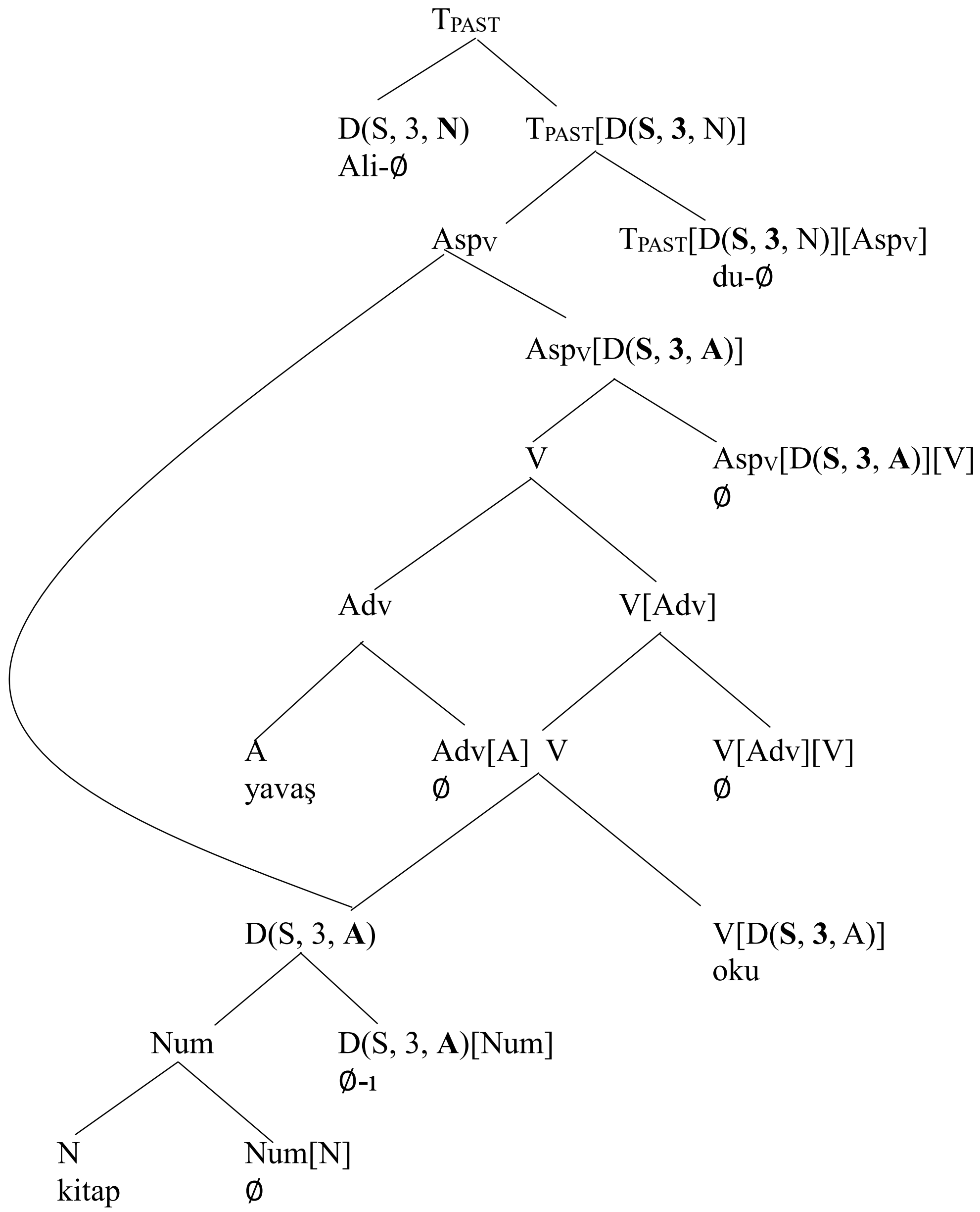
In this structure we must first induce the Agreement operation between the categories *D(S, 3, **A**)* and *Asp_V[D(□, □, □)]* so that we can label the whole structure. (Recall that there are no rules of realization for object agreement in Turkish)

(24)



The derivation of the sentence in (19) continues as before:

(25)

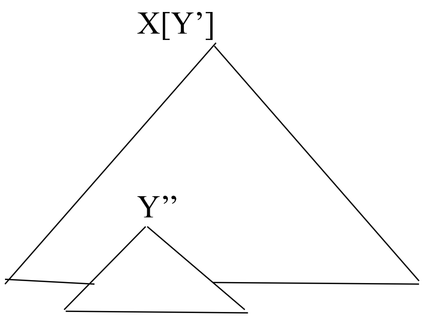


Let us express Internal Merge more formally.

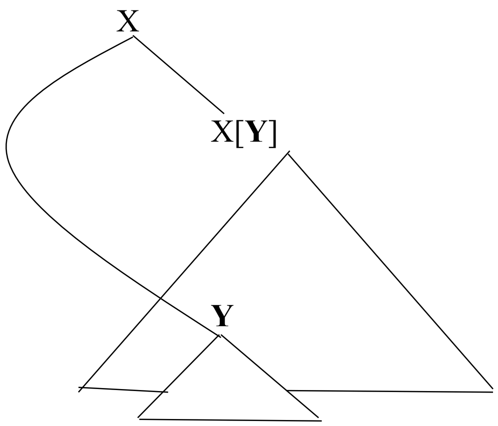
(26) The Internal Merge Operation

where Y’’ is a category whose graph is **contained** in the graph labeled with X[Y’] and the Agreement Operation is defined between X[Y’] and Y’’

INPUT



OUTPUT



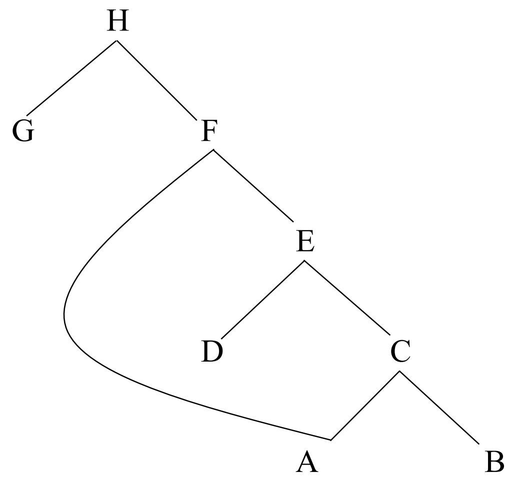
where **Y** is the new category obtained after feature sharing between Y’ and Y’’.

Notice that we use the term graph rather than tree. The reason is that the structure obtained after Internal Merge is no longer a tree. Trees do not contain nodes with two distinct mothers. From this point on, most of what we draw as representations of Turkish will be syntactic graphs rather than syntactic trees.

The operation of Internal Merge presupposes scanning into a graph to find a sub-graph, a graph contained in another graph. Since we are working with graphs that may have nodes with two mothers, we need to define the concept of **containment** accordingly:

(27) Containment
(1) Every graph contains itself.
Whenever two graphs X and Y are distinct,
(2a) If a graph labeled with Y has one mother,
Then a graph labeled with X contains Y if
the node X is the mother of the node Y or
the node X is the mother of the node Z and Z contains Y.
(2b) If a graph labeled with Y has more than one mother,
Then a graph labeled with X contains Y if
X contains every graph which is labeled with the mothers of Y.

To see this definition at work, consider the small graph given in (28):

(28)


Does the graph labeled with F (the graph F) contain the graph labeled with C (the graph C)? C has only one mother (E), which means we will only take the condition (2a) of the definition into consideration. We ask: Is the node F the mother node of the node C? No. The node F is the mother node of the E. So we start the procedure one more time, this time for E. We ask: Is the node E the mother node of the node C? Yes. This means that there is a node such that F is the mother node of this node and the graph labeled with this node contains the graph C, from which we conclude the graph F contains the graph C.

Does the graph H contain the graph A? Well, A has two distinct mothers: C and F. We must consult the condition (2b) in the definition in (27). We now ask: Does the graph H contain the graph C? Does the graph H contain the graph F? Yes and Yes. This means that the graph H contains the graph A.

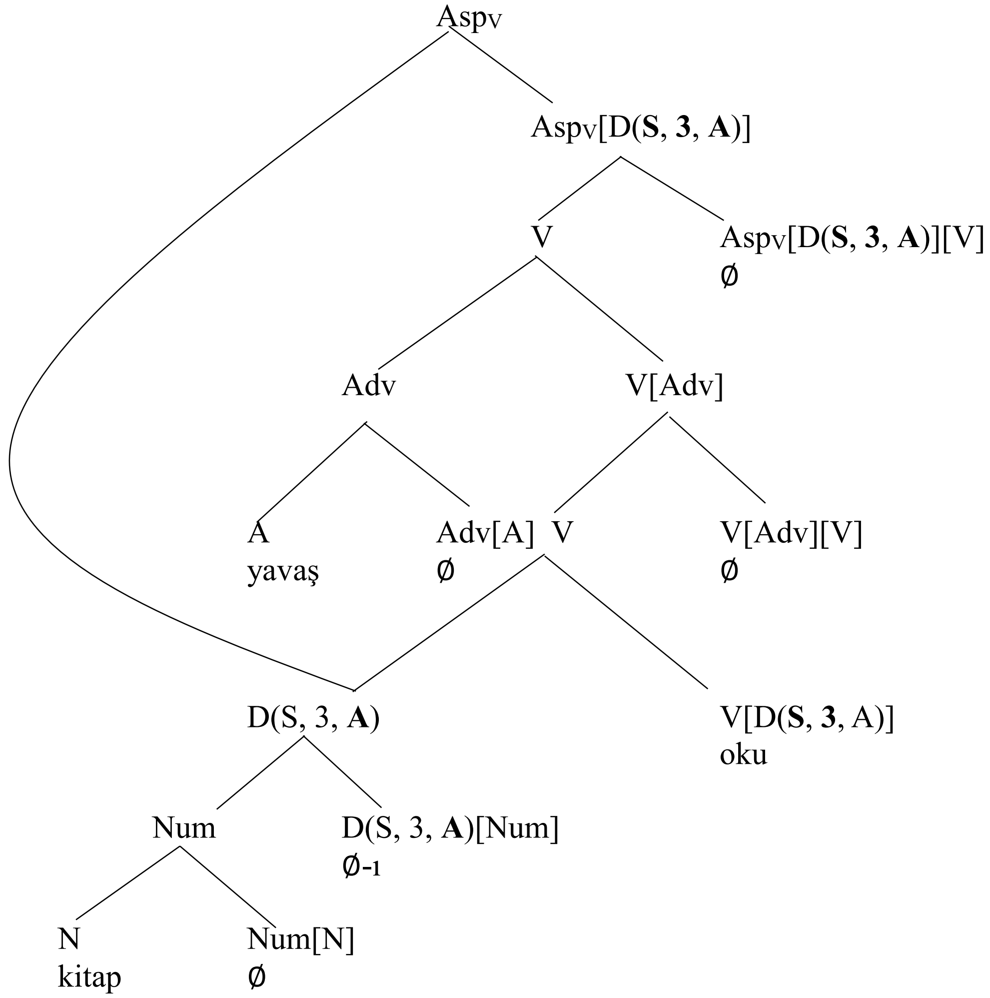
Does the graph F contain the graph A? Again, A has two distinct mothers: C and F. Does the graph F contain the graph C? Yes, as we have just shown two paragraphs ago. Does the graph F contain the graph F? Yes, every graph contains itself. In short, the graph F contains the graph A.

Finally, does the graph E contain the graph A? No. For this containment to hold, it must be that the graph E contain the graph F, which is one of the mother nodes of the node A. This is not the case.

There are two graphs in (28) that contain the graph A: the graph H and the graph F. We can say that the graph F is the smallest graph containing the graph A that is distinct from the graph A itself. That is because, there is no graph X ≠ F contained in the graph F such that X contains the graph A (distinct from the graph A itself). The graph H contains the graph A but it is not the smallest graph doing that. That is because, there is a graph distinct from H and contained in H such that it contains the graph A, which is the graph F.

There is one final aspect of Internal Merge that we need to address: the phonological realization of the morphemes under the node with more than one mother. In (25), the graph *D(S, 3, **A**)* has two mothers. In general, we want a graph with more than one mother to be phonologically realized in the highest of its positions and only there. This is the only way we can get the object in (25) to be phonologically realized preceding the adverb. To do so, we must first find the smallest graph that contains the graph labeled with *D(S, 3, **A**)* distinct from the graph *D(S, 3, **A**)* . Looking at (25), we see the graphs labeled with T_{PAST}, T_{PAST}*[D(**S**, **3**, N]* and Asp_V all contain the graph *D(S, 3, **A**)*. Of these three graphs, the graph Asp_V is the smallest graph containing the graph *D(S, 3, **A**).* We reproduce this graph below.

(29)



To get the terminal nodes (nodes with no branches) in the graph *D(S, 3, A)* to be phonologically realized in the highest position available to *D(S, 3, A*), we rely on the rule (30).

(30) The rule of externalization for Internal Merge

Let X be a node that has more than one mother and let Y be the label of the smallest graph containing the graph X that is distinct from the graph X,
Then terminal nodes in X are phonologically realized in the sister position to Z,
where Z is the daughter of Y distinct from X.

Looking at the graph in (29), we see that *D(S, 3, A)* has more than one mother and *Asp$_V$* is the label of the smallest graph containing the graph *D(S, 3, A)* that is distinct from the graph *D(S, 3, A)*. By (30), the morphemes in the terminal nodes in *D(S, 3, A)* must be phonologically realized in the sister position to *Asp$_V$[D(**S**, 3, A)]*, which is the daughter of *Asp$_V$* that is distinct from *D(S, 3, A)***.** As a result, the case-marked object in (29) precedes the adverb *yavaş* 'slowly'. This is as it should be.

It has been something of a hustle to get a grasp of how Internal Merge works and how it changes the basics of our syntactic analyses. It is important, however, to be as clear about formal underpinnings of an operation as possible. This is a type of rigor that is required to really know how complex the operations we employ are and it is fair, we believe, to say that the addition of the operation of Internal Merge does complicate our syntactic analyses. In fact, it is the first time in the book where we need more information than the categories of the daughter nodes of a mother node. So be it!

6.3. Case markers and postpositions

Lexical verbs differ in how many participants they require for the event or situation they express. The verb *horla* 'snore' requires only one participant, the one who snores. The verb *kır* 'break' requires at least two arguments: the one who breaks and the thing that gets broken. In (31), the object *kapıyı* 'the door' is the thing that gets broken, hence an argument of the verb. There are also phrases in a sentence that are not strictly required by the verb. In (31), for instance, there is an ablative marked noun in the sentence, which is an optional phrase, also called an **adjunct**.

(31) Ali öfke-den kapı-yı kır-dı-∅
Ali anger-ABL door-ACC break-PST-3SG
'Ali broke the door out of anger'

There are ways of distinguishing adjuncts in a sentence from arguments of the lexical verb in the sentence. In Turkish, arguments can at times go missing when they are retrievable from the context. There are, however, some rules governing whether an argument can be deleted. For instance, when a sentence is understood to be an answer to a general question such as 'What happened?' or 'What was the case?', which is called an **out-of-blue context**, deleting the arguments of a verb leads to unacceptability. Deleting the adjuncts of a sentence, however, has no effect on the acceptability of the sentence. Consider the sentences in (32a) and (32b), which are derived from (31), as responses to the question 'What happened yesterday?' While the argument *kapıyı* 'the door' is obligatory in this context, the adjunct *öfkeden* 'out of anger' is not.

(32) a. Ali kapı-yı kır-dı (in an out-of-blue context)
Ali door-ACC break-PAST
'Ali broke the door
b. *Ali öfke-den kır-dı (in an out-of-blue context)
Ali anger-ABL break-PAST
Int. 'Ali broke it out of anger'

This asymmetry in (32) should not lead us to conclude that ablative-marked nouns are always adjuncts. There are also cases in which they are required by the properties of the verb. In (33), for instance, the verb *hoşlan* 'take a liking to' requires two participants: one who takes a liking to someone and one who is the target of this crush. This means that the phrase *geçen yıla kadar* 'up to the last year' and the adverb *çok* 'a lot' are adjunct whiles the ablative marked noun is an argument.

(33) Zeynep geçen yıl-a kadar Ali-den çok hoşlan-ıyor idi-Ø
Zeynep last year-DAT up.to Ali-ABL a.lot like-IMPV PST-3SG
'Zeynep had a big crush on Ali up to last year'

This should mean in an out-of-blue context, deleting the ablative-marked proper name should lead to unacceptability. Nevertheless, we should be able to delete the two adjuncts and still get acceptable sentences. This is precisely what we find.

Judgments in an out-of-blue context ("What was the case?")

(34) a. Zeynep Ali'den çok hoşlanıyor idi-Ø
Zeynep Ali-ABL a.lot like-IMPV PST-3SG
'Zeynep had a big crush on Ali.'
b. Zeynep geçen yıl-a kadar Ali'den hoşlanıyor idi-Ø
Zeynep last year-DAT up.to Ali-ABL like-IMPV PST-3SG
'Zeynep had a crush on Ali up to last year'
c. Zeynep Ali-den hoşlan-ıyor idi-Ø
Zeynep Ali-ABL like-IMPV PST-3SG
'Zeynep had a crush on Ali.'
d. *Zeynep geçen yıl-a kadar çok hoşlan-ıyor idi-Ø
Zeynep last year-DAT up.to a.lot like-IMPV PST-3SG
e. *Zeynep çok hoşlan-ıyor idi-Ø
Zeynep a.lot like-IMPV PST-3SG
f. *Zeynep geçen yıl-a kadar hoşlan-ıyor idi-Ø
Zeynep last year-DAT up.to like-IMPV PST-3SG
g. *Zeynep hoşlan-ıyor idi-Ø
Zeynep like-IMPV PST-3SG

Comparing the ablative-marked nouns in (31) and (33), we find that they are of different nature. The ablative-marked noun in (33) is an argument of the verb while in (31) what we are dealing with is an adjunct. This implies that ablative marking is not always a function of case assignment. What this means is that we need to find a way to represent the ablative-marked noun in (31).

We suggest that the ablative marker in Turkish is ambiguous between a case marker (as in (33)) and a postposition (as in (31)). Similar to verbs, postpositions in Turkish assign case to the category they select for, i.e. to the members of the D category. Upon merger with an element of the determiner category, Postpositions project the category P, which is short for the postposition category.

(35)

a. on-un	**için**	b. on-un	**gibi**	c. on-un	**ile**
s|he/it-GEN	for	s|he/it-GEN	like	s|he/it-GEN	with
'for him/her/it'		'for him/her/it'		'for him/her/it'	
c. on-a	**göre**	d. ora-ya	**kadar**	e. on-a	**doğru**
s|he/it-DAT	according to	there-DAT	up.to	s|he/it-DAT	towards
'according to him/her/it'		'up to that (point)'		'towards him/her/it'	
f. on-dan	**beri**	g. on-dan	**ötürü**	h. on-dan	**itibaren**
s|he/it-ABL	since	s|he/it-ABL	because.of	s|he/it-ABL	beginning
'since then'		'because of him/her/it'		'beginning with that'	

The step-by-step derivation of the postpositional phrase (35a) is shown in (36)

(36)

D(S, 3, □) P[D(□, □, G)]
on için

D(S, 3, **G**) P[D(**S**, **3**, G)]
on-un için

P

D(S, 3, **G**) P[D(**S**, **3**, G)]
on-un için

Turkish do not exhibit number and person agreement on adpositions (This is a term covering both prepositions and postpositions). There are no rules of realization for adposition agreement in Turkihs. There are, however, languages such as Welsh that have such rules.

(37)

a. arno	fo	b. arni	hi	c. arnyn	nhw
on.3.S.M	he	on.3.S.F	she	on.3.P	they
'on him'		'on her'		'on them'	

It is an idiosyncratic fact about Turkish that postpositions do not exhibit agreement with the pronoun they are merged with, but this is not a deep fact about the nature of human grammar.

When an ablative-marked nominal element is used as an adjunct, it assigns nominative case to the D-element that it selects for and project the category P.

(38)

	P	
	D(S, 3, **N**)	P[D(**S**, **3**, N)]
	o(n)	-dan

We now need to understand how postpositional adjuncts are integrated into our syntactic graphs. One property that postpositional expressions share with adverbs is that we can add as many of them into a sentence as we desire.

(39)

a. Zeynep kek pişiriyor idi-Ø
Zeynep cake cook-IMPV PST-3SG
'Zeynep was baking a cake.'

b. Zeynep sen-in için kek pişir-iyor idi-Ø
Zeynep you-GEN for cake cook-IMPV PST-3SG
'Zeynep was baking a cake for you.'

c. Zeynep sabah-tan beri sen-in için kek pişir-iyor idi-Ø
Zeynep morning-ABL since you-GEN for cake cook-IMPV PST-3SG
'Zeynep had been baking a cake for you since the morning.'

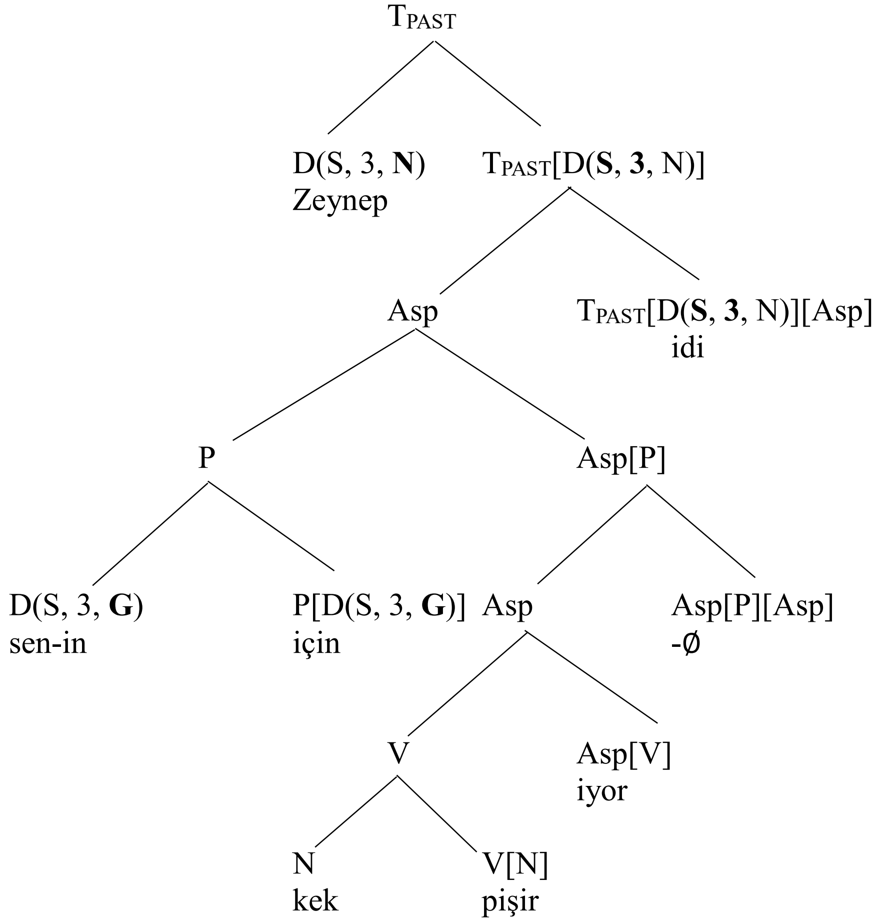
…

It, then, makes sense to introduce postpositional phrases into our syntactic representations the same way as we represented adverbs in our graphs, i.e. via the help of linkers.

(40) The category of a postpositional linker is "Asp_V[P][Asp_V]" or "Asp[P][Asp]" (whichever works)

The syntactic representation of a sentence such as (39b) is given in (41). In the exercises you will be asked to draw graphs with case-marked objects.

(41)



We have a formal definition of the term adjunct. An adjunct is the second argument of a linker. Linkers are functional categories that end up giving as output whatever the category of their first argument happens to be. In other words, linkers instantiate the schema *X[Y][X],* where *X* and *Y* are some categories.

We now know how to distinguish between case-marked and non-case-marked objects. We also have some understanding of how adverbs and postpositional phrases are integrated into the structure. The next section is about the representation of subjects. As it turns out, their final position in the graph is not the position they are introduced to the graph.

Exercises:

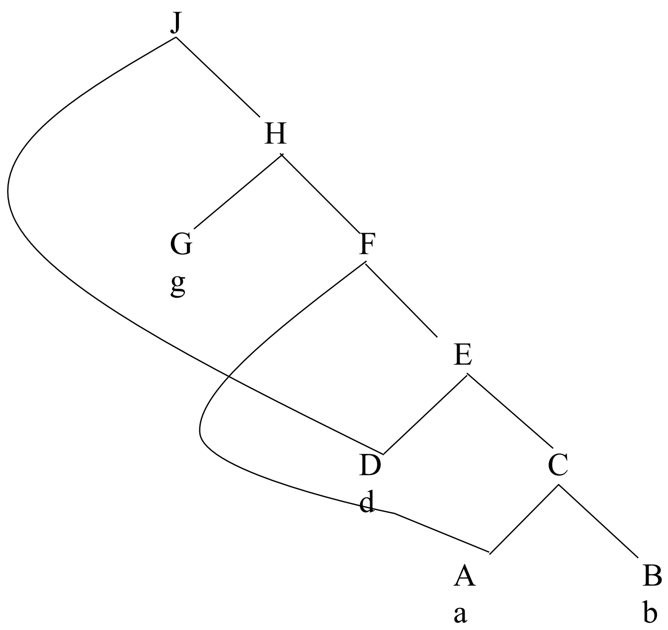
1. Draw syntactic graphs for the following expressions. Analyze the expressions *hızlı* 'fast' and çok 'very/a.lot' as adverbs.

a. Zeynep sen-den ötürü üç kitab-ı hızlı oku-Ø-du-Ø
Zeynep you-ABL because.of üç book-ACC fast read-PRV-PST-3SG
'Because of you, Zeynep has read the three books fast.'

b. Ben-im arkadaş-ım sen-in için sabah-tan beri kek pişir-iyor idi-Ø
I-GEN.1SG friend-1SG you-GEN for morning-ABL since cake cook-IMPV PST-3SG
'My friend had been baking a cake for you since the morning.'

c. Zeynep geçen yıl-a kadar Ali-den çok hoşlan-ıyor idi-Ø
Zeynep last year-DAT up.to Ali-ABL a.lot like-IMPV PST-3SG
'Zeynep had a big crush on Ali up to last year'

2. Consider the graph below. Answer the following questions by making reference to the definition of containment in (27)



2.1. Does the graph J contain the graph A?
2.2. Does the graph H contain the graph E?
2.3. Does the graph H contain the graph D?
2.4. Does the graph J contain the graph D?
2.5. Does the graph F contain the graph A?
2.6. Does the graph E contain the graph A?
2.7. What is the smallest graph that contains the graph A that is distinct from the graph A itself?
2.8. What is the smallest graph that contains the graph D that is distinct from the graph D itself?
2.9. What is the smallest graph that contains the graph E that is distinct from the graph E itself?
2.10. What is the phonological linearization of {a, b, d, g}, given (30)?

3. 1. Define a category for the adverbial linker that takes participles, i.e. *Asp*, as its first input. Using this linker, explain the acceptability of (a) and the unacceptability of (b). The category of the adverbializer (ADV) suffix *–DIr* is “Adv[Num]”.

a. Zeynep saat-ler-dir araba-yı yavaş sürü-yor idi-∅
Zeynep hour-PLU-ADV car-ACC slow drive-IMPV PST-3SG
‘Zeynep has been driving the car slowly for hours.’
b. *Zeynep saat-ler-dir araba-yı yavaş sür-∅-dü-∅
Zeynep hour-PLU-ADV car-ACC slow drive-PRV-PST-3SG

References and Selected Readings: see Diesing (1992), Zidani-Eroğlu (1997), Kelepir (2001) Arslan-Kechriotis (2006) on object movement in Turkish. The idea that internal Merge results in a node that has two distinct mothers is developed in Epstein et al. (1998), Starke (2001), Johnson (2012), among many others. The Welsh example of preposition agreement is from Borsley (2009).

Chapter 7: Internal Merge of subjects

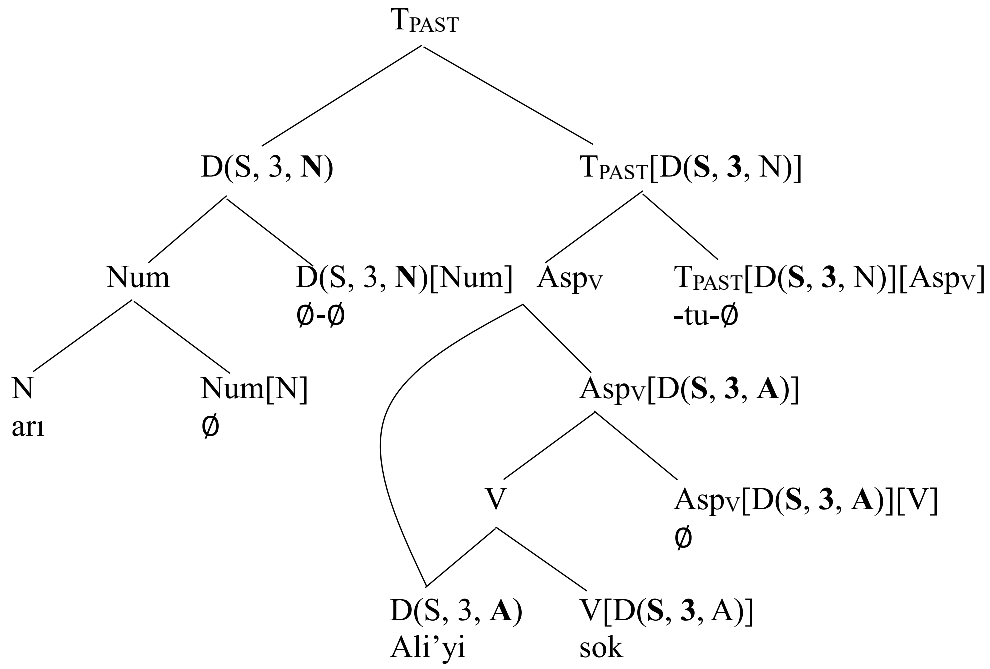
This section is about the representation of subjects inside sentences. We provide evidence that, similar to objects, subjects have an initial low position as an argument of the verb. Their high position in the graph is made by possible by the operation Internal Merge. In other words, Internal Merge has a role to play in the representation of subjects, too.

7.1. Subjects in two positions

Thus far in the book, we have taken it for granted that subjects are in the sister node of the tense node, *T_{PAST}[D(□, □, N)]*, which is how they get to have the nominative case. In this way, we also explain how it is that the number and person features of the subject are reflected on the tense marker. This syntactic representation of a sentence such as (1) is given in (2):

(1) Arı Ali-yi sok-tu-Ø
Bee Ali-ACC sting-PST-3SG
'The bee stung Ali.'

(2)



There is a problem, though. It is possible for the noun *arı* 'bee' to come between the object and the verb with notable interpretive differences compared to (1), in which we are taking about a specific bee which happened to sting Ali. In (3), on the other hand, we are talking about a bee-stinging event that may have involved one bee or many bees.

(3) Ali-yi arı sok-tu
Ali-ACC bee sting-PST
'Ali got stung by (a) bee(s)'

We have seen such an interpretative distinction before in the context of grammatical objects. Recall that while a case-marked object denotes a single specific entity, in the absence of case-marking as in (4a), we get a non-specific reading, in which Ali may have read one book or many books.

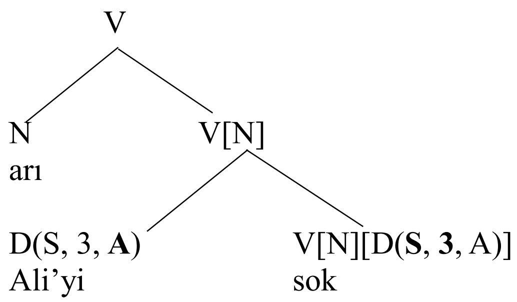
(4) a. Ali kitap oku-du-Ø
 Ali book read-PST-3SG
 'Ali read (a) book(s)'
 b. Ali kitab-ı oku-du-Ø
 Ali book-ACC read-PST-3SG
 'Ali read the book.'

We took this as evidence that non-case-marked object lacks Num and D projections. A similar claim can be made about the subjects that are adjacent to the verb as in (3). The lack of specificity reading and number specification for the subject in (3) suggests that this subject is simply a noun projection.

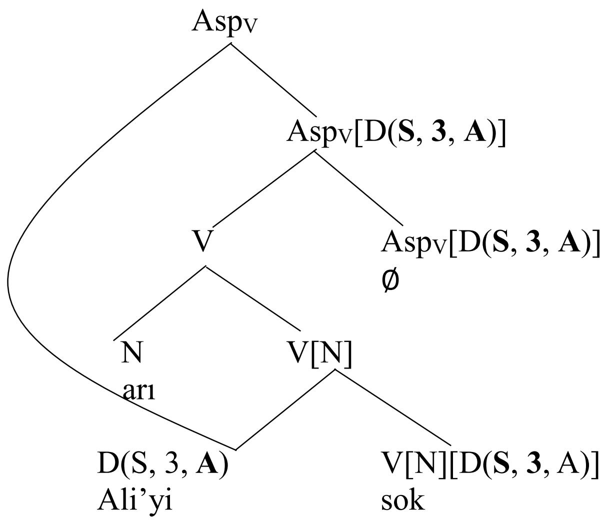
This suggests that similar to non-case-marked objects, subjects occupy a relatively low position in the syntactic representation when they are in the noun category. We take it that subjects start out as an argument of the verbs. The verb *sok-* 'sting' has the following categorical property.

(5) The category of the verb *sok-* is "V[N][N]", "V[N][D(▢, ▢, A)]", "V[D(▢, ▢, ▢)][N]" or "V[D(▢, ▢, ▢)][D(▢, ▢, A)]" (whichever works)

The representation of the sentence in (3) starts out as in (6).

(6)


We, then, add the aspectual morpheme, which triggers internal merge of the object. This is how the object gets to precede the non-specific subject.

(7)


We revise the category of the tense morpheme so that the sentence is acceptable in the absence of any subject in the sister position of the T category.

(8) The category of the past tense marker "-DI" is

"T_{PAST}[D(□, □, N)][Asp_V]"	if there is a subject in the category D in the tree
"T_{PAST}[Asp_V]"	otherwise

The category of the past tense marker "idi" is

"T_{PAST}[D(□, □, N)][Asp]"	if there is a subject in the category D in the tree
"T_{PAST}[Asp]"	otherwise

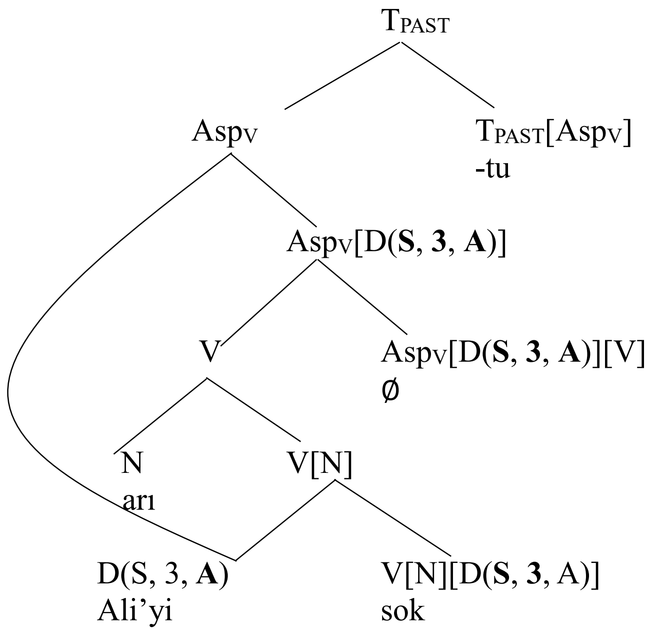
The category of the null present tense morpheme is

"T_{PRES}[D(□, □, N)][ASP]	if there is a subject in the category D in the tree
"T_{PRES}[ASP]"	otherwise

(where ASP is one of {Asp, Asp_V})

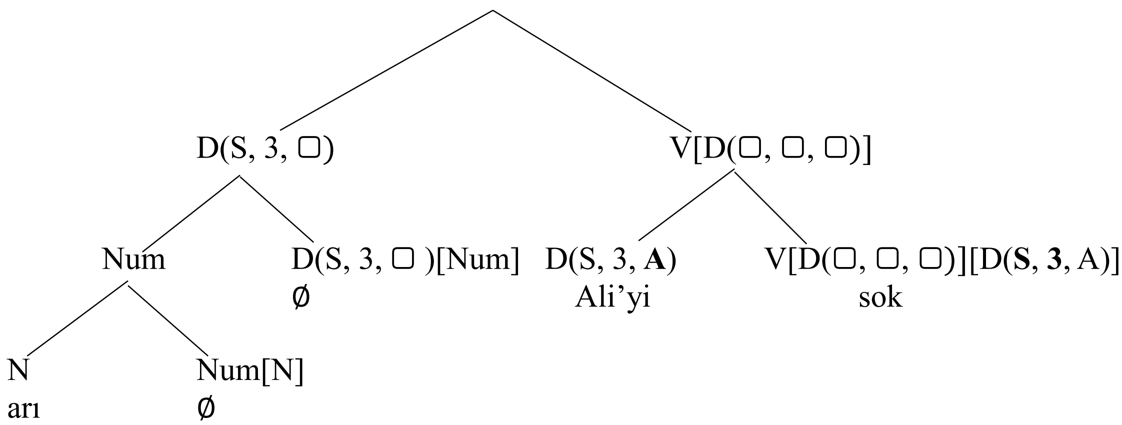
The representation of the complete graph for (3) is given in (9):

(9)



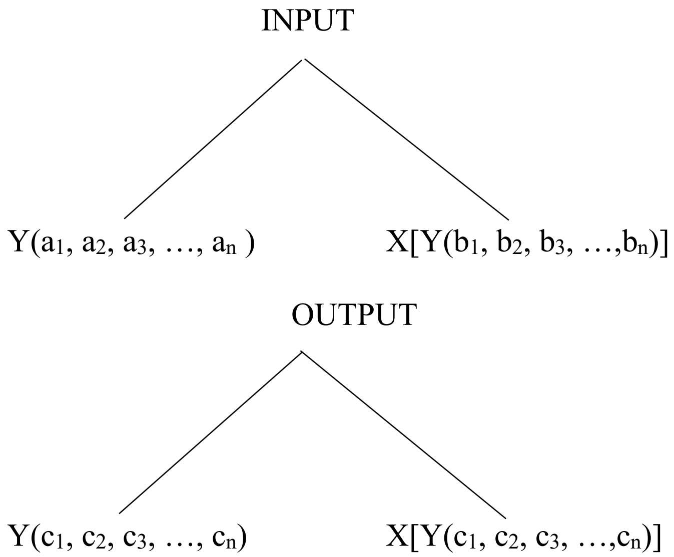
In those cases where the subject is of the category D, with the specific bee reading, the subject undergoes Internal Merge with the Tense category. The subject in the D category is first to be merged as the second argument of the verb.

(10)



We have got a problem here. Earlier we defined the Agreement operation so that there would be complete feature-swapping between sister categories. However, when we look at the merger of the subject and the lexical verb in (10), we see that neither has any case feature to share. What this means is that we need to relax the Agreement operation in such a way that it is acceptable to not share features when the target categories both lack the relevant feature. The revised Agreement Operation, where the newly added part is **bolded**, is defined in (11)

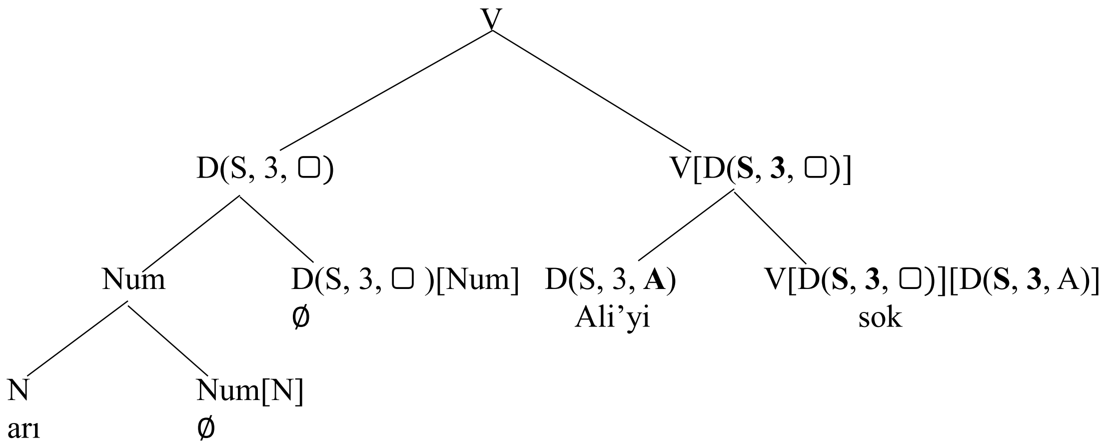
(11) for any j such that $1 \leq j \leq n$
The Agreement Operation
is defined when at least one of a_j and b_j is "□"
when defined,



where **c_j =□ only if a_j = □ and b_j =□**
otherwise $c_j \neq$ □ and [$c_j = a_j$ or $c_j = b_j$]

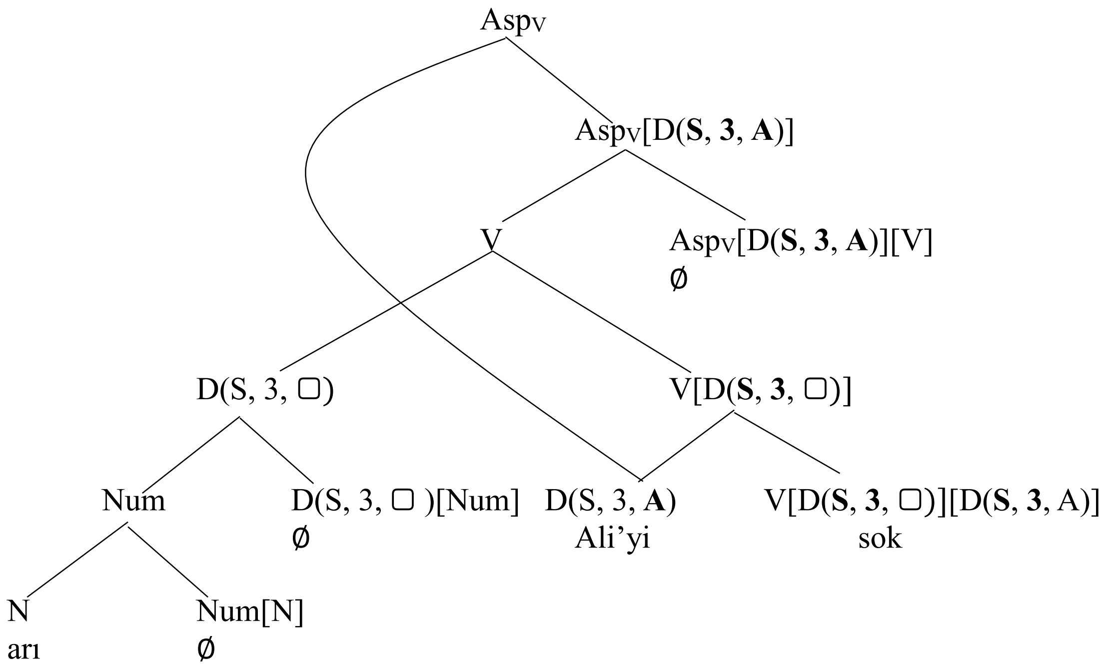
We now let a feature remain unvalued precisely when it is unvalued in both categories that undergo the Agreement operation. With this new definition of Agreement, we can label the graph in (10) as *V*. The crucial observation to made about the graph in (12) is that the case feature of the subject remains unvalued at this point.

(12)



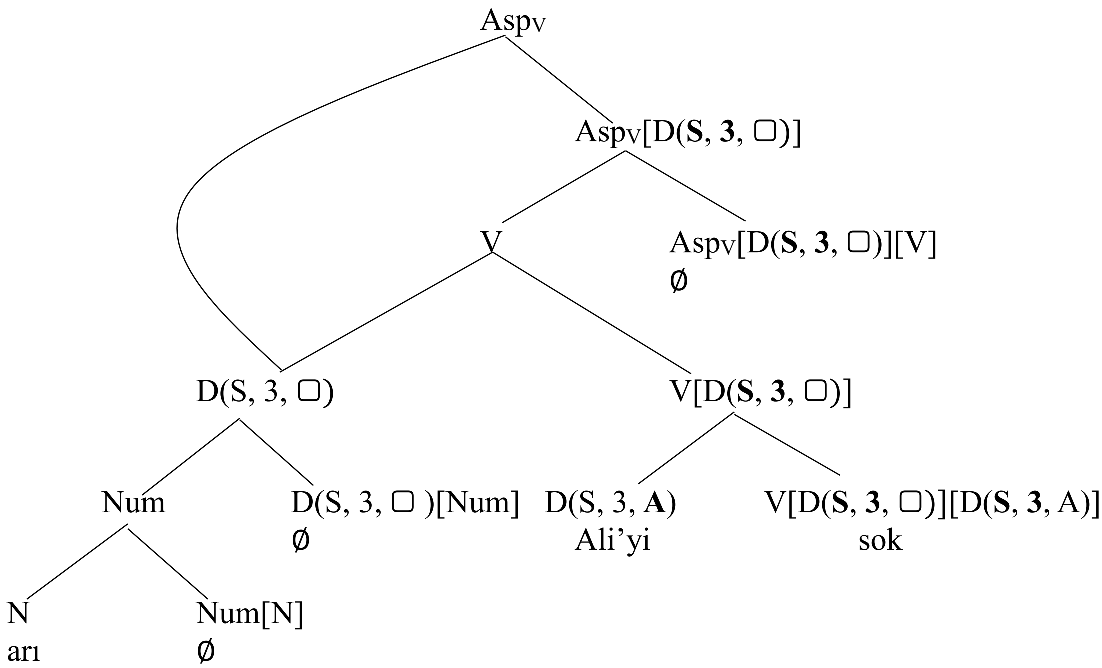
We then insert the Aspect head, which triggers Internal Merge with the object.

(13)



Here is a question. Why is it we have applied the Internal Merge to *Asp(V)[D(▢, ▢, ▢)]* and the object *D(S, 3, **A**)* and not to *Asp(V)[D(▢, ▢, ▢)]* and the subject *D(S, 3, ▢)***.** As a matter of fact, the revised version of the Agreement operation makes it possible to apply the Internal Merge to Aspect and the subject. Such a graph would look as in (14):

(14)



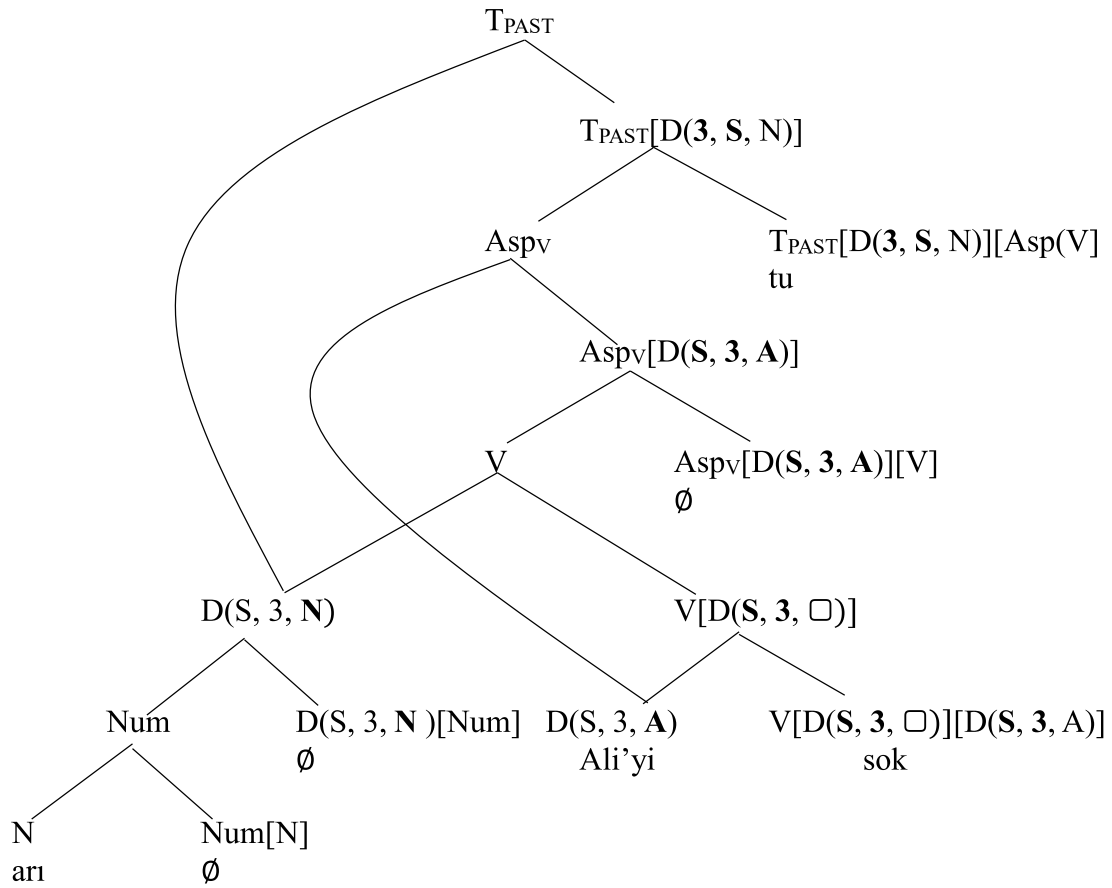
What is wrong with the graph in (14)? What is wrong with applying Internal Merge to *Asp(V)[D(□, □, □)]* and *D(S, 3, □)*? When we apply the Agreement operation to these two categories, what we get is *Asp(V)[D(**S**, **3**, □)]* and *D(S, 3, □)* respectively, where the case feature remains unvalued. We stipulate that we should apply the syntactic operations in a way that reduces the number of unvalued features as much as possible in each application. We call this the Principle of Value Maximization.

(15) The Principle of Value Maximization
Operations are geared towards eliminating as many unvalued features as possible at each step.

The Principle of Value Maximization makes the object a better candidate than the subject for the application of Internal Merge at the level of Aspect. This means that we will continue our derivation with (13) and not with (14).

We now add the Tense category, which triggers the Internal Merge operation with the subject. This gives us the syntactic representation of the sentence with the subject *arı* 'bee' in the determiner category.

(16)



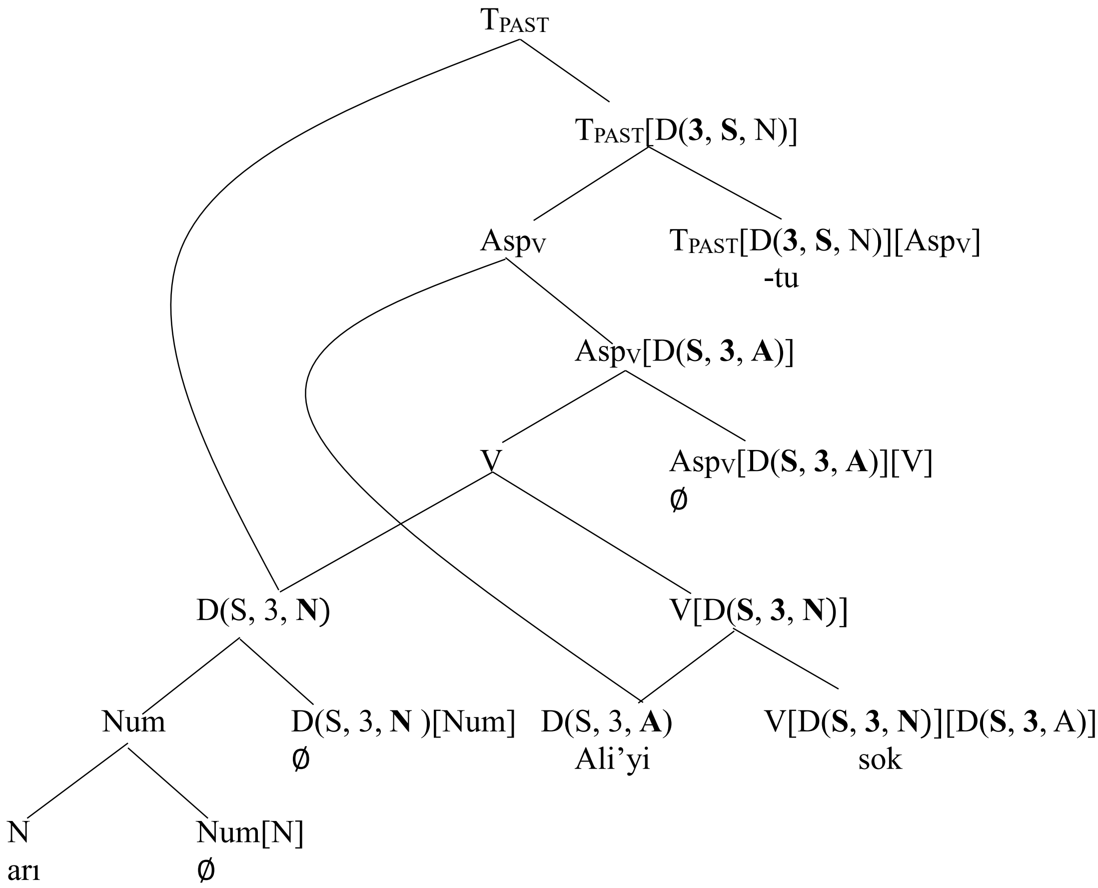
There is one final issue we need to take care of before we can leave the issue of the syntactic representation of the subject. Observe that, the case feature of the subject remains unvalued on the verbal category. In the graph (16), we have *V[D(**S**, **3**, □)]* and *V[D(**S**, **3**, □)][D(**S**, **3**, A)]* as categories that lack the case feature, which we wish to provide with a value in the spirit of the *Principle of Value Maximization*. This can be done by extending the Copy Rule to sisters in addition to daughter nodes. (The newly added text is bolded.)

(17) The Copy Rule

whenever X ≠ X’ **or Y ≠ Y’**

Applying the Copy Rule to the graph in (16) twice, we derive the graph in (18) as the representation of the sentence in (1).

(18)



We are done!

7.2. Corroborative evidence for the low initial position of subjects

Distinguishing objects that are in the D category from those that are not is straightforward: We just look at the case marker on the object; the case markers that can appear on the object of a sentence are all visible. Distinguishing between subjects that are D-elements and those that are not is slightly trickier. The reason is that the nominative case in Turkish is a null morpheme. Even though (19a) and (19b) differ in whether the subject has a case marker, the silence of the nominative case marker makes this distinction somewhat more difficult to diagnose.

(19) a. Arı-∅-∅ Ali-yi sok-tu-∅
bee-SING-NOM Ali-ACC sting-PST-3SG
'The bee stung Ali.'
b. Ali-yi arı sok-tu
Ali-ACC bee sting-PST
'Ali got strong by (a) bee(s).'

Our analysis of the noun *arı* 'bee' (19b) as the V-level subject relies on the fact that it must come adjacent to the verb in order to have the number-neutral, non-specific reading. Once we put the subject in its sentence-initial position (19a), the singular, specific reading becomes obligatory. One might wonder if we could find independent evidence for the claim that the subjects in (19a) and (19b) differ in case-marking that does not rely exclusively on word order and semantic intuitions. The good news is that we can. The relevant evidence comes from nominalized embedded clauses in Turkish, whose structure we shall study in this chapter. Suppose that we want to say that we heard that (19a) is true (but, say, we are not committed to its truth). To do so, we must embed (19a) inside another sentence of the form *I heard that*…. One way of doing that is given in (20).

(20) Ben arı Ali-yi sok-tu-∅ diye duy-du-m
I bee Ali-ACC sting-PST-3SG that hear-PST-1SG
'I heard that the bee stung Ali.'

Here we simply take the sentence in (19a) and embed it inside the string "Ben *X* diye duydum." in the place of X. We can do something similar to (19b).

(21) Ben Ali-yi arı sok-tu-∅ diye duy-du-m
I Ali-ACC bee sting-PST-3SG that hear-PST-1SG
'I heard that Ali got stung by (a) bee(s).'

In either case, the embedded subject *arı* 'bee' has no visible case marker. By our analysis, the embedded subject (20) has a case-marker that is null but the embedded subject in (21) has no case marker at all. This strategy of embedding does not really help us distinguish between null case and no case.

Luckily, there is a second strategy of sentence embedding in Turkish. In this strategy, the embedded clause behaves like an element in the *D* category (where *-DIK* is some kind of embedded determiner to be discussed later). It gets to be case-marked depending on the verb the way an object would be case marked.

(22)	a. Ben	[arı-nın	Ali-yi	sok-tuğ-un]-**u**	duy-du-m
	I	bee-GEN	Ali-ACC	sting-DIK-3SG-**ACC**	hear-PST-1SG
	'I heard that the bee stung Ali.'				
	b. Ben	[arı-nın	Ali-yi	sok-tuğ-un]-**a**	inan-dı-m
	I	bee-GEN	Ali-ACC	sting-DIK-3SG-**DAT**	believe-PST-1SG
	'I believed that the bee stung Ali.'				

Observe that in each case, the embedded subject *arı* 'bee' is in the genitive. In fact, we can think of the genitive as corresponding to the nominative case in simple (non-embedded) clauses. This should mean that when we embed (21) inside a nominalized clause, there should be no case marker on the low subject. This is, indeed, what we find.

(23)	a. Ben	[Ali-yi	arı	sok-tuğ-un]-u	duy-du-m
	I	Ali-ACC	bee	sting-DIK-3SG-ACC	hear-PST-1SG
	'I heard that Ali got stung by (a) bee(s).'				
	b. Ben	[Ali-yi	arı	sok-tuğ-un]-a	inan-dı-m
	I	Ali-ACC	bee	sting-DIK-3SG-DAT	believe-PST-1SG
	'I believed that Ali got stung by (a) bee(s).'				

When we force the low subject that is not case-marked into the sentence initial position, what we get is complete unacceptability.

(24)	a. *Ben	[arı	Ali-yi	sok-tuğ-un]-u	duy-du-m
	I	bee	Ali-ACC	sting-DIK-3SG-ACC	hear-PST-1SG
	'I heard that the bee stung Ali.'				
	b. *Ben	[arı	Ali-yi	sok-tuğ-un]-a	inan-dı-m
	I	bee	Ali-ACC	sting-DIK-3SG-DAT	believe-PST-1SG
	'I believed that the bee stung Ali.'				

There is, after all, some independent evidence for the distinction between null case and no case, which is quite remarkable. To make these observations into a concrete argument, we must provide an analysis of simple and nominalized embedded clauses in Turkish. This is what we are to do in the next section.

7.3. Sentences inside sentences

Now that we have some understanding of how to syntactically represent sentences, there are some options about what to do with them. We can assert them as being true of the world we live in or we can ask others if our sentences describe the world correctly. In Turkish, a sentence is turned into a question by inserting the morpheme *mI* at the end of it. The absence of this morpheme indicates that the sentence is intended as an assertion and sentences of assertion are called declaratives.

(25) a. Ali kitap oku-du-Ø
Ali book read-PST-3SG
'Ali read a book'
b. Ali kitap oku-du-Ø mu
Ali book read-PST-3SG Q
'Did Ali read a book?'

The distinction between a question and a declarative has syntactic consequences. There are some verbs that can only be merged with sentences with a question marker (27). There are also verbs that select for a declarative sentence and cannot be merged with questions (26).

(26) a. Ben Ali kitap oku-du-Ø diye düşün-dü-m
I Ali book read-PST-3SG that think-PST-1SG
'I thought that Ali read a book.'
b. *Ben Ali kitap oku-du-Ø mu düşün-dü-m
I Ali book read-PST-3SG Q think-PST-1SG
(27) a. *Ben Ali kitap oku-du-Ø diye merak.ed-iyor-Ø-um
I Ali book read-PST-3SG that wonder-IMPV-PRES-1SG
b. Ben Ali kitap oku-du-Ø mu merak.ed-iyor-Ø-um
I Ali book read-PST-3SG that wonder-IMPV-PRES-1SG
'I wonder if Ali read a book.'

The morphemes that indicate whether a sentence should be understood as a question or an assertion are called **Complementizers**, shortened to C. In (26) and (27), the absence of the complementizer leads to unacceptability.

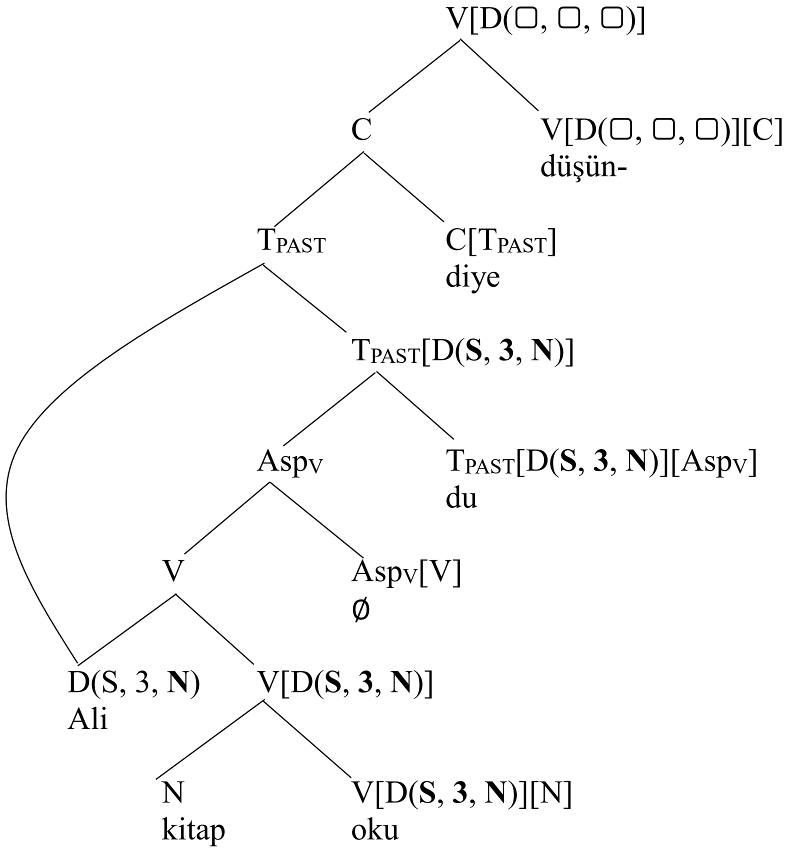
(28) a. *Ben Ali kitap oku-du-Ø düşün-dü-m
I Ali book read-PST-3SG think-PST-1SG
'I thought that Ali read a book.'
b. *Ben Ali kitap oku-du-Ø merak.ed-iyor-Ø-um
I Ali book read-PST-3SG wonder-IMPV-PRES-1SG
'I wonder if Ali read a book.'

This means that the verb *merak.et-* 'wonder' selects for a complementizer that has the Question feature, shortened the feature Q. The verb *düşün-* 'think' selects for a complementizer that lacks the Q feature, just a C. The categories of the complementizers in Turkish can be shown as in (29):

(29) The category of the complementizer *-mI* is "$C_Q[T_{PAST}]$" or "$C_Q[T_{PRES}]$"
The category of the complementizer *diye* is "$C[T_{PAST}]$" or "$C[T_{PRES}]$"
The category of the verb *merak.et* 'wonder' is "V[D(□, □, □)][C_Q]"
The category of the verb *düşün* 'think' is "V[D(□, □, □)][C]"

Under these assumptions, the syntactic representation of the graph projected by the verb *düşün-* 'think' can be shown as in (30):

(30)



As we have noted in the previous section, this is not the only strategy for embedding sentences inside sentences in Turkish. We can also use a nominalized embedded clause to express the thought associated with (26a).

(31)	Ben	Ali-nin	kitap	oku-duğ-un-u	düşün-dü-m
	I	Ali-GEN	book	read-DIK-3SG-ACC	think-PST-1SG
	'I thought that Ali read a book.'				

There are various properties of such sentences that need to be highlighted. Firstly, such embedded clauses are case-marked the way D-elements are case-marked. The case marker on the embedded clause depends on the verb that selects for it.

(32)	a. Biz	[Ali-nin	kitap	oku-duğ-un]**-u**	**düşün**-dü-k
	we	Ali-GEN	book	read-DIK-3SG**-ACC**	think-PST-1PL
	'We thought that Ali read a book.'				
	b. Biz	[Ali-nin	kitap	oku-duğ-un]**-a**	**inan**-dı-k
	we	Ali-GEN	book	read-DIK-3SG-**DAT**	think-PST-1PL
	'We believed that Ali read a book.'				
	c. Biz	[Ali-nin	kitap	oku-duğ-un]**-da**	**karar.kıl**-dı-k
	we	Ali-GEN	book	read-DIK-3SG-**LOC**	conclude-PST-1PL
	'We concluded that Ali read a book.'				
	d. Biz	[Ali-nin	kitap	oku-duğ-un]**-dan**	**şüphelen**-di-k
	we	Ali-GEN	book	read-DIK-3SG-**ABL**	doubt-PST-1PL
	'We suspected that Ali read a book.'				

Secondly there is number and person agreement between the embedded genitive-marked subject and the embedded predicate.

(33) a. Zeynep [**ben-im** kitap oku-duğ-**um**]-u düşün-dü-Ø
Zeynep **I-GEN.1SG** book read-DIK-**1SG**-ACC think-PST-3SG
'Zeynep thought that I read a book.'
b. Zeynep [**sen-in** kitap oku-duğ-**un**]-u düşün-dü-Ø
Zeynep **you-GEN.2SG** book read-DIK-**2SG**-ACC think-PST-3SG
'Zeynep thought that you read a book.'
c. Zeynep [**on-un** kitap oku-duğ-**un**]-u düşün-dü-Ø
Zeynep **s|he-GEN.3SG** book read-DIK-**3SG**-ACC think-PST-3SG
'Zeynep thought that s|he read a book.'
d. Zeynep [**biz-im** kitap oku-duğ-**umuz**]-u düşün-dü-Ø
Zeynep **we-GEN.1PL** book read-DIK-**1PL**-ACC think-PST-3SG
'Zeynep thought that we read a book.'
e. Zeynep [**siz-in** kitap oku-duğ-**unuz**]-u düşün-dü-Ø
Zeynep **you-GEN.2PL** book read-DIK-**2PL**-ACC think-PST-3SG
'Zeynep thought that you (all) read a book.'
f. Zeynep [**onlar-ın** kitap oku-duk-**ların**]-ı düşündü
Zeynep **they-GEN.3PL** book read-DIK-**3PL**-ACC think-PST-3SG
'Zeynep thought that they read a book.'

The genitive case-marking on the embedded subject pronoun and the agreement marker on the embedded predicate is identical to the ones we have seen in possessive constructions that we studied in *Chapter 4*.

(34) a. **ben-im** sorun-**um**
I-GEN.1SG problem-**1SG**
'my problem'
b. **sen-in** sorun-**un**
you-GEN.2SG problem**-2SG**
'your problem'
c. **on-un** sorun-**u**
s|he-GEN.3SG problem-**3SG**
'his/her problem'
d. **biz-im** sorun-**umuz**
we-GEN.1PL problem-**1PL**
'our problem'
e**. siz-in** sorun-**unuz**
you-GEN.2PL problem-**2PL**
'your problem'
f**. onlar-ın** sorun-**ları**
they-GEN.3PL problem-**3PL**
'their problem'

The presence of the genitive marker on the embedded noun and the possessive agreement on the embedded predicate suggests we are dealing with some kind of possessive construction, which is an element in the D category. This is also compatible with the fact that the whole embedded clause is treated as some type of D element that needs to be case-marked.

We first need to explain the fact that the embedded subject is in the genitive case and not in the nominal case as in simple clauses. To do so, we shall assume that the T head inside a nominalized embedded clause do not select for a D-element (similar to what we have observed with the tense category when the subject is not a D-element but a noun).

(35) The category of the (embedded) tense morpheme is “T[$\mathrm{Asp_V}$]” or “T[Asp]”

We suggest that the suffix *–K* functions as a complementizer that selects for a tense category. Crucially, this selected temporal category is “T” and not “$\mathrm{T_{PAST}}$” or “$\mathrm{T_{PRES}}$”. This guarantees that the subject is not assigned any case feature at the T-level. Secondly, the complementizer that selects for the temporal category *T* must, then, be selected by a determiner. To distinguish the complementizer *–K* from other complementizers we have seen in (29), we shall use the category C_N for it (as a shorthand for the nominal complementizer).

(36) The category of the nominal complementizer *-K* is “$\mathrm{C_N}$[T]”

The possessive determiner in embedded clauses selects first for a nominal complementizer and then for a D-element that occupies the embedded subject position. This possessive determiner is also responsible for assigning the genitive case to the embedded subjects. This is a type of determiner

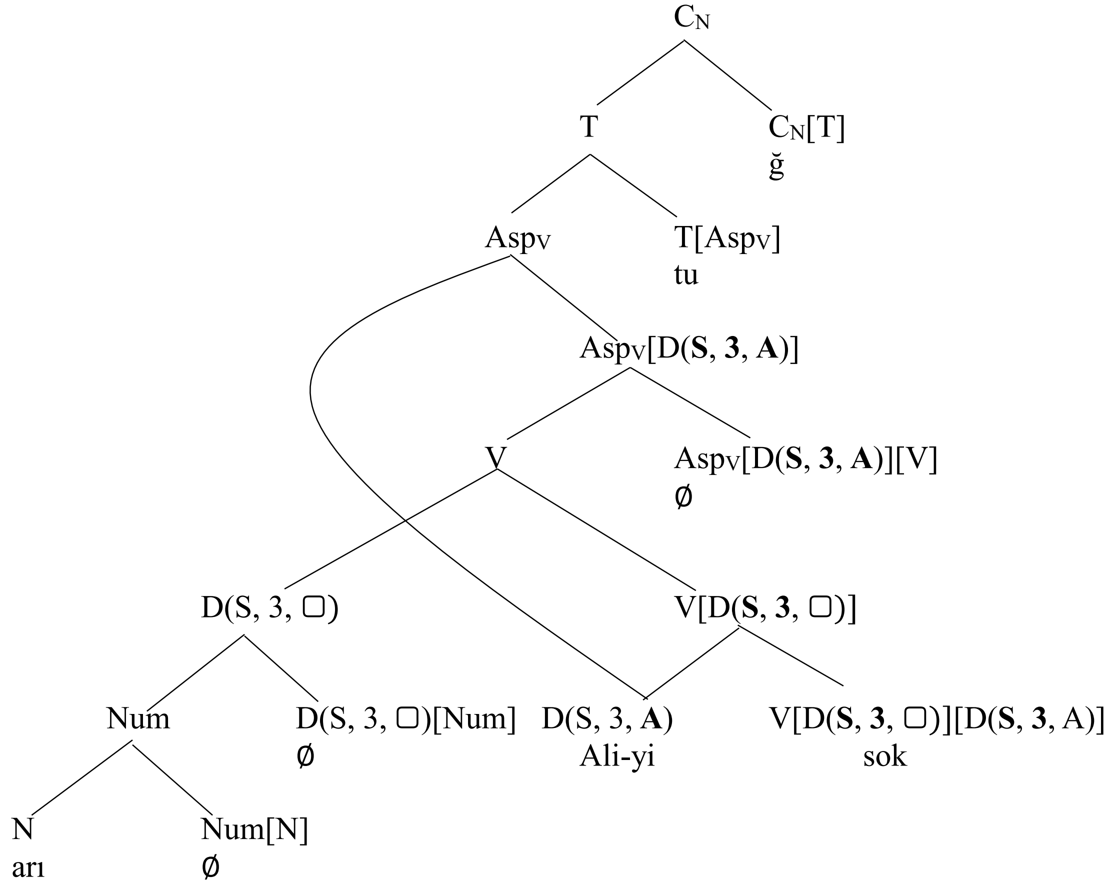
(37) The category of the clausal possessive determiner is “D(S, 3, □)[D(□, □, G)][$\mathrm{C_N}$]”
The category of the clausal determiner *–I(n)* is “D(S, 3, □)[$\mathrm{C_N}$]”

Let us see all these claims at work in the representation of the sentence in (22a), repeated in (38) for convenience.

(38)	Ben	[arı-nın	Ali-yi	sok-tuğ-un]-u	duy-du-m
	I	bee-GEN	Ali-ACC	sting-DIK-3SG-ACC	hear-PST-1SG
	‘I heard that the bee stung Ali.’				

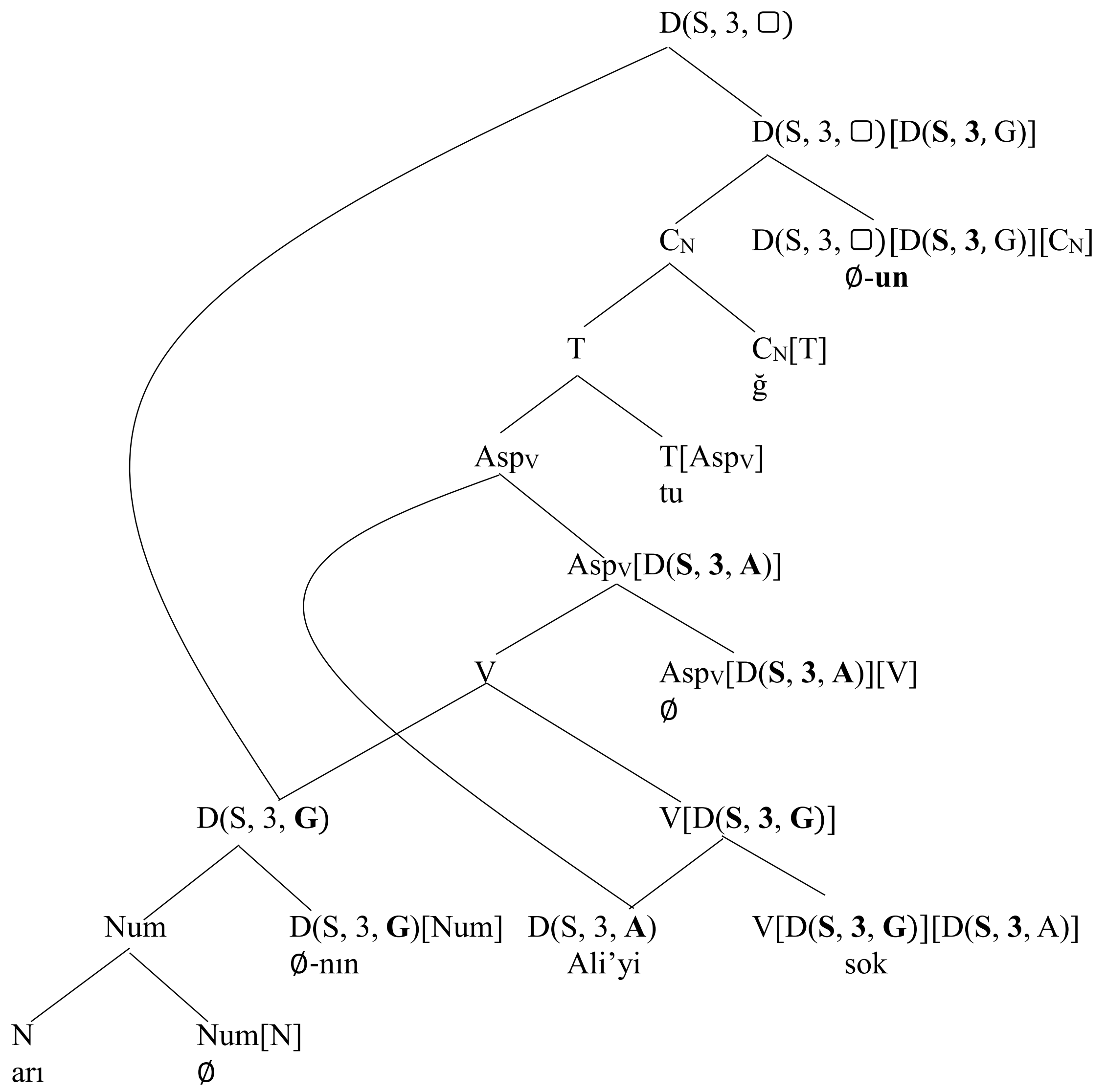
The representation of the graph before the insertion of the possessive determiner in the embedded sentence is shown in (39).

(39)



Notice that at this point in the derivation, the subject has not yet been assigned a case feature. In embedded clauses with the complementizers *diye* or *mI*, this could be done at the Tense layer. In nominalized embedded clauses, however, this option is not available. We then insert the clausal possessive determiner to the graph, thanks to which we can assign the case feature to the subject.

(40)

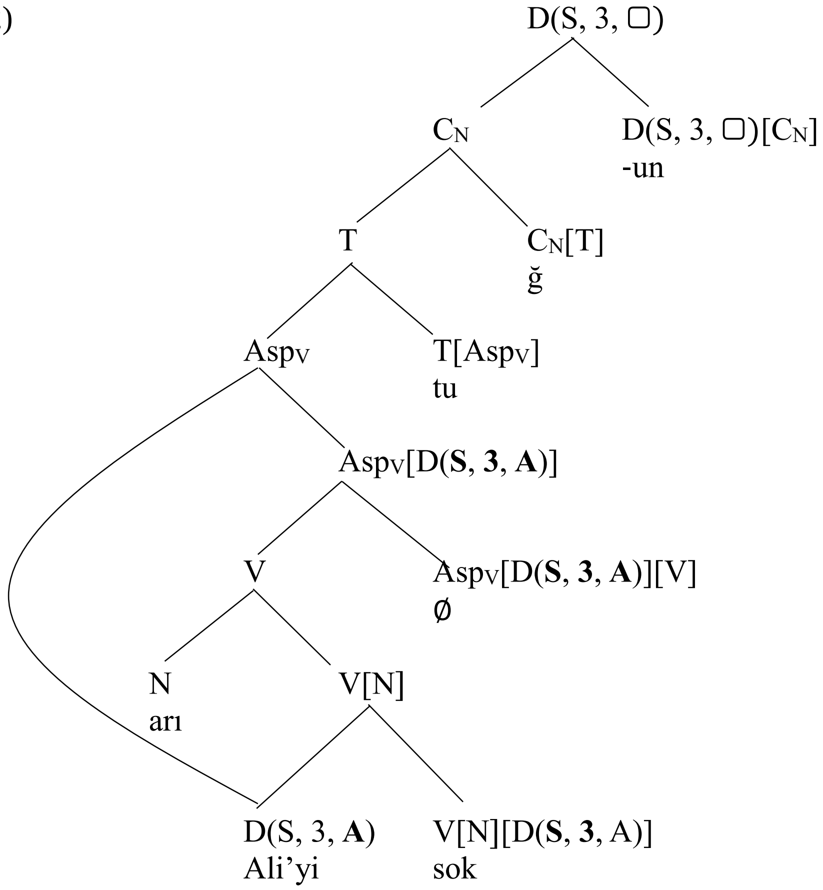


The end result is a graph labeled with the category *D(S, 3, ▢)*. When merged with a verb, this category can get its case feature directly from the verb. This provides an explanation for the fact that the case of nominalized embedded clauses is determined by the verb that selects for them. The agreement marker that appears on the clausal possessive determiner is governed by the same rules that are active possessive agreement. These rules of realization are repeated from Chapter 4:

(41) The rules of realization for the possessive agreement suffixes
where NUM is one of {S, P},
… D(NUM, 3, ▢)[D(**S**, **1**, G)] … ⇔ -(I)m
… D(NUM, 3, ▢)[D(**S**, **2**, G)] … ⇔ -(I)n
… D(NUM, 3, ▢)[D(**S**, **3**, G)] … ⇔ -(s)I(n)
… D(NUM, 3, ▢)[D(**P**, **1**, G)] … ⇔ -(I)mIz
… D(NUM, 3, ▢)[D(**P**, **2**, G)] … ⇔ -(I)nIz
… D(NUM, 3, ▢)[D(**P**, **3**, G)] … ⇔ -lArI

In those cases where the embedded subject *arı* 'bee' is a nominal element, we need to use the clausal determiner *–I(n)* that selects only for the nominal complementizer.

(42)



In this way, we can account for the fact that when the embedded subject is not case-marked, it must come between the object and the verb.

(43) a. Ben [Ali-yi arı sok-tu-ğ-un]-u duy-du-m
I Ali-ACC bee sting-TNS-C-D-ACC hear-PST-1SG
'I heard that Ali got stung by (a) bee(s).'
b. *Ben [arı Ali-yi sok-tu-ğ-un]-u duy-du-m
I bee Ali-ACC sting-TNS-C-D-ACC hear-PST-1SG
Int. 'I heard that Ali got stung by (a) bee(s).'

The final issue we will explore in this book is the status of embedded subjects with the category *Num*. Recall that verbs can select for objects in the category *Num*.

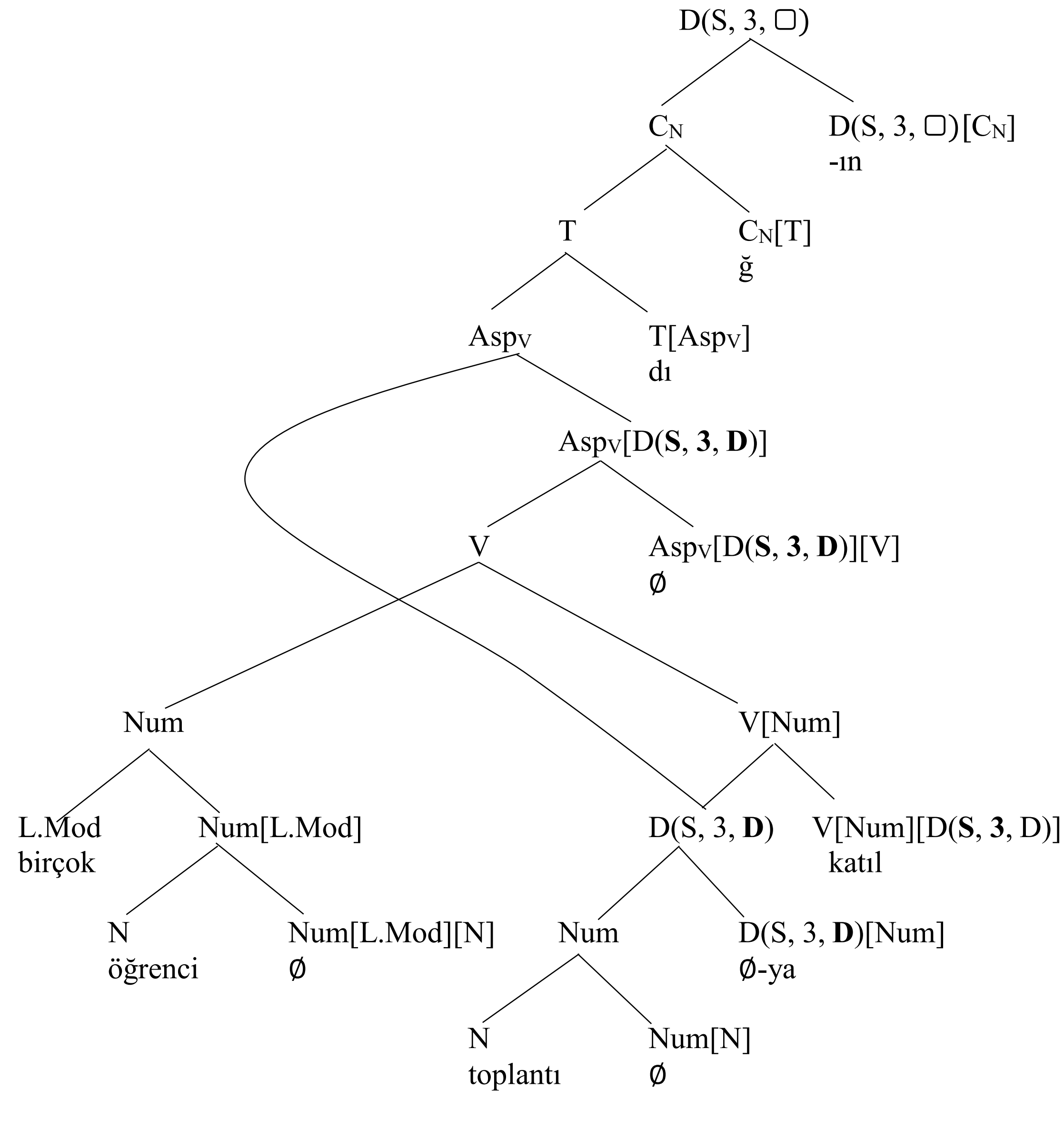
(44) a. Ali üç kitap oku-du-Ø
Ali three book read-PST-3SG
'Ali read three books.'
b. Ali birçok kitap oku-du-Ø
Ali many book read-PST-3SG
'Ali read many books.'
c. Ali birkaç kitap oku-du-Ø
Ali several book read-PST-3SG
'Ali read several books.'

We could then expect verbs to also select for a subject in the category *Num*. This is, indeed, the case. We discussed the relevant examples already in *Chapter 1*.

(45)	a. Ben toplantı-ya	birçok öğrenci		katıl-dı-ğ-ın-ı	duy-du-m
	I meeting-DAT	many student		attend-TNS-C-D-ACC	hear-PST-1SG
	‘I heard that many students attended the meeting.’				
	b. *Ben birçok öğrenci		toplantı-ya	katıl-dı-ğ-ın-ı	duy-du-m
	I many student		meeting-DAT	attend-TNS-C-D-ACC	hear-PST-1SG

In drawing the graph for (45a), we must first note that the category of the verb *katıl* ‘attend’ is either “V[D(□, □, □)][D(□, □, D)]”, “V[Num][D(□, □, D)]” or “V[N][D(□, □, D)]”. Given this fact, the embedded clause in (45a) can be represented as in (46)

(46)



In the exercises you will be asked to explain why it is that replacing the low modifier *birçok* 'many' with the high modifier *çoğu* 'most' leads to unacceptability.

(47) a. *Ben toplantı-ya çoğu öğrenci katıl-dı-ğ-ın-ı duy-du-m
I meeting-DAT most student attend-TNS-C-D-ACC hear-PST-1SG
'I heard that many students attended the meeting.'
b. *Ben çoğu öğrenci toplantı-ya katıl-dı-ğ-ın-ı duy-du-m
I most student meeting-DAT attend-TNS-C-D-ACC hear-PST-1SG

This is all that we have to say about the syntactic representation of subjects in Turkish.

Exercises:

1. Draw syntactic graphs for the following expressions. Analyze *hızlı* 'fast', *saatlerdir* 'for hours' and *çok* 'a.lot' as adverbial expressions and analyze *gir* 'enter' as a verb with two arguments.

a. Biz Zeynep-in Ali-ye bak-tı-ğ-ın-ı san-mış idi-k
we Zeynep-GEN Ali-DAT look-TNS-C-3SG-ACC think-ANT PST-1PL
'We thought that Zeynep looked at Ali.'
b. Onlar saat-ler-dir Zeynep-in hızlı araba sür-dü-ğ-ün-ü
They hour-PLU-ADV Zeynep-GEN fast car drive-PST-C-3SG-ACC
düş-üyor idi
think-IMPV PST-3SG
'For hours, they thought that Zeynep drove a car fast'
c. Ben oda-ya üç öğrenci gir-di-ğ-in-e
I room-DAT three student enter-TNS-C-D-DAT
çok inan-dı-m
a lot believe-PST-1SG
'I really believed that three students entered the room.'
d. Biz siz-in onlar-ın birçok kitap oku-du-k-ların-ı
we you.all-GEN they-GEN many book read-TNS-C-3PL-ACC
bil-di-ğ-iniz-i düşün-dü-k
know-TNS-C-2PL-ACC think-PST-1PL
'We thought that you knew that they read many books'

2. Explain whether our analyses in this chapter can account for the judgments associated with the following sentences.

a. *Ben toplantı-ya çoğu öğrenci katıl-dı-ğ-ın-ı duy-du-m
I meeting-DAT most student attend-TNS-C-D-ACC hear-PST-1SG
Int. 'I heard that most students attended the meeting.'
b. *Ben çoğu öğrenci toplantı-ya katıl-dı-ğ-ın-ı duy-du-m
I most student meeting-DAT attend-TNS-C-D-ACC hear-PST-1SG
Int. 'I heard that most students attended the meeting.'
c. Ben çoğu öğrenci-nin toplantı-ya katıl-dı-ğ-ın-ı duy-du-m
I most student-GEN meeting-DAT attend-TNS-C-D-ACC hear-PST-1SG
'I heard that most students attended the meeting.'
d. *Ben çok Ali-nin oda-ya gir-di-ğ-in-e inan-dı-m
I a.lot Ali-GEN room-DAT enter-TNS-C-D-DAT believe-PST-1SG
Int. 'I really believed that Ali entered the room.'

Selected Readings and References:
The idea that subjects have an initial low position is known as VP-Internal Subject Hypothesis (Speas, 1986; Koopman & Sportiche, 1991). The subject-related data in Turkish we have relied on in this chapter were discussed in Cagri (2005); Kornfilt (2008) and Öztürk (2009).

Partial Answer Key:

Chapter 2 Exercise Solutions:

3. Only (c) should be acceptable and the meaning should be "the status/property of lacking a book-shelf." As a matter of fact, this is the only interpretation available for *kitaplıksızlık*.
4. (zıs)^n-kitap-(sız)^m where $n \leq m \leq n+1$
5. (a, 1); (b, 2); (c, 3); (d, 3); (e, 6).

Chapter 3 Exercise Solutions:

2. (In the exercise, we are asked to assume that the rules discussed in Chapter 3 exhaust all the rules of nominal syntax in Turkish)
   We cannot explain the unacceptability of (a), (b) and (g).
   We cannot explain the acceptability of (d).
   We can explain the unacceptability of (c) and (h)
   We can explain the acceptability of (e) and (f)
3. (a) is expected to be unacceptable and it is.
   (b) is expected to be unacceptable but it is acceptable.
   (c) is expected to be unacceptable and it is.
   (d) is expected to be unacceptable but it is acceptable.
   (e) is expected to be unacceptable and it is.

Chapter 4 Exercise Solutions:

2. The category of the silent determiner in Turkish is
   "D(NUM, 3, □)[Num]" or **"D(NUM, 3, □)[H.Mod][Num]"** (whichever works)
   where *NUM* is one of *{S, P}* (reflecting the feature on the selected category *Num*).
   The category of the silent possessive determiner in Turkish is
   "D(NUM, 3, □)[D(□, □, G)][Num]" or **"D(NUM, 3, □)[D(□, □, G)][H.Mod][Num]"** (whichever works),
   where *NUM* is one of *{S, P}* (reflecting the feature on the selected category *Num*)

Chapter 5 Exercise Solutions:

2. (a) The category of "için" is "P[D(□, □, G)]"
   (b) The category of "doğru" is "P[D(□, □, D)]"
   (c) The category of "kadar" is "P[D(□, □, D)]"
   (d) The category of "beri" is "P[D(□, □, B)]"
   (e) The category of "göre" is "P[D(□, □, D)]"
   (f) The category of "ötürü" is "P[D(□, □, B)]"

Chapter 6 Exercise Solutions:

2. (2.1, Yes); (2.2, Yes); (2.3, No); (2.4, Yes); (2.5, Yes); (2.6, No); (2.7, F); (2.8, J); (2.9, F); (2.10, dgab)
3.1. the adverbial linker is of the category Asp[Adv][Asp]. This can explain the unacceptability of (b), where Aspect is verbal, i.e. Asp$_V$

Chapter 7 Exercise Solutions:

(2a) is expected to be unacceptable and it is.
(2b) is expected to be unacceptable and it is.
(2c) is expected to be acceptable and it is.
(2d) is expected to be unacceptable and it is.

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